

NiCE IA & QC v12.5 Fundamentals

NiCE Interaction Analytics & Quality Central v12.5

Global Education Services

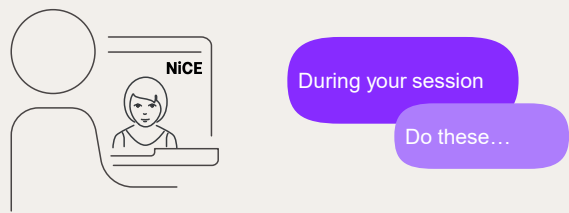


Course Overview

NiCE Nexidia Analytics & Quality Central
Fundamentals

Create a
NiCE..
world 

Training Expectations



Setup Out of Office

An icon representing email, showing an envelope with a checkmark and an '@' symbol.

Be on Time

An icon of a clock face with hands.

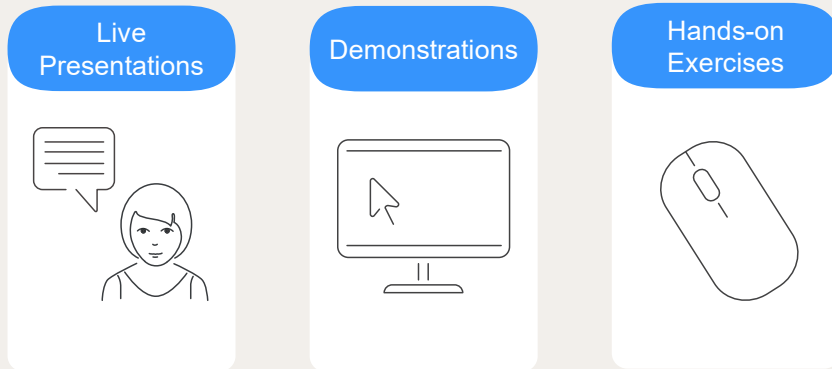
Ask Questions

An icon of two speech bubbles, one overlapping the other.

NiCE

Even though this is a virtual training environment, it should be treated like a classroom environment. Setup your email out of office and update your voicemail that you will be unavailable during training. Be punctual when returning from the breaks and lunches. Active participation is appreciated and contributes to the learning process.

Instructor-led Training



This training is delivered over the web as 'Virtual Classroom Training' which combines live presentations and demonstrations.

Every day during this training course, we will open a Teams session which will remain open throughout the day.

This is your platform to communicate with the trainer, ask questions and comment. It will also be used for the live presentations and demos.

NICE Training Environment

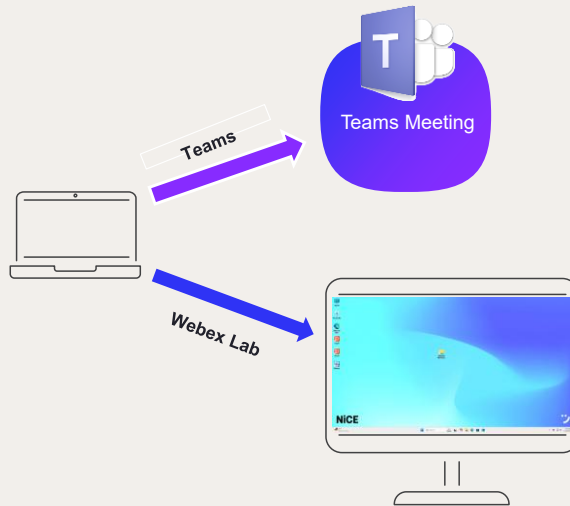
Training Environment consists of 2 sessions (Teams + Webex)

Teams Meeting for Trainer Presentations, Demos

Webex Lab Session for Hands-on Exercises



Two monitors are required so both sessions can be visible at the same time



- This is a diagram of the virtual environment. A Teams session will be used for the audio conference. A second Webex connection will be made to computer.

Navigating the Teams Session

01:34

Take control

Pop out

Chat

People

Raise

React

View

Rooms

Apps

More

Camera

Mic

Share

Leave

Pop-out for Full Screen

Participants

React

Mute / Unmute

NiCE Product Intro and Application Design

NiCE Training

Create a NiCE world

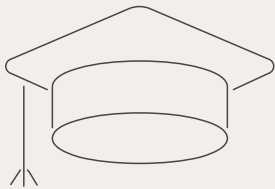
Participant 1

Participant 2

NiCE

A stylized logo icon consisting of two blue dots above a black curved line, resembling a smile or a checkmark.

Certification



Need 80%
or better



Practical



Theoretical

To receive a certification, you must have a scores above 80% on the practical and theoretical exams.

Course Materials



Course Book



Exercise Book

Let's take a look!



agenda

Day 1

- Using NIA
- NIA Architecture Part 1
- NIA Architecture Part 2

Day 2

- Introduction to QC
- QC System Configuration
- Using QC
- QC Architecture

Day 3

- Certification





Before we begin...

- What is your experience with NiCE Solutions?
- Where are you located?
- What do you want to learn about in this course?





Thank You



INTRODUCTION TO ANALYTICS

NiCE Interaction Analytics & Quality Central Fundamentals

Create a
NiCE..
world 

objectives

- Determine how Analytics impacts NiCE customers
- Describe Analytics main products: Search and Discovery
- Identify different speech Analytics technologies



Generating value from Analytics



Analytics (NIA)



Quality Management (QC)

- Tailored usage depending on operational, customer, or strategic focus
- Eliminate repeat calls
- Optimize Average Handle Time (AHT)
- Reduce agent knowledge gaps
- Identify root causes of dissatisfaction
- Predict customer churn
- Leverage cross-sell opportunities

Uses of Quality Management & Interaction Analytics

•Role-Based Insights

- Different stakeholders analyze the same interactions for varied insights
- Tailored usage depending on operational, customer, or strategic focus

•Operational/Contact Center Level

- Eliminate repeat calls
- Optimize Average Handle Time (AHT)
- Reduce agent knowledge gaps

•Customer Loyalty & Retention

- Identify root causes of dissatisfaction
- Predict customer churn

•Marketing & Sales

- Leverage cross-sell opportunities
- Measure market awareness and brand perception

•Strategic Intelligence

- Gain insights into competitive threats
- Discover new market opportunities

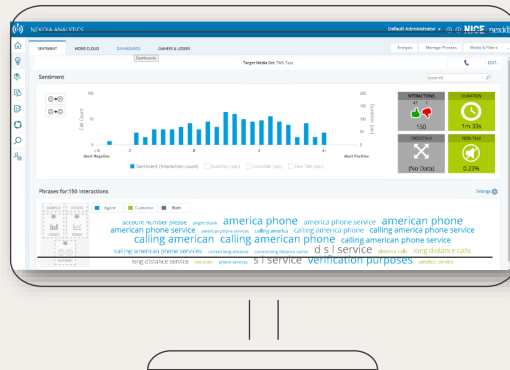
Those are the main Interaction Analytics products, in this course we will cover the Interactions analytics and the Quality Central products



Introduction To Analytics



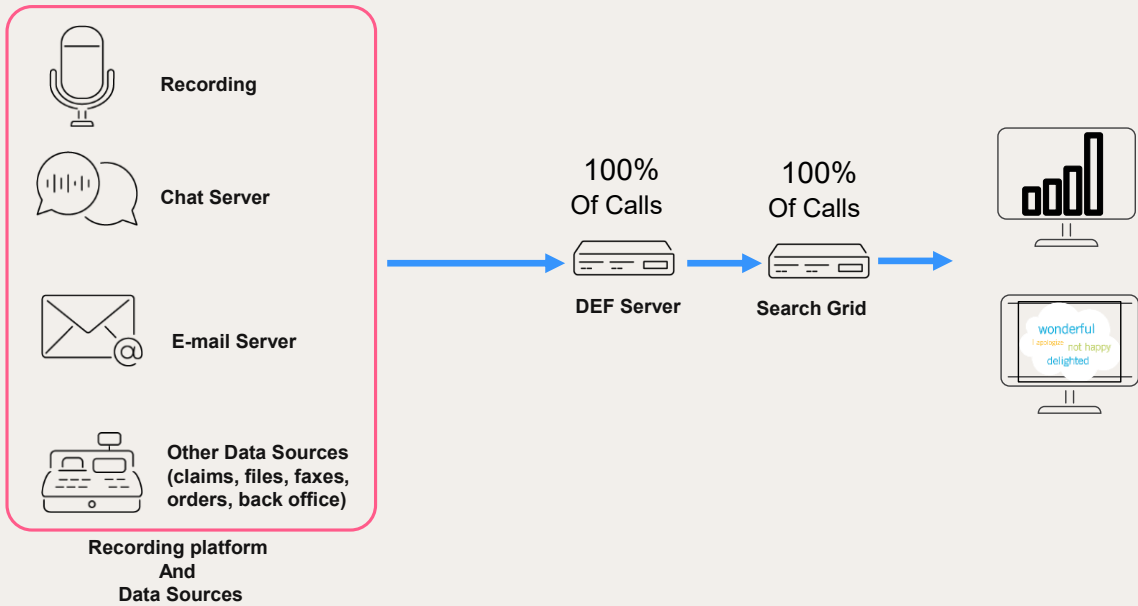
Interaction Analytics



Interaction Analytics is an interactions, analytics, and quality management platform. It imports agent information, audio files, text interactions, and related metadata from the recording system. The analytics engine processes and analyzes all interactions, and all of the agents are evaluated and scored on their performance. Those actions provide you with business insights. For example, you can see what your customers are saying about their product and service level, who the top agents are, and which agents need additional training.



Data Exchange Framework (DEF)



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Interaction Analytics is not a recording platform; it is an interaction analytics and quality management platform to analyze the calls; we first need to import them from the recording system.

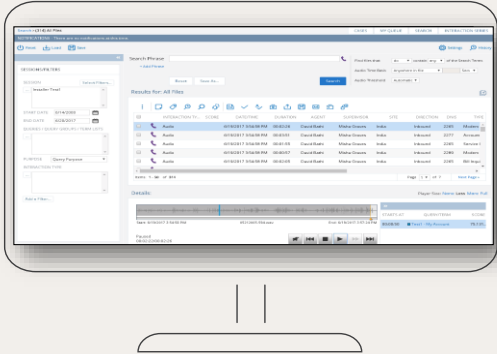
For that purpose, Interaction Analytics has the Data Exchange Framework tool, which extracts the audio files, text interactions, and related metadata from the recording system. DEF processes the data according to a data structure expected by Interaction Analytics' application server, such as QC & IA, and copies them to the Interaction Analytics servers.

All interactions are ingested into the system and are processed by the analytics engines.

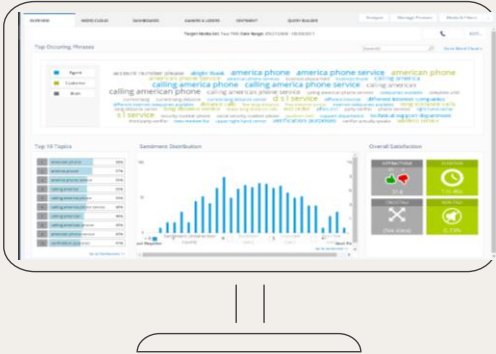


Analytics Products

Search



Discovery



Interaction Analytics has two products:
Search
Discovery



Analytics Components



Query Builder

Application used to build queries to search Words and Phrases



Search Grid

Analytic Engine - performs the search for defined words & phrases. Creates the phonetic index file & transcription



Language Pack

Unique for each language, it is the set of phonetic model files used to perform phonetic analysis



Phrase Pack

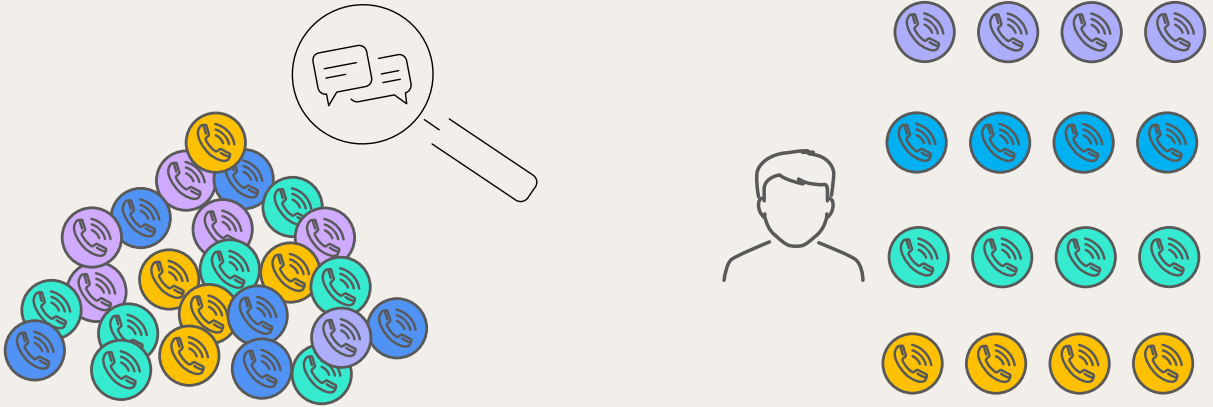
Contains both the language dictionary and words and phrases that are not part of the standard language; industry specific phrases\common phrases



Search



What our customers are saying?



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After we imported the interactions from the recorder, we have many interactions that we wish to sort while we are inserting them into the system, we would like to find all the interactions that are related to Billing questions; all the interactions where the customer was angry.

All the interactions related to Tech support.

All the interactions where the customer called to cancel the service, etc.



Visual Query Builder

The screenshot displays the Visual Query Builder interface. At the top, there are tabs for 'EDIT' and 'TEST'. Below the tabs, there are fields for 'Name' (Angry Customers_8.6.0.143339 (1)), 'Language' (NAEnglishTeleUniversal-8.7.7.172728), 'Type' (Audio), 'Version (Internal)' (2.0), and 'Invert Final Score' (checkbox). There is also a 'Description' field and a 'Target Media Set' dropdown (TMS2). A 'Data Set' field shows '100 files' and a 'Show Unused Items' checkbox. Below these fields, there are two main sections: 'PHRASE UNITS' and 'EXPRESSIONS'. The 'PHRASE UNITS' section has a list of phrases with a '+' icon and a '6d' icon: 'shocked', 'frustrated', 'this is ridiculous', 'that is crazy', 'I want help immediately', 'no need to get angry', and 'I'm always holding'. The 'EXPRESSIONS' section has a search bar and a list of operators: 'AND', 'OR', 'BEFORE', 'AND NOT', 'SUBSET', and 'TIMING'. The main query area shows a sequence of phrases: 'experience' followed by an equals sign, 'OR', 'shocked', 'frustrated', 'this is', 'that is crazy', and 'I want help'. Below this, there are more phrases: 'no need to', 'I'm always', and a 'DROP ITEM HERE' button. The interface is clean and modern, with a light blue and white color scheme.

To sort the interactions, we need to define what words and phrases we would like to search for in the content of each interaction.

Here is an example of a query that searches for all the interactions where the customer was angry.

We would look for phrases such as: “I want help immediately, ““this is crazy “ etc.



Query

Example: Cancellation-Intent Query

Cancel = OR ("cancel my service", "close my account", "why would you like to cancel", \ "why do you want to close")
Final = FIRST_60 (Cancel)

Example: Cancellation-Actual Query

Cancel = OR ("cancellation confirmation number", "cancellation number", \ "legal disclaimer regarding your cancellation")
Final = LAST_90(Cancel)

Let us look at the following example:

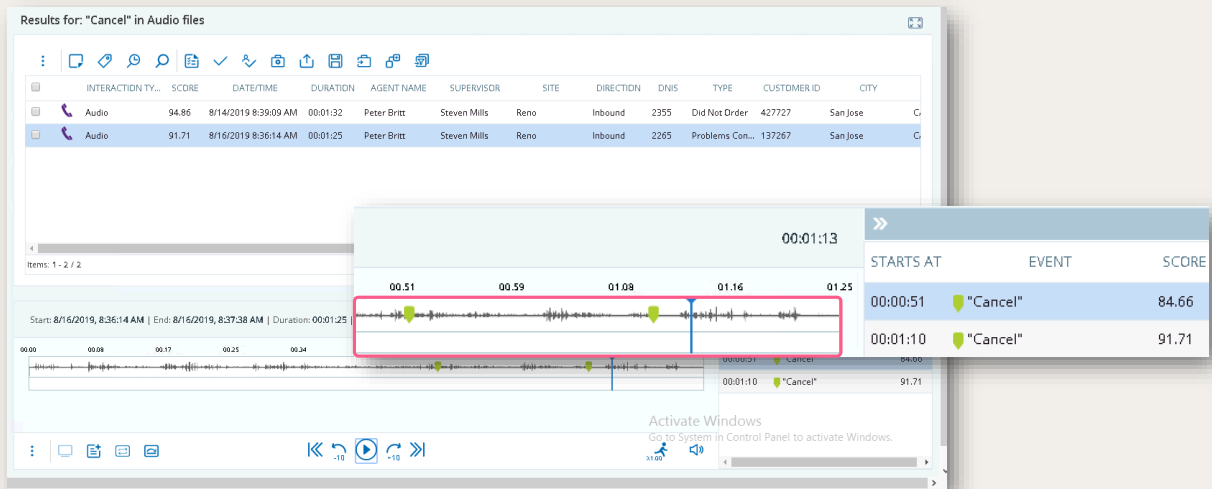
1. We want to find all the interactions where the customer called with the intend to cancel their account. We will look for phrases such as “cancel my service”, “close my account”, and we will look for it at the beginning of the call.
2. We want to find all the interactions where the customer canceled their account. We will look for phrases such as “cancellation confirmation number”, “cancellation number”, and we will look for it at the end of the call.

For example: if interaction appears in the first group but not in the second, meaning the customer called with the intent of cancelling their account, but at the end they did not cancel.

We can listen to the calls and learn how the agent handled the case and use this interaction as a coaching example to all other agents.

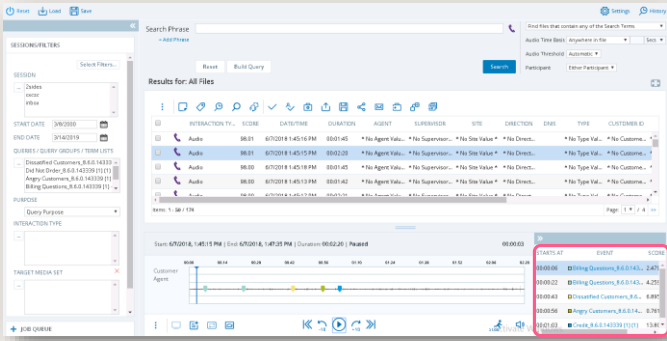


Search



Once the queries are built, interactions matching the results appear in our search results with indications of where the phrases are found in the call and a score determining how close of a match it is to our phrase.

Search



Details Dialog - Google Chrome
nia/niA/Search/QueryDetailsDialog.aspx?ingestId=15&resultId=27

Query: Repeat Calls_8.7.7.172728

exp1 = "called in before"
exp4 = "called last month"
exp5 = "person I spoke with"
exp3 = "second time I've called"
exp2 = "they said"
exp0 = "this is the third time I've called"
repeat = OR(exp0, exp1, exp2, exp3, exp4, exp5)

Properties

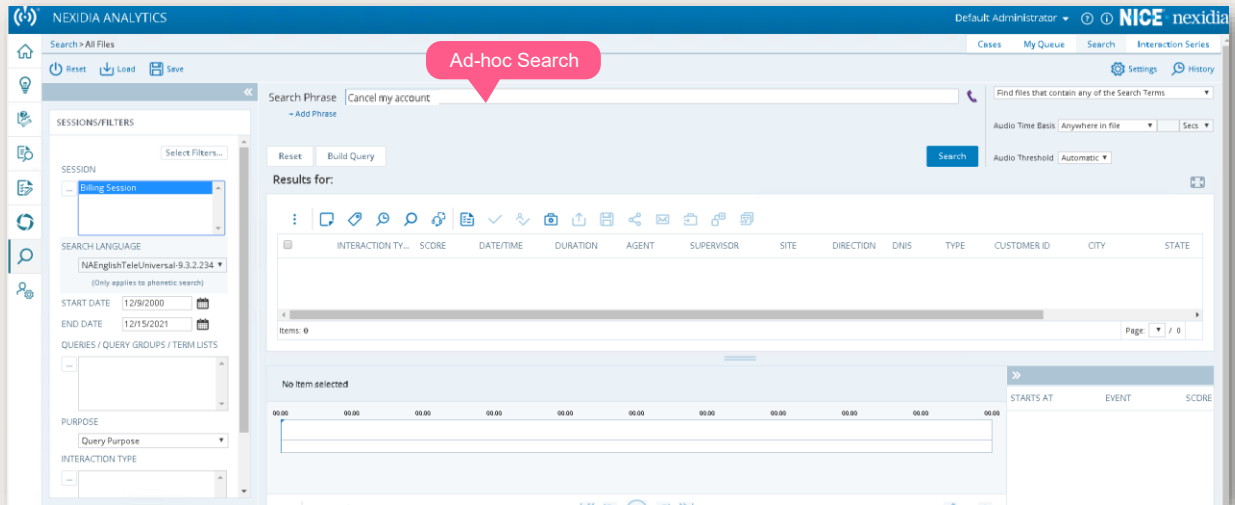
Name: Repeat Calls_8.7.7.172728
Language Pack: NAEEnglishTeleUniversal-8.7.7.172728
Creation Date: 1/8/2018 4:31:01 PM
Description:
Created By: superuser
Modified Date: 1/8/2018 4:31:01 PM
Last Modified By: superuser
Version: 1
Thresholds:
0.82 (Targeted Listening - High)
0.82 (Categorization/Reporting - Medium)
0.82 (Compliance - Low)
70.00 (Custom)
Has Search Model: No

We can see on the player the location of the query detection in the call duration.

We do not know which word or phrases were said (we will know it only by listening to the audio), by clicking on the query name, a window with the query definitions will appear.



Search

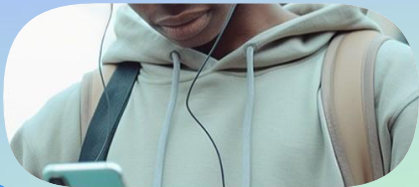
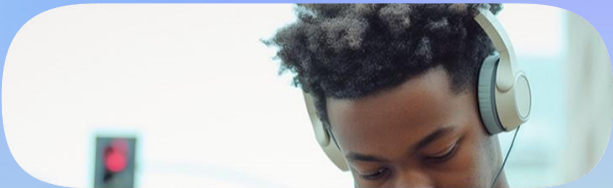


We can also perform an ad-hoc search. This is a search for a specific topic without building a complex query. These can only be performed on simple phrases.

For example, we want to find all the interactions where a customer called to complain about a specific product. We can create an ad-hoc search for the name of that product.



Phonetics Indexing



What is Phonetics?

Washington

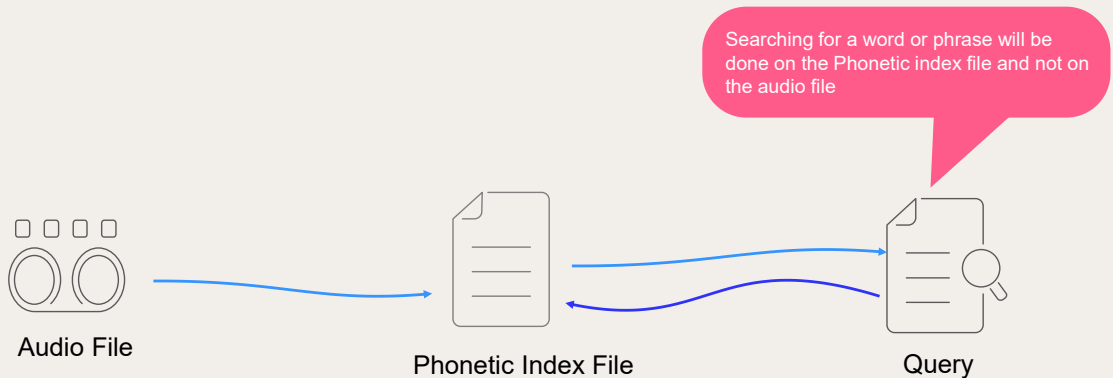
w-aa-sh-in-ng-t-ah-n

Interaction Analytics uses phonetics to break down words into distinct sounds called phonemes -sequence of sounds that make up words)

NOTE: these are only representations and not actual phonemes.



How Phonetics Works



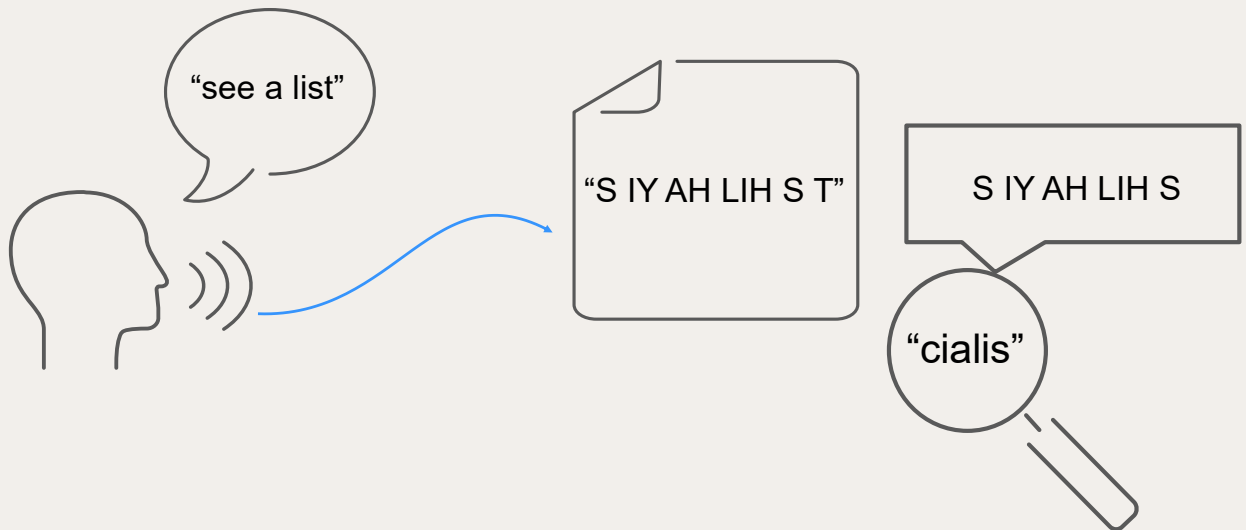
The audio file is converted into phonemes, and these are compiled into a phonetic index file. When a search is performed, the query criteria is also converted into phonemes. The phonemes in the search criteria can then be compared to the Phonetic Index Files and matches will be returned in the results.

NOTE: that once the indexing has been completed, the original media files are not involved in the search process.

Phonetics requires a language pack to understand the spoken language. The language pack provides phonemes of words in that language; for example, UK English sounds different from USA English.

An example of a phonetics search. During the call, the customer says, “transfer me to the Washington office.” the call is converted into a phonetic Index file containing the relevant phonemes. Now we want to search for the word “Washington.” The search term is also converted to a phonetics representation, and we search for the phonemes sequence in the index file to see if there are any matches. Each resulting hit is given a score based on the confidence level in the match

How Phonetics Works – Example 2



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The greatest advantage of a phonetic recognition system is that words that are not in a predefined vocabulary can still be found, provided the phonemes are recognizable. For instance, when searching for the name of the drug "Cialis", the term may still be found in the text if that sequence of phonemes exists, "S IY AH LIH S." The disadvantage is that, since there are many possible sequences in the index file, the term may be found in many places where it was never actually said, for instance, if the actual words were "see a list." The phonemes are very similar, but it's a false positive match.

Phonetics systems are also typically faster at processing (turning the phone calls into data), mostly because the size of the "vocabulary" being used is very small as phonetics relies only on the sounds of the language, and there are only a few unique phonemes in most languages. However, the search process is much slower since phonemes cannot be efficiently searched the same way that words can be searched.

Phonetic Search

INDEX

- Audio is processed with a phonetic recognizer to generate a searchable index file that stores the phonetic content of the speech.

SEARCH

- Search term (word or phrase) is defined, and a number of phonetic index files are searched to retrieve matches for the search term.

NOTE

- Once indexing has been completed, the original media files are not involved in the search process

Audio mining is a technique used to search audio for occurrences of spoken words or phrases. This is a 2 steps process:

1. Index – PAT file - Phonetic Access Track; index file is only created once.
2. In the second stage (search stage), a search term is defined (e.g., a word or phrase), and one or more index files are searched for all occurrences that match the specified search term.

Here's some additional information about phonetic searches
It does not use a dictionary – this makes it easy to search for uncommon words



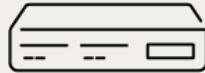
Discovery



Discovery Flow



 Target Media Set



Search Grid



Discovery Application

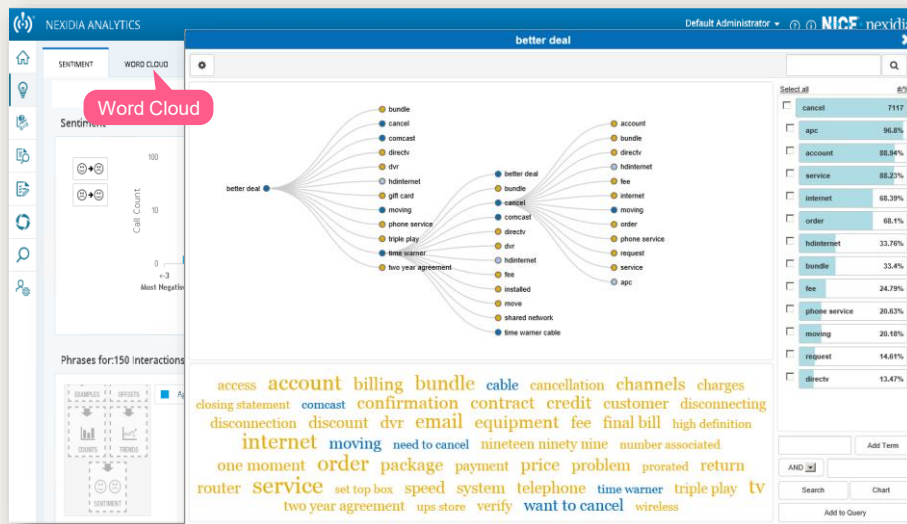
Unlike the Search, the Discovery product is fully automated. We do not need to build complex queries, and in most cases, we don't have to know what we are trying to search. Discovery will answer the question of "Why?"

The word cloud will show us what is happening on a specific population\set of calls.

There is no need to build complex queries; all we need to do is define which interactions we want to process.

The analytic engine will process those interactions, and the application will present to us what our customers are saying.

Discovery Flow- Word Cloud



As we can see, the words cloud is color-coded to differentiate between the agent and the customer. The larger the font represents the more frequently used words.

The related phrases functionality shows relationships between topics, providing context for root cause analysis. Related phrases are the most frequently spoken words/phrases that occur 30 seconds before or after the phrase that was selected.

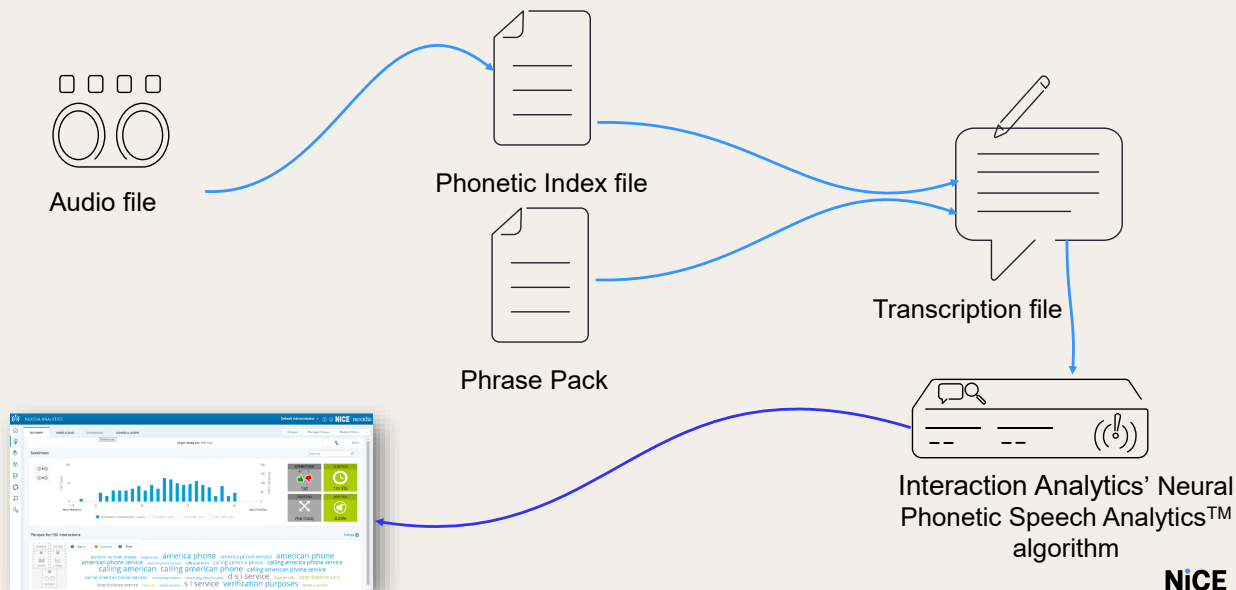
Discovery helps you identify new topics – uncovering what is occurring on your calls. Discovery will help you answer the question “What should I investigate?” and helps provide insight into what should be analyzed further.



How Discovery Works? Using Transcription



How Transcription Works



Large Vocabulary Continuous Speech Recognition (LVCSR) begins by recognizing phonemes much like a phonetic system, but then applies a dictionary or language model of potentially thousands of words and phrases to produce a full transcription. In LVCSR, every word is recognized and nothing is thrown away or skipped. While the initial process of recognizing the full transcript, not just the individual phonemes, requires more processing power than phonetics only recognition, the resulting transcript makes it much easier and faster for contact centers to search and use the gathered information. The initial processing of the audio also takes a bit longer than with a phonetic approach because of the large vocabulary that needs to be analyzed, even though search time is faster and more accurate.

Phrase Pack – Dictionary + words and phrases that are not part of the standard language, such as industry specific phrases and terminology. For example, a pharmaceutical company will have a different phrase pack than an insurance company. So, for example, Cialis is a new drug, and it is not part of the phrase pack, so the engine will not know this word and will not be able to transcribe it (it is also not a known word in the English language). Therefore, there is a need to maintain & update the phrase pack constantly.

Once we have the transcription file, a text-mining algorithm is applied to the transcribed file. It counts how many times each word appears in the file, what are the words that are adjacent to it (phrases relations) etc. The outcome of this algorithm is the word cloud. There is no need to define which words & phrases to search for; the algorithm will present to us what our customers are saying (what are the most common words and phrases in those interactions).



Speech Analytics Technologies

Search Product

Phonetic Indexing

- Faster audio file processing
- No dictionary dependency
- 100% of the calls are processed

Discovery Product

Transcription

- More accurate
- Need Dictionary
- Word Cloud
- Not every call is processed

There are two main speech analytics technologies – **Phonetic Indexing** and **Transcription (Speech to Text)**

Phonetic indexing is much faster than transcription and can analyze any word, without dependency on a dictionary. Transcription provides root cause analysis that is not available with phonetic indexing.

Interaction Analytics combines both into a powerful methodology that provides the best of both worlds:

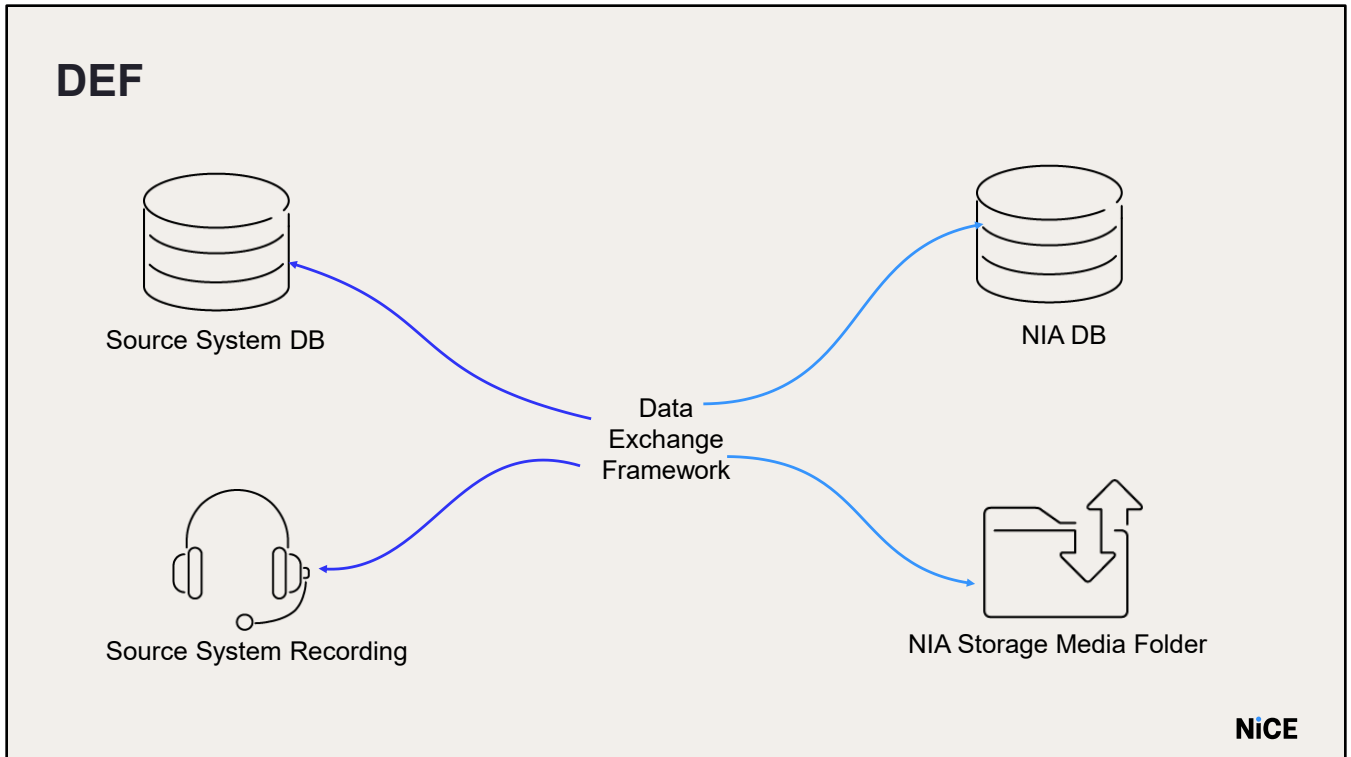
1. Analyzing 100% of calls
2. Fast time-to-insight
3. Highest business accuracy

With phonetic search, you need to know what to look for. Discovery will show you what your customers are saying, no need to search for a specific phrase.

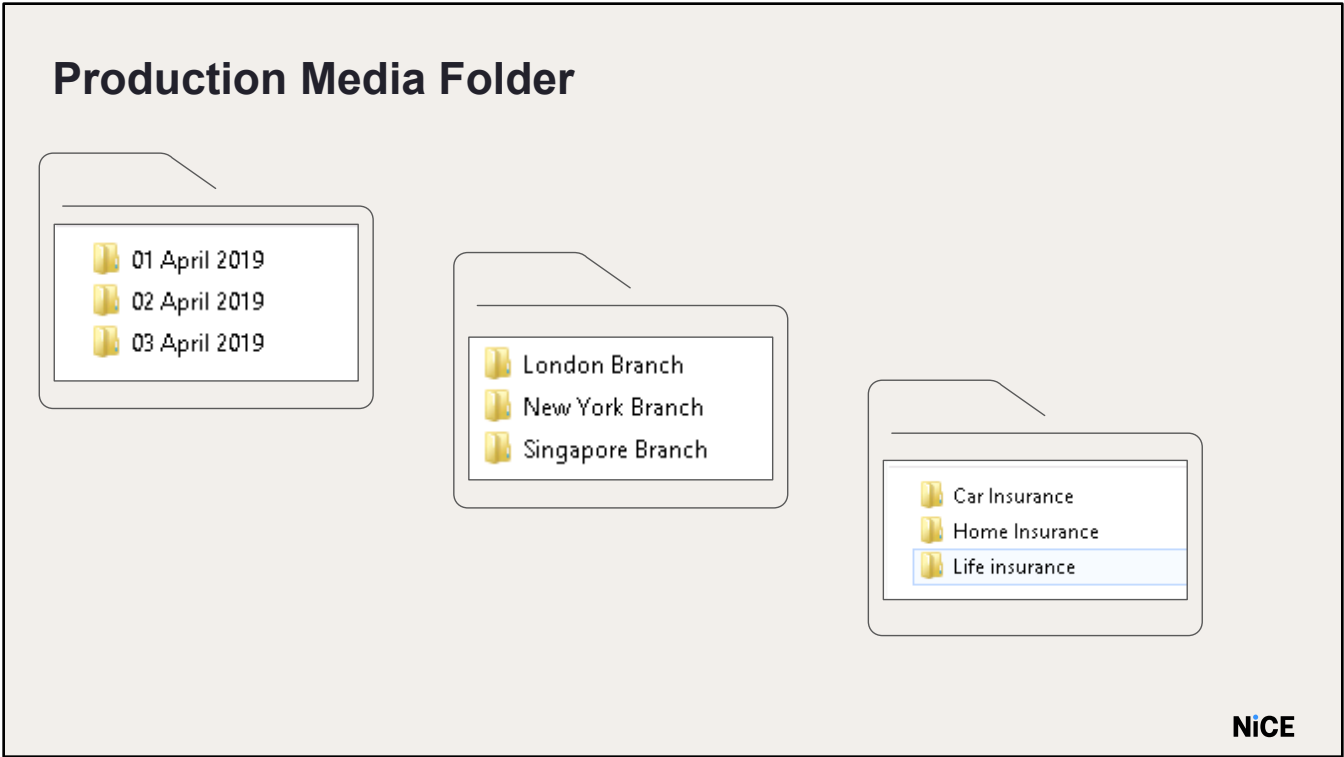


High Level Flow





1. DEF - automates the retrieval of media files and metadata from external sources.
2. DEF insert the metadata into NIA database
3. DEF place the data in Staging folders for media (under the Production Media root directory), the folders structure is based on rules for example by date\region\line of business,



DEF placed the interactions that it extracted from the recording system in the Production Media root directory. The folder structure is based on rules, for example, by date\region\line of business.



Call Metadata

```
<?xml version="1.0" encoding="UTF-8"?>
<CallData>
  <HubId>1</HubId>
  <InteractionID>6536201056524763141</InteractionID>
  <HasVideo>True</HasVideo>
  <IsH264Embedded>True</IsH264Embedded>
  <AgentName>Peter Britt</AgentName>
  <AgentId>11945</AgentId>
  <Group>West</Group>
  <Supervisor>Steven Mills</Supervisor>
  <Site>Reno</Site>
  <DNIS>2265</DNIS>
  <Extension>2946</Extension>
  <CallDirection>Inbound</CallDirection>
  <CallType>Did Not Order</CallType>
  <CustomerId>507901</CustomerId>
  <CustomerCity>San Jose</CustomerCity>
  <CustomerState>CA</CustomerState>
  <CallDateTime>2019-10-5T11:05:09+05:30</CallDateTime>
</CallData>
```

Here is an example of an interactions metadata XML file that is generated by DEF and imported into NIA servers.

It contains the following information:

Interaction ID

Agent Name

Agent ID

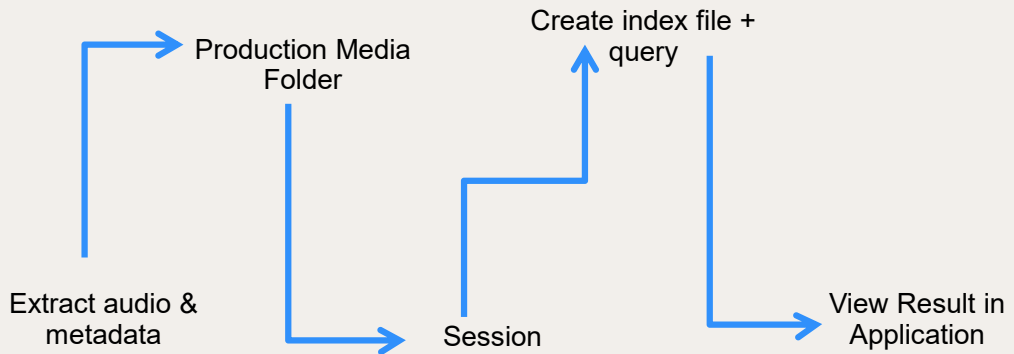
Supervisor name

Site

Call Date



Interaction Analytics Flow



1. DEF - automates the retrieval of media files and metadata from external sources.
2. DEF place the data in Staging folders for media
3. Session is pointing to the media that needs to be process \ingested into the system
4. The search grid (analytic engine)create a phonetic index file for each interaction and search for words and phrases.
5. Viewing the result in the application

summary

- Determine how Interaction Analytics impacts NiCE customers
- Describe Interaction Analytics main products: Search and Discovery
- Identify different speech analytics technologies





Thank You



INTRODUCTION TO INTERACTION ANALYTICS

NiCE Interaction Analytics & Quality Central
Fundamentals

Create a
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objectives

- Describe how to configure the following:
 - Define Term List
 - Create a Session
 - Perform a Search
 - Define Target Media Set
 - Use the Discovery Application



Initial
Configuration



Configuration Steps

Add Language Pack

Define Search Term\Query

Create Session

Search Application

Create Target Media Set (TMS)

Discovery Application



Leave the Maintenance mode through Management Console before testing the system

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Step 1: Install the Language Pack

Begin by installing the Language Pack on the Search Grid. These packs enable language-specific processing and are essential for the application to function correctly.

Step 2: Define Search Terms

Specify the words and phrases you want to search for within the interaction content. These are your search terms.

Step 3: Select Interactions to Search

Use a session to define which interactions should be included in the search. This helps narrow down the scope to relevant conversations.

Step 4: Create a Target Media Set (TMS)

To use the Discovery application, create a Target Media Set. The TMS defines the population of calls you want to analyze, ensuring your insights are focused and actionable.

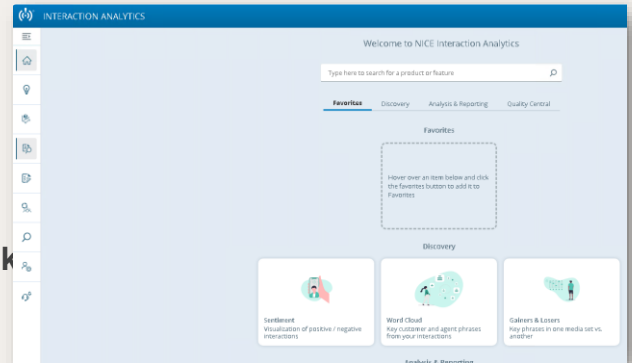


Add Language Pack



Steps to add Language Pack

- Exit **Maintenance Mode** via the **Management Console**
- Log in to NX by either:
 - Double-clicking the **Interaction Analytics** icon
 - or
 - Visiting <http://NX/NxIA/Main/>
- Navigate to **Administration**
- Go to **Installed Languages** and select **Add Language**
- Choose the appropriate **language pack**



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- Before we begin, exit Maintenance Mode using the Management Console to ensure restricted access is disabled.
- Log in to NX by either double-clicking the Interaction Analytics icon or visiting <http://NX/NxIA/Main/>.
- Navigate to the Administration section, click on **Installed Languages**, and select **Add Language**.
- Choose the appropriate language pack and click **OK**.
- Once completed, you'll see that the language pack has been successfully added.

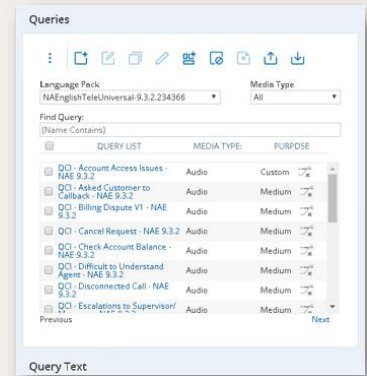


Import Query



Defining Term to Search & Steps to Import Queries

- From the Administration menu, select Term Lists and click New
- Type the term you want to search for and click Add
- Set the required Access Rights and Save
- Extract **QCI Queries – NA English 9.3.2** to the **nicetech** folder
- Ensure **query and language pack versions match**
- In NX Administration:
 - Click **Queries**
 - Select the language pack and click **Import Queries**
 - Browse to the query folder, select all files, and click **Open**



From the Administration menu select the “Term Lists” option and click “New”.

Type the term to search and click “Add”

In training, enter “My Account” as a term to search for testing.

Provide the required Access Rights and then “Save”.

Ensure the QCI Queries – NA English 9.3.2 are extracted in the nicetech folder

Confirm that the query version matches the language pack version

To import queries:

Click Queries in the Administration menu

Select the installed language pack and click Import Queries

Browse to the query location, select all files, and click Open

In the training lab, the 9.3.2.x.x.x queries are saved in the nicetech folder

Make sure the files are unzipped before importing

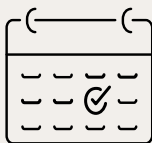


Session



What is a Session?

Session defines which interactions should be analyzed and what words & phrases should be searched



Time Period



Language Pack



Media source



Kindly refer the NiCE Interaction Analytics & Quality Central User guide for additional information

Session defines which interactions to analyze and what words & phrases to search in the interaction’s content.



Create a New Session

- From the Administration menu, click Session Management → Create New
- On the New Session page, go to the Basic Settings tab
- Set the start and/or end date for the session
- Enter a Name and Description for the session
- Select the appropriate Language Pack
- Click Media File Paths → Add → Browse to production media folder → Add
- In the Queries/Groups/Term List tab, select relevant queries to include
- Under Access Rights tab, grant access to the session information to profiles and / or specific users.

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- Start/End Date Boundaries limit which media files are included based on their creation date
- In the lab, leave these fields blank to ingest all media files
- Ensure the Language Pack matches the media, term lists, and queries used in the session
- Nexidia.Enterprise.ServerSideBrowseStartPath must point to Production Media to enable folder selection
- Under Access Rights, assign session access to specific users or profiles and click Save
 - Without access, users won't see session results in Search, Workflows, or Dashboards
 - Permissions can also be set at the User/Profile level
- When prompted, click Yes to activate the session



Job Status and Session Status

Job Status

JOB NAME	CURRENT STATE	LAST RESULT	LAST SUCCESSFUL UPDATE	NEXT SCHEDULED UPDATE	ENABLED	LAST EXECUTION DATE	LAST EXECUTION DURATION (SEC)	REQUEST
Core Update	Pending	Successful	12/15/2021 1:53:23 AM	12/15/2021 1:54:00 AM	Yes	N/A	N/A	
AggrCleanupQueueProcessor/AggrCleanupQueueProcessor	Running	Successful	12/15/2021 12:53:31 AM	N/A	Yes	N/A	N/A	
AutoDiscoveryAggrCleanupQueueProcessor/AggrCleanupQueueProcessor	Running	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryModelGenerator/ModelGenerator	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/AggrCleanupQueueProcessor	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/Cleanup	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/DataCompliance	Running	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/PreProcessing	Idle	N/A	N/A	N/A	Yes	N/A	N/A	
Core Cleanup	Unknown	N/A	N/A	12/15/2021 11:00:00 PM	Partial	N/A	N/A	
Core DM_Maintenance	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
Core ES_Maintenance	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
Core PCR	Unknown	N/A	N/A	12/15/2021 10:00:00 PM	Yes	N/A	N/A	
SQL DataCompliance	Idle	Successful	12/15/2021 1:00:19 AM	12/15/2021 1:05:00 AM	Yes	12/15/2021 1:00:19 AM	16	

Session Status

NAME	VERSION	OWNER	STATUS	DATE FILTERS	STARTED	HOURS	FILES	LAST REPROCESSED	REPROCESS STATUS
Billing Session	1	superuser	Waiting	[open] - [open]	12/15/2021 1:03:00 AM	00:00:00	0		n/a

Waiting

Ingesting

Queued

Inactive

NICE

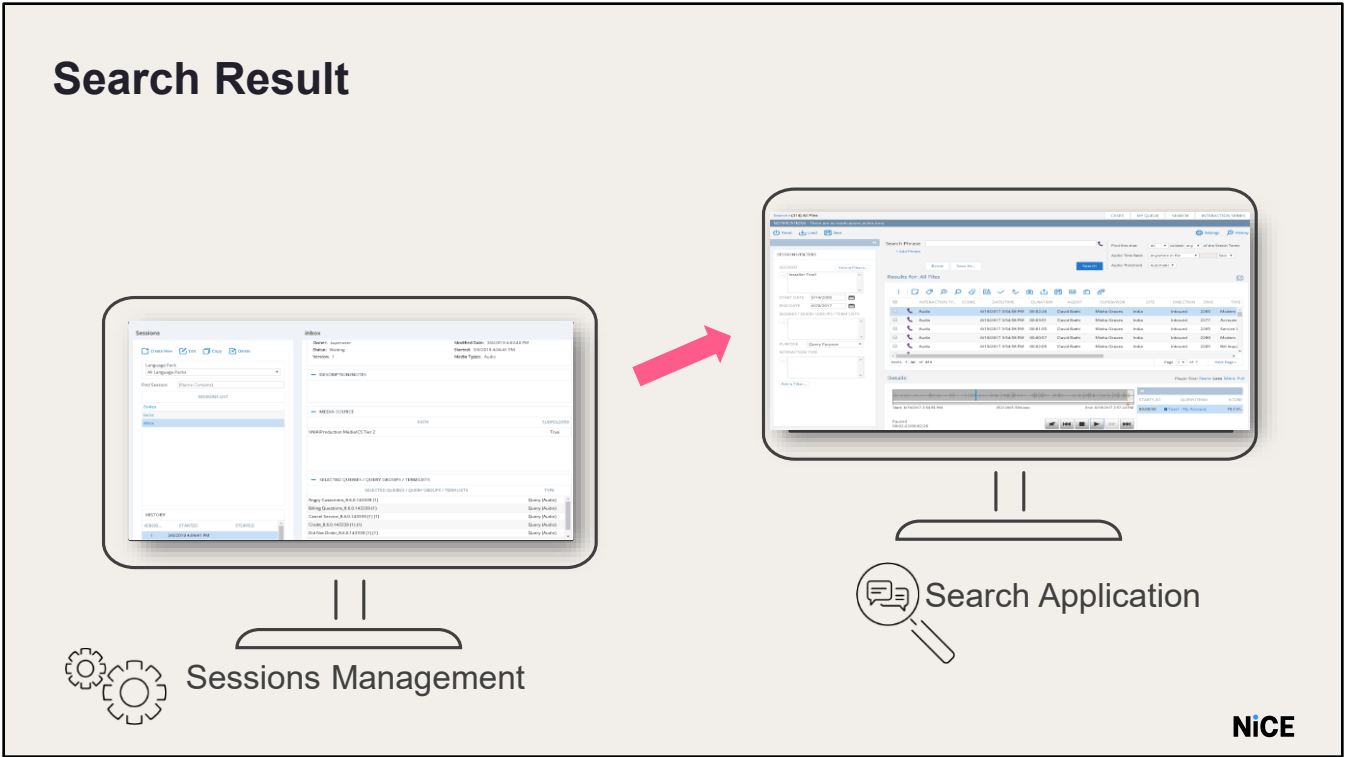
- Click Job Status to view all SQL job statuses
- If jobs are Disabled, open NX Configuration Manager and select Enable All Jobs
- Check Session Status to view ingested files for each session
- All calls are ingested, but only those in a session are viewable in the application. We have the following Session Status Types:
 - Waiting – Session is active and will auto-process new audio
 - Ingesting – Audio is being processed and .PAT files are created
 - Queued – Audio is waiting to be processed
 - Inactive – Session is stopped and won't process new audio
- If files aren't ingesting, restart the Ingest and Background Worker services
- To reprocess, click the Reprocess icon & review



Search Application



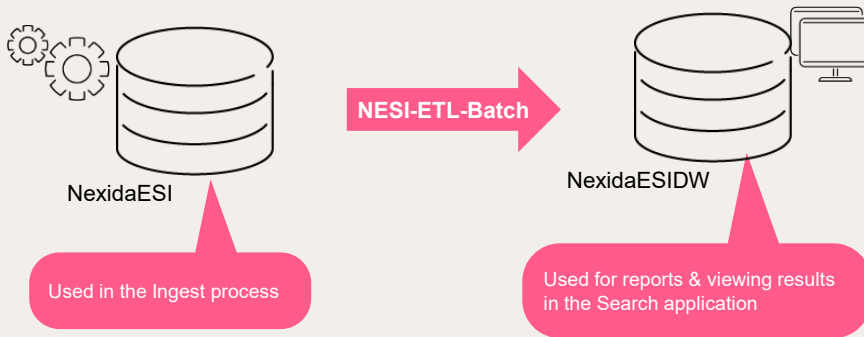
Search Result



The Search application displays the result of the processing of the session



Search - NX Database



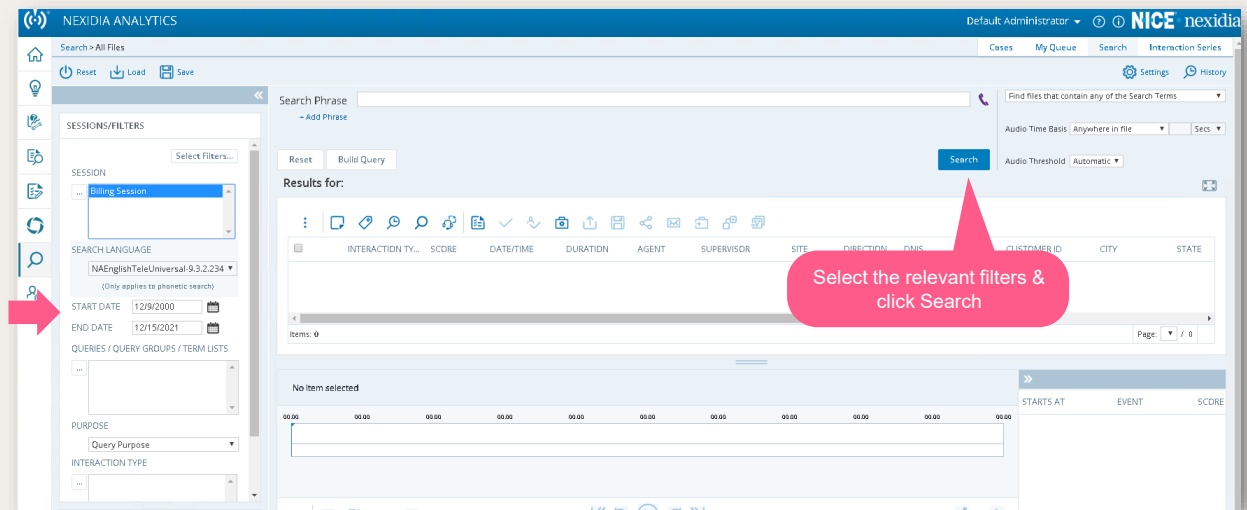
NexidiaESI - The results database records individual search terms and query results against the processed media.

NexidiaESIDW – Provides rapid response at scale, an ETL (Extract – Transform – Load) process is used to build a search database.

The search server accesses this database to identify any pre-existing results against user-requested search.

When you activate a session, the ingest process begins. The Reporting Database must be updated before you can view, search, and analyze your results.

Search



The Search function allows you to quickly investigate media.

You can perform in-depth searches of your media to analyze crucial data to solve business challenges.

You can investigate media from multiple sessions, build queries using Query Builder or Visual Query Builder, listen to media, and build queries based on ad hoc searches.

Select the Search tab on the main page.

- Select the session to use for your search.
- Click Search.

NOTE: You must have at least one session associated with a current language pack before you can use the Search functionality.

In the training lab: set the start date to the year 2000



Search Result

The screenshot displays the NEXIDIA ANALYTICS search results page. The top navigation bar includes 'Default Administrator', 'NICE nexidia', and links for 'Cases', 'My Queue', 'Search', and 'Interaction Series'. The left sidebar contains a 'SESSIONS/FILTERS' section with a 'Select Filters...' button and a 'Billing Session' filter selected. Below this is a 'SEARCH LANGUAGE' section with a dropdown menu set to 'NAEnglishTeleUniversal-9.3.2.234'. The main results area shows 'Results for: All Files' and a table of search results. The table has columns for 'INTERACTION TY...', 'SCORE', 'DATE/TIME', 'DURATION', 'AGENT', 'SUPERVISOR', 'SITE', 'DIRECTION', 'DNIS', 'TYPE', 'CUSTOMER ID', 'CITY', and 'STATE'. Two rows of results are visible, both for 'Audio' interactions. A red arrow points to the 'Billing Session' filter in the sidebar, and another red arrow points to the 'Items: 1 - 50 / 150' text below the results grid.

When the session is the only filter in the search, the result grid will display all the interactions associated with the session.

All the interactions available in the media folder will be shown. In this example 150 interactions. When we add queries as a filter to the search, the result grid will display only those interactions where the given queries had a match.

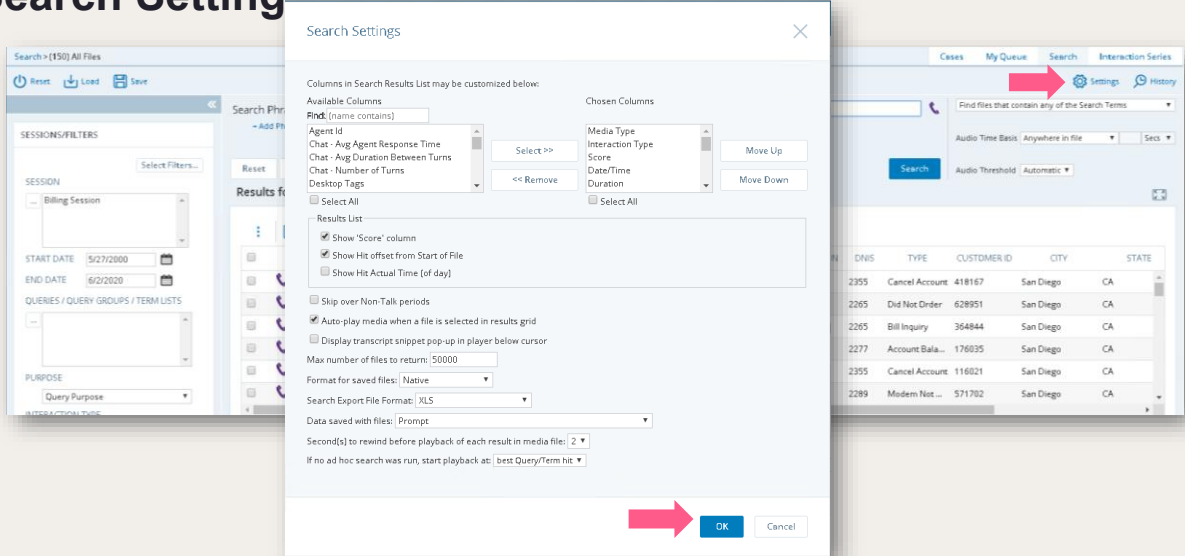
In this example, we added Cancel Service as a query and only 2 out of the 150 interactions in this session had these words matches. You can specify the language to use in the Search application and turn on or off Accessibility.

To specify account settings:

1. Click the Default Administrator (or logged in user) drop-down on the top right of the window.
2. Select the Account Settings link.
3. Select a Language.
4. Click OK.



Search Setting

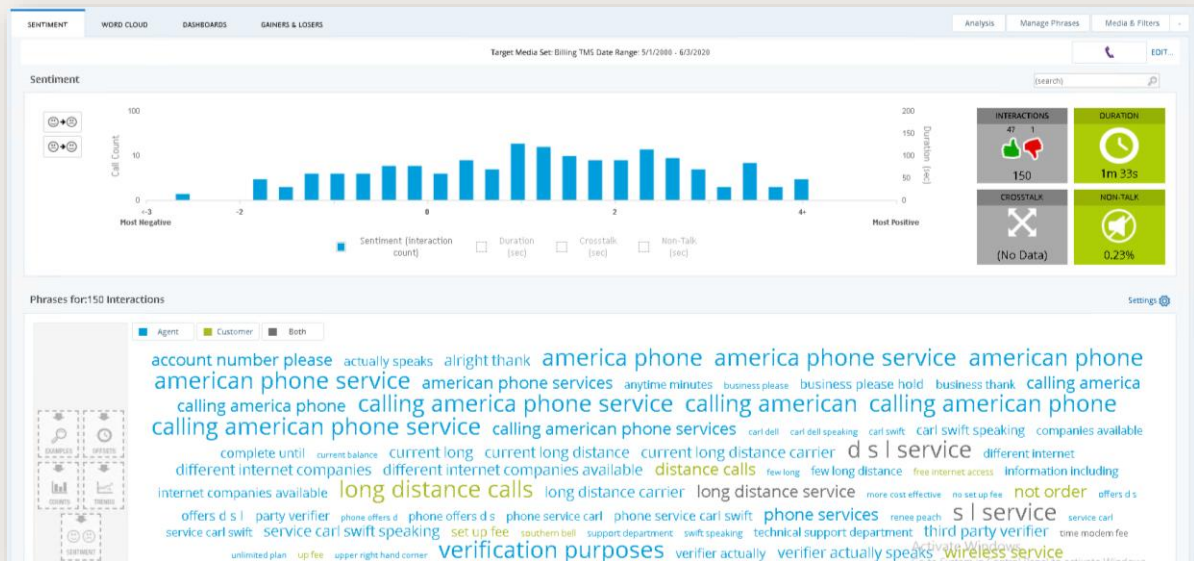


- You can customize the columns to display in your search results.
- Click Settings in the top right corner.
- Customize your Search Settings.

Discovery – Define TMS



Discovery – What You Don't Know



Search uses “speech to phonetics” and Discovery uses “speech to text”. Discovery results are not based on query results, the insights provided are based on the transcriptions of calls. Therefore, we don’t need to build a query up front. Instead, the word cloud will simply show the phrases occurring on a specific set of calls.

Discovery Result



Discovery is a process intensive flow - Phrase recognition consumes more CPU and memory resources than other Search Grid functions, therefore not all interactions will be sent for Discovery.

We use the TMS to reduce limit the number of calls sent to Discovery.

Define Target Media Set (TMS)

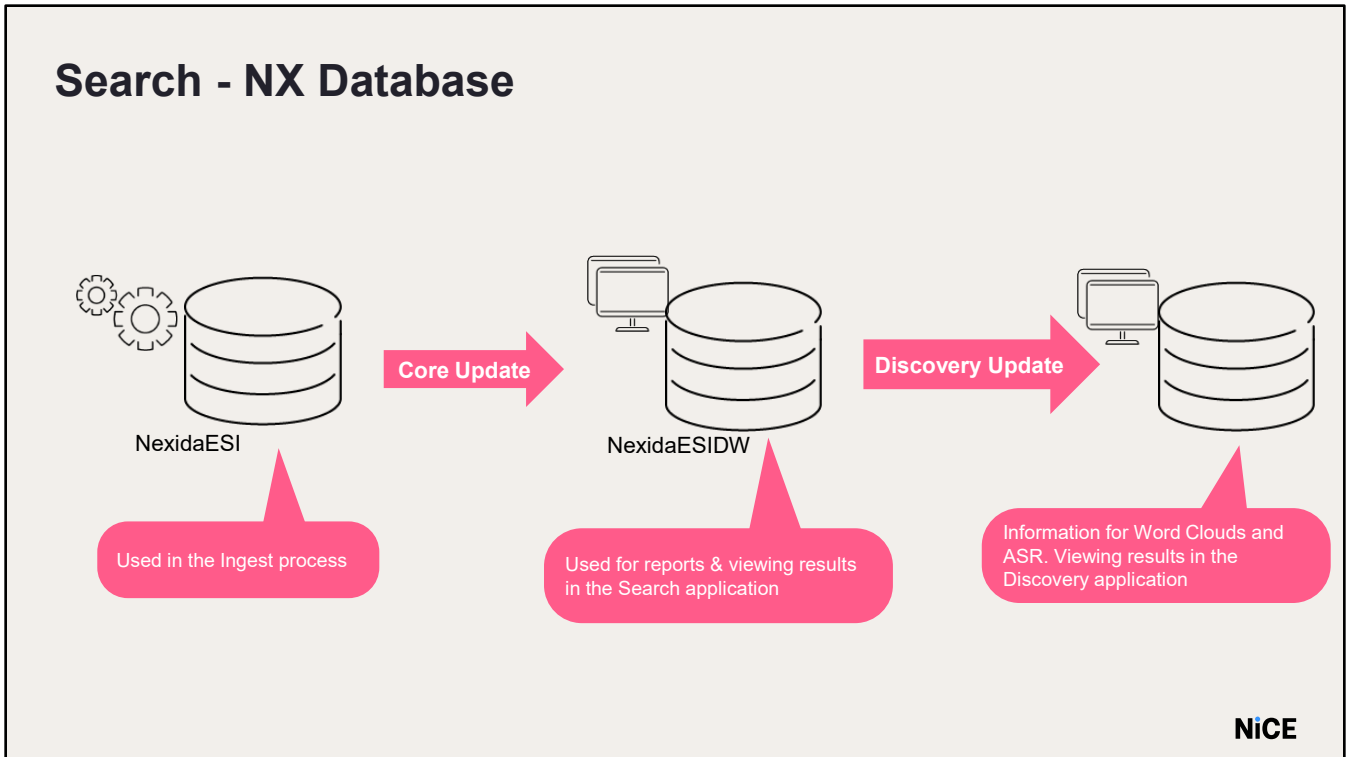
The screenshot displays the 'Define Target Media Set (TMS)' interface in the NEXIDIA ANALYTICS application. The sidebar on the left contains navigation options: STATUS, SESSION MANAGEMENT, and TARGET MEDIA SETS (highlighted with a red arrow). The main area is titled 'Target Media Set Details' and includes a 'Save' button and a 'Cancel' button. The form contains the following fields:

- Name:** Billing TMS (highlighted with a red box)
- Session:** Billing Session (highlighted with a red box)
- Purpose:** Audio Query Purpose (highlighted with a red box)
- Description/Notes:** (empty text area)

Below the form, there are sections for 'SEARCH FILTERS' and 'METADATA FILTERS'. The 'SEARCH FILTERS' section includes 'Include files that match' and 'Exclude files that match' dropdowns. The 'METADATA FILTERS' section includes an 'Add/Remove Filter(s)' button and a list of filters (e.g., Interaction Duration (seconds) is greater than 30). The 'REPORT METRICS' section at the bottom right lists various metrics (e.g., Average Duration (Sec), Average Interactions per Day) with checkboxes for selection. A red arrow points to this section.

- Go to Administration → Target Media Sets, then click New
- Important: You must have at least one Session with Read-Only or Full Control access rights defined before you can create a TMS. You can create a TMS without using any filters if desired.
- Ensure at least one session has Read-Only or Full Control access before creating a TMS
- Enter a name for the Target Media Set
- Select a session from the drop-down list
- Choose a query purpose (e.g., targeted listening for higher accuracy, lower volume)
- Select a metric (e.g., average interactions/day, non-talk time)
 - If Discovery shows no results, change the metric and use Total Interaction Count
- Interactions are scored by the phonetics engine based on detection accuracy

Search - NX Database



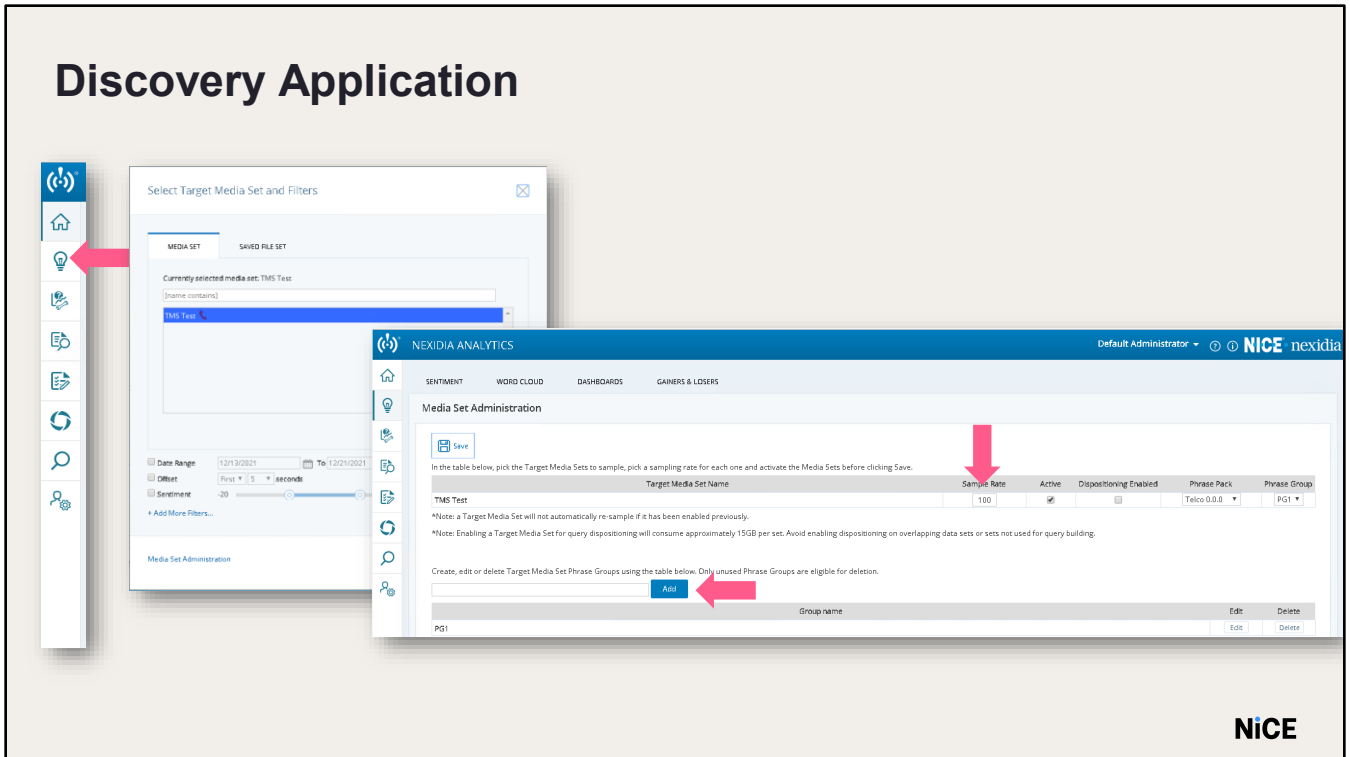
NexidiaESI - The results database records individual search term and query results against the processed media.

NexidiaESIDW – Provides rapid response at scale, an ETL (Extract

– Transform – Load) process is used to build a search database.

The search server accesses this database to identify any pre-existing results against user-requested searches.

Discovery Application



- Click the Discovery tab
- If you see a "No TMS is associated" error:
 - Restart the Nexidia Discovery Service in NX
 - Run NX-Core Update and NX-Discovery Update SQL jobs in NXDB1
 - Discovery Update is a sub job of the Core Update
- In the Select Target Media Set and Filters window:
 - Choose a Date Range (defaults to past 7 days)
 - If unchecked, all calls in the Media Set will be used
- Click the Media Set Administration button
- Enter the phrase group name and Add.
- Phrase Group is an alias for Phrase Pack; in training, it's named Discovery Group
- Click Add to include the group name in the list

- Phrase Groups control ignored phrases and custom respellings for all TMSs in the group
- These changes do not affect media sets in other Phrase Groups
- In the Discovery word cloud, you can:
 - Remove phrases
 - Change phrase spellings
- Changes are stored in the Phrase Group linked to the selected TMS
- Useful for organizations with multiple Lines of Business (LOBs) to tailor word clouds
- Example:
 - Media Sets A & B → Phrase Group One
 - Media Set C → Phrase Group Two
 - Ignoring a phrase in Group One affects A & B, but not C



Define the Population of Calls to Investigate

Default Administrator
NICE nexidia

SENTIMENT
WORD CLOUD
DASHBOARDS
GAINERS & LOSERS

Media Set Administration

Save

In the table below, pick the Target Media Sets to sample, pick a sampling rate for each one and activate the Media Sets before clicking Save.

Target Media Set Name	Sample Rate	Active	Dispositioning Enabled	Phrase Pack	Phrase Group
TMS Test	100	☒	☐	Telco 0.0.0	PG1

*Note: a Target Media Set will not automatically re-sample if it has been enabled previously.

*Note: Enabling a Target Media Set for query dispositioning will consume approximately 15GB per set. Avoid enabling dispositioning on overlapping data sets or sets not used for query building.

Create, edit or delete Target Media Set Phrase Groups using the table below. Only unused Phrase Groups are eligible for deletion.

Add

Group name	Edit	Delete
PG1	Edit	Delete

If you don't see the Phrase Pack in the drop down menu, restart the Discovery Service

Since the transcription is process intensive, not all calls may need to be indexed for Discovery.

Here we define the sampling percentage for the selected TMS. The possible values are 1 through 100.

Choose the Phrase Pack from the drop-down menu.

Choose the Phrase Group.

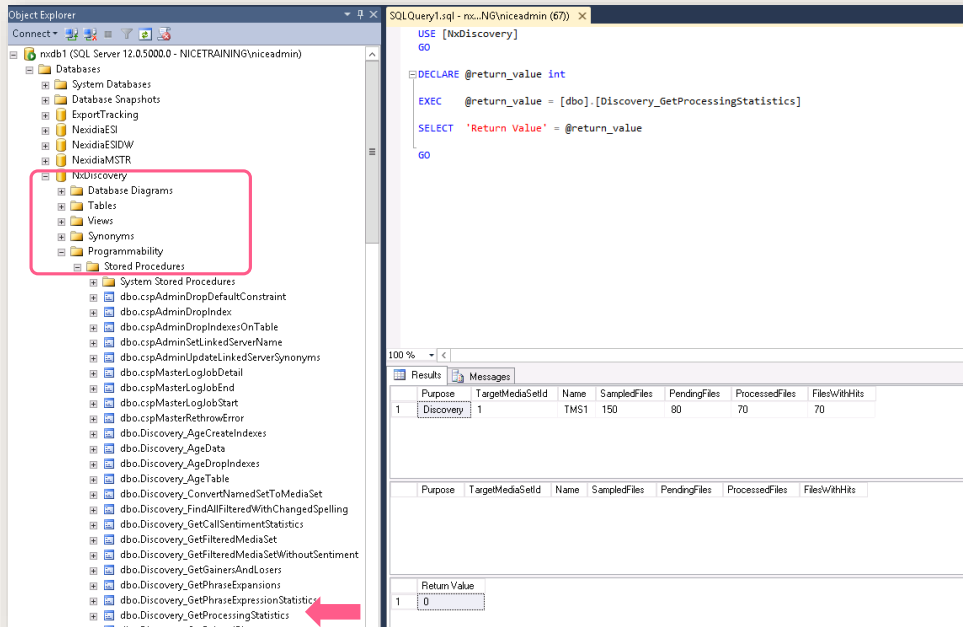
Mark the active column.

Click Save.

Note: This is the percent of interactions to be sampled from the selected TMS.

Click the Discovery/Sentiment tab again.

Discovery Progress



The screenshot shows the SQL Server Enterprise Manager interface on the left and a SQL Query window on the right. In the Enterprise Manager, the 'NdxDiscovery' folder under 'Databases' is highlighted with a red box. The 'Stored Procedures' folder under 'NdxDiscovery' is also highlighted, and a red arrow points to the 'dbo.Discovery_GetProcessingStatistics' procedure. The SQL Query window shows the following code:

```
USE [NdxDiscovery]
GO

DECLARE @return_value int

EXEC @return_value = [dbo].[Discovery_GetProcessingStatistics]

SELECT 'Return Value' = @return_value

GO
```

The Results pane shows a single row of data:

Purpose	TargetMediaSetId	Name	SampledFiles	PendingFiles	ProcessedFiles	FilesWithHits
1	Discovery	1	TMS1	150	80	70

The Messages pane shows the following message:

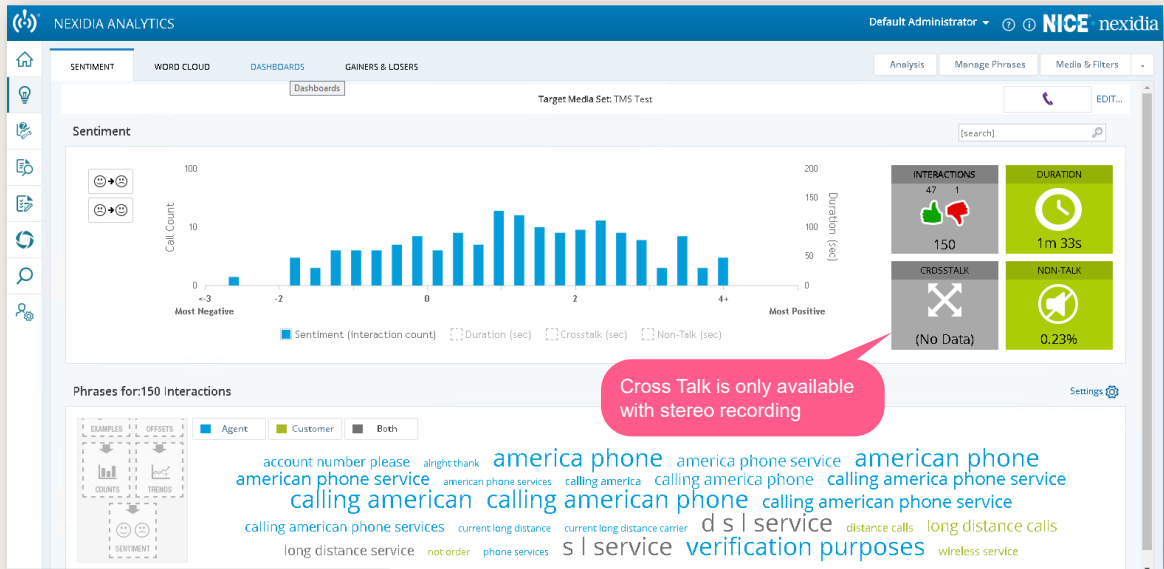
Purpose	TargetMediaSetId	Name	SampledFiles	PendingFiles	ProcessedFiles	FilesWithHits
1	Discovery	1	TMS1	150	80	70

The Return Value pane shows the value 0.

To check the progress of the discovery task from the Discovery DB, run the `Discovery_GetProcessingStatistics` Stored Procedure.

- Tip: if you don't see any result - restart the discovery service, run the discovery update job, and try changing the Date Range on the Discovery Media & filters window.

Testing Discovery



NiCE

We can see the result in the Discovery Application.

Right-click on a phrase to view the context menu – from here you can choose to remove the phrase from the phrase group, change the phrase spelling etc.

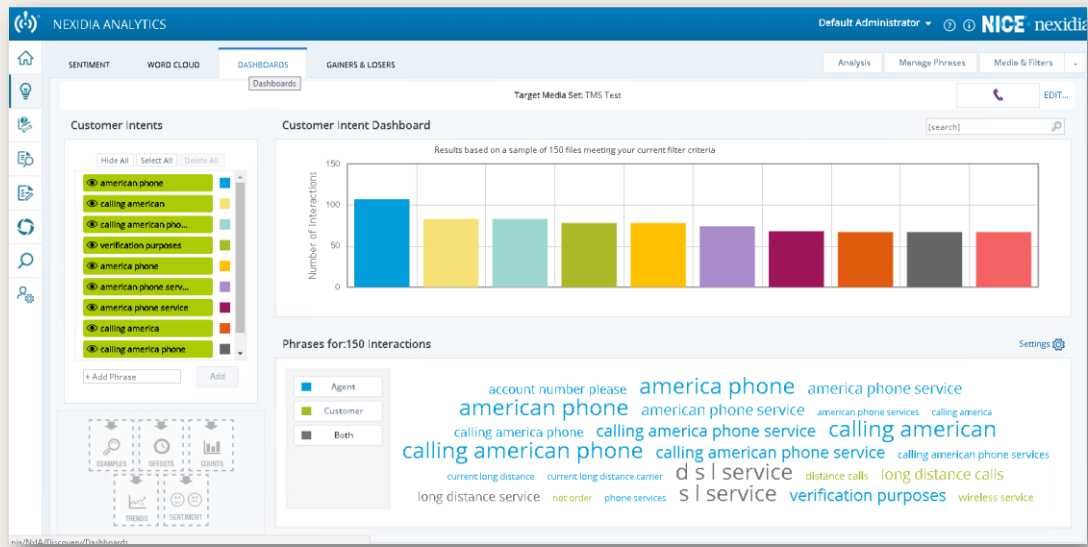
Double click on a phrase to view the transcription.

In the Phrase Pack there is a Stop Words file, which contains words that will appear in the transcription file but will not be part of the Word Cloud.

Cross Talk is only available with stereo recording



Discovery Dashboard



This is an example of a Dashboard.



Review Questions



Review Question #1

What is a Phrase Pack?

Phrase pack contains words that are not part of the standard language and required to create a phonetic index for these “unknown” words.

Phrase pack contains words that are not part of the standard language and required to transcribe these “unknown” words

Phrase pack contains words that are not part of the standard language and required to create sentiment modeling for these “unknown” words.

None of the answers are correct



Review Question #2

What is a Session?

Session defines which interactions we would like to analyze and what words & phrases we would like to search for in the interaction's content

Session defines the population of calls to be analyzed by the Discovery module

Session is a set of logical conditions and phrases used for performing searches on the interaction's content

None of the answers are correct



Do it Yourself

- Refer to exercise 1:
- Create a new session and point it to the “**Billing**” media folder, select the queries to apply to the interactions
- Run the session via the Session status tab
- Create a TMS and point it Session created in the previous step
- View results in the Discovery application



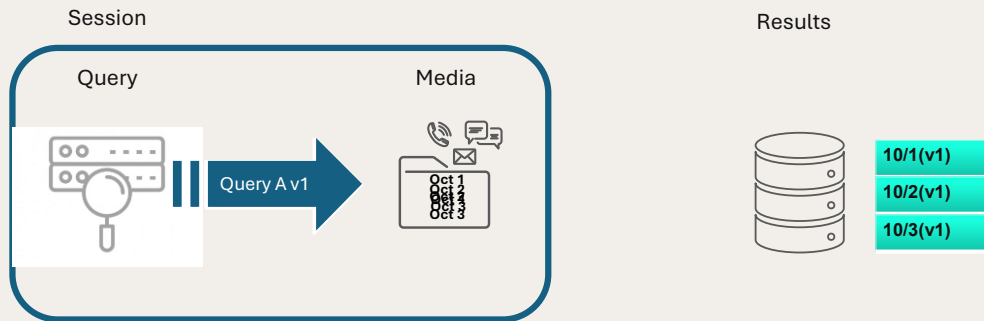


Thank You

Appendix: Query Processing



Normal query processing



- Query A v1 was published
- Results will be generated and stored in the database.

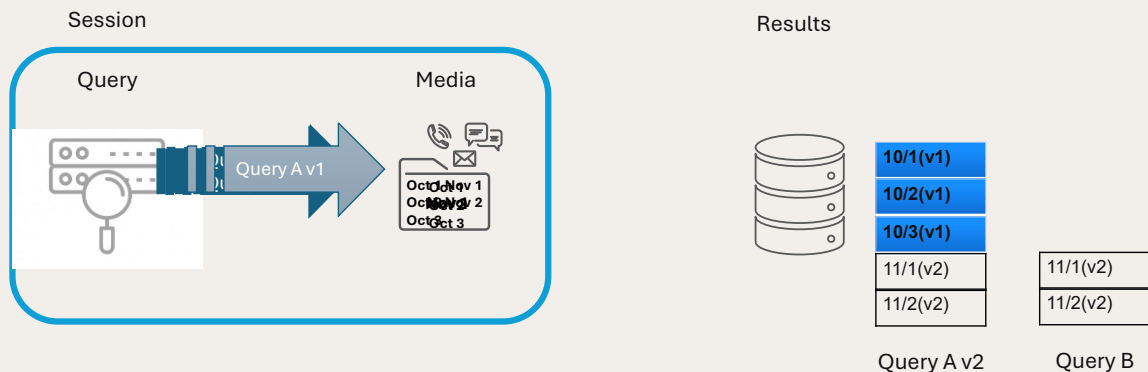
When we create a session, we can add queries for that session and activate the same.

Assume, Query A version 1 was created on October 1 and associated with a session.

This query will run against the media dated October 1 onwards.

Results will be generated and stored in the database.

Updating a Query – Day Forward



- Query A v2 or Query B was published
- Only newly added media will be processed, and results will be stored in the database for both Query A v2 or Query B

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When the query is updated and republished,

- the query can be renamed; therefore, not having any effect on version 1 or
- the query can overwrite the existing query and becomes version 2

Now, consider on November 1 you have an updated query-Query A v2, or a created a new query -Query B to associate with the existing session.

When a new media is added to the directory , Query A v2/ Query B will run against the media and results are written to the database.

In this example:

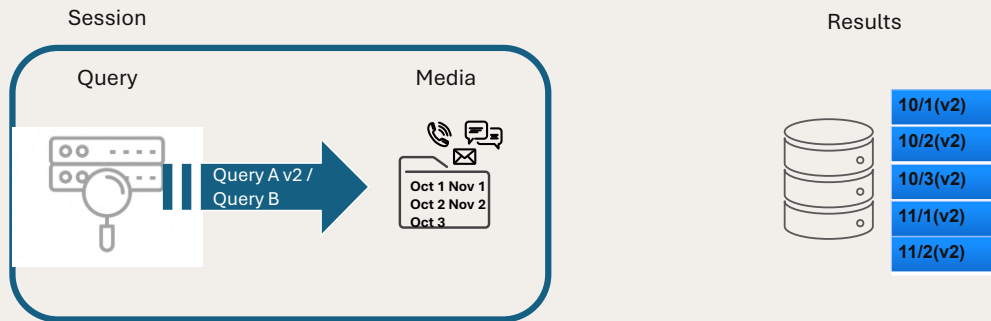
- results processed from October 1st- 3rd (till 31st) (using Query A v1) remain in the database
- results processed from November 1st- November 2nd (Query A v2 /Query B) are added to the database

So, the media that is already processed(say all media till October 31st) will not be reprocessed by the session unless we manually do it.

Rather the session will be processing if there is any new media.

So, the Query A v2 will have results from Query A v1 and the new results
Query B will have results from Nov 1st onwards

Updating a Query – Session Reprocessing



- Query A v2 / Query B was published
- Reprocessing will remove the results from Query A v1 from the database
- Run query against all audio, generate results and write to the database

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Consider a situation where in you change an existing query, then you must decide if you want the query to be used day forward or want to go back and reprocess the session that already used this query.

We should consider the number of media you have while reprocessing.

You should not reprocess , for example if a customer has 13 months retention and millions of files.

A query we are writing today may not have a meaning 13 months earlier.

So, when we have the all calls - all queries session, those if you add new queries to a session, simply turn the session off and turn it on to reload the new queries and processes day forward.

If we chose to reprocess, it means that you are running the query against all the audio.

Its not a good practice to reprocess huge sessions.





INTRODUCTION TO INTERACTION ANALYTICS

NiCE Interaction Analytics & Quality Central
Fundamentals

Create a
NiCE
world 

objectives

- Describe how to configure the following:
 - Define Term List
 - Create a Session
 - Perform a Search
 - Define Target Media Set
 - Use the Discovery Application



Initial
Configuration



Configuration Steps

Add Language Pack

Define Search Term\Query

Create Session

Search Application

Create Target Media Set (TMS)

Discovery Application



Leave the Maintenance mode through Management Console before testing the system

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Step 1: Install the Language Pack

Begin by installing the Language Pack on the Search Grid. These packs enable language-specific processing and are essential for the application to function correctly.

Step 2: Define Search Terms

Specify the words and phrases you want to search for within the interaction content. These are your search terms.

Step 3: Select Interactions to Search

Use a session to define which interactions should be included in the search. This helps narrow down the scope to relevant conversations.

Step 4: Create a Target Media Set (TMS)

To use the Discovery application, create a Target Media Set. The TMS defines the population of calls you want to analyze, ensuring your insights are focused and actionable.



Add Language Pack



Steps to add Language Pack

- Exit **Maintenance Mode** via the **Management Console**
- Log in to NX by either:
 - Double-clicking the **Interaction Analytics** icon
 - or
 - Visiting <http://NX/NxIA/Main/>
- Navigate to **Administration**
- Go to **Installed Languages** and select **Add Language**
- Choose the appropriate **language pack**

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- Before we begin, exit Maintenance Mode using the Management Console to ensure restricted access is disabled.
- Log in to NX by either double-clicking the Interaction Analytics icon or visiting <http://NX/NxIA/Main/>.
- Navigate to the Administration section, click on **Installed Languages**, and select **Add Language**.
- Choose the appropriate language pack and click **OK**.
- Once completed, you'll see that the language pack has been successfully added.

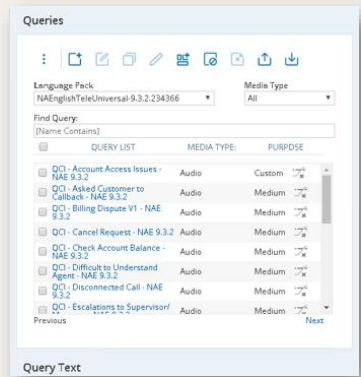


Import Query



Defining Term to Search & Steps to Import Queries

- From the Administration menu, select Term Lists and click New
- Type the term you want to search for and click Add
- Set the required Access Rights and Save
- Extract **QCI Queries – NA English 9.3.2** to the **nicetech** folder
- Ensure **query and language pack versions match**
- In NX Administration:
 - Click **Queries**
 - Select the language pack and click **Import Queries**
 - Browse to the query folder, select all files, and click **Open**



From the Administration menu select the “Term Lists” option and click “New”.

Type the term to search and click “Add”

In training, enter “My Account” as a term to search for testing. Provide the required Access Rights and then “Save”.

Ensure the QCI Queries – NA English 9.3.2 are extracted in the nicetech folder

Confirm that the query version matches the language pack version

To import queries:

Click Queries in the Administration menu

Select the installed language pack and click Import Queries

Browse to the query location, select all files, and click Open

In the training lab, the 9.3.2.x.x.x queries are saved in the nicetech folder

Make sure the files are unzipped before importing

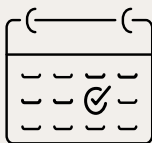


Session



What is a Session?

Session defines which interactions should be analyzed and what words & phrases should be searched



Time Period



Language Pack



Media source



Kindly refer the NiCE Interaction Analytics & Quality Central User guide for additional information

Session defines which interactions to analyze and what words & phrases to search in the interaction’s content.



Create a New Session

- From the Administration menu, click Session Management → Create New
- On the New Session page, go to the Basic Settings tab
- Set the start and/or end date for the session
- Enter a Name and Description for the session
- Select the appropriate Language Pack
- Click Media File Paths → Add → Browse to production media folder → Add
- In the Queries/Groups/Term List tab, select relevant queries to include
- Under Access Rights tab, grant access to the session information to profiles and / or specific users.

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- Start/End Date Boundaries limit which media files are included based on their creation date
- In the lab, leave these fields blank to ingest all media files
- Ensure the Language Pack matches the media, term lists, and queries used in the session
- Nexidia.Enterprise.ServerSideBrowseStartPath must point to Production Media to enable folder selection
- Under Access Rights, assign session access to specific users or profiles and click Save
 - Without access, users won't see session results in Search, Workflows, or Dashboards
 - Permissions can also be set at the User/Profile level
- When prompted, click Yes to activate the session



Job Status and Session Status

Job Status

JOB NAME	CURRENT STATE	LAST RESULT	LAST SUCCESSFUL UPDATE	NEXT SCHEDULED UPDATE	ENABLED	LAST EXECUTION DATE	LAST EXECUTION DURATION (SEC)	REQUEST
Core Update	Pending	Successful	12/15/2021 1:53:23 AM	12/15/2021 1:54:00 AM	Yes	N/A	N/A	
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AutoDiscoveryAggrCleanupQueueProcessor/AggrCleanupQueueProcessor	Running	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryModelGenerator/ModelGenerator	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/AggrCleanupQueueProcessor	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/Cleanup	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/DataCompliance	Running	N/A	N/A	N/A	Yes	N/A	N/A	
AutoDiscoveryPreProcessing/DataCompliance	Idle	N/A	N/A	N/A	Yes	N/A	N/A	
Core Cleanup	Unknown	N/A	N/A	12/15/2021 11:00:00 PM	Partial	N/A	N/A	
Core DM_Maintenance	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
Core ESI_Maintenance	Unknown	N/A	N/A	N/A	Yes	N/A	N/A	
Core PCR	Unknown	N/A	N/A	12/15/2021 10:00:00 PM	Yes	N/A	N/A	
BEI DataCompliance	Idle	Successful	12/15/2021 1:00:19 AM	12/15/2021 1:05:00 AM	Yes	12/15/2021 1:00:19 AM	16	

Session Status

Start All Stop All Export Refresh

NAME	VERSION	OWNER	STATUS	DATE FILTERS	STARTED	HOURS	FILES	LAST REPROCESSED	REPROCESS STATUS
Billing Session	1	superuser	Waiting	[open] - [open]	12/15/2021 1:03:00 AM	00:00:00	0		n/a

Waiting

Ingesting

Queued

Inactive

NICE

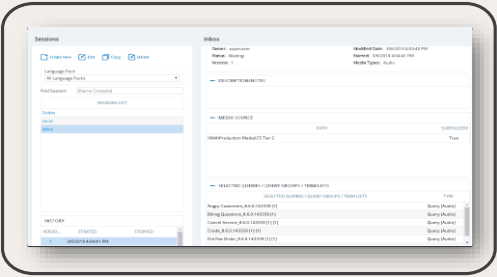
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 - Inactive – Session is stopped and won't process new audio
- If files aren't ingesting, restart the Ingest and Background Worker services
- To reprocess, click the Reprocess icon & review



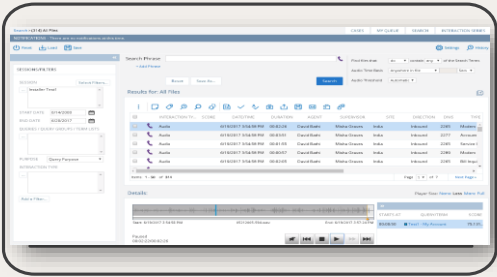
Search Application



Search Result



Sessions Management

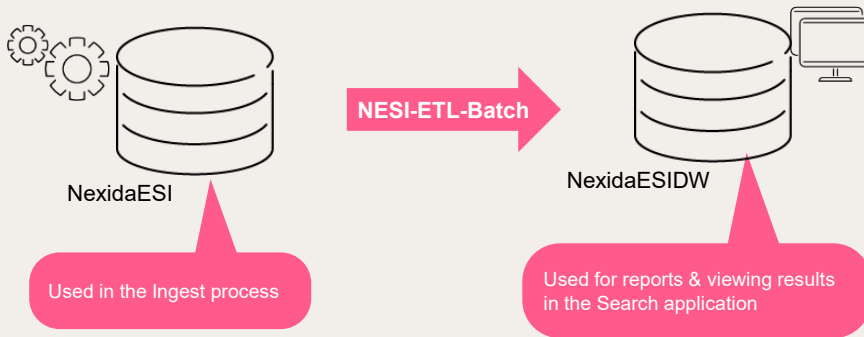


Search Application

The Search application displays the result of the processing of the session



Search - NX Database



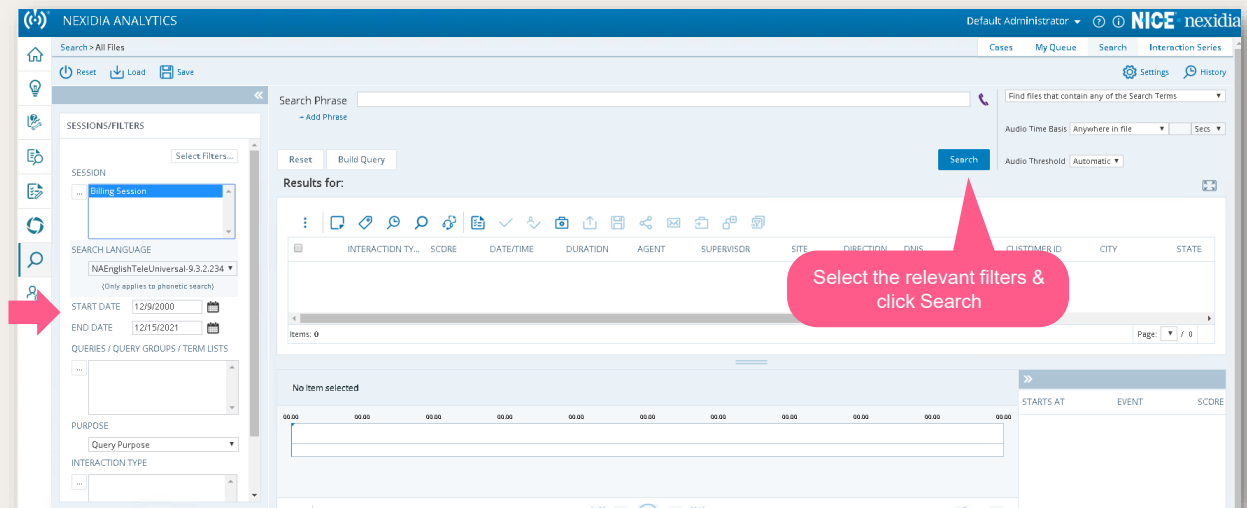
NexidiaESI - The results database records individual search terms and query results against the processed media.

NexidiaESIDW – Provides rapid response at scale, an ETL (Extract – Transform – Load) process is used to build a search database.

The search server accesses this database to identify any pre-existing results against user-requested search.

When you activate a session, the ingest process begins. The Reporting Database must be updated before you can view, search, and analyze your results.

Search



The Search function allows you to quickly investigate media.

You can perform in-depth searches of your media to analyze crucial data to solve business challenges.

You can investigate media from multiple sessions, build queries using Query Builder or Visual Query Builder, listen to media, and build queries based on ad hoc searches.

Select the Search tab on the main page.

- Select the session to use for your search.
- Click Search.

NOTE: You must have at least one session associated with a current language pack before you can use the Search functionality.

In the training lab: set the start date to the year 2000



Search Result

The screenshot displays the NEXIDIA ANALYTICS search results page. The top navigation bar includes 'Default Administrator', 'NICE nexidia', and links for 'Cases', 'My Queue', 'Search', and 'Interaction Series'. The left sidebar contains various icons for navigation. The main search area includes a 'Search Phrase' field, 'Reset', 'Build Query', and 'Search' buttons. Below the search bar, the results are displayed in a table format. The table has columns for 'INTERACTION TY...', 'SCORE', 'DATE/TIME', 'DURATION', 'AGENT', 'SUPERVISOR', 'SITE', 'DIRECTION', 'DNIS', 'TYPE', 'CUSTOMER ID', 'CITY', and 'STATE'. Two rows of data are visible, both for 'Audio' interactions. The first row is dated '11/5/2016 1:00:00 PM' and the second is dated '11/5/2016 12:00:00 P...'. The table also shows 'Items: 1 - 50 / 150'. On the left sidebar, under 'SESSIONS/FILTERS', the 'Billing Session' filter is selected, indicated by a red arrow. Another red arrow points to the 'Items: 1 - 50 / 150' text in the results area.

When the session is the only filter in the search, the result grid will display all the interactions associated with the session.

All the interactions available in the media folder will be shown. In this example 150 interactions. When we add queries as a filter to the search, the result grid will display only those interactions where the given queries had a match.

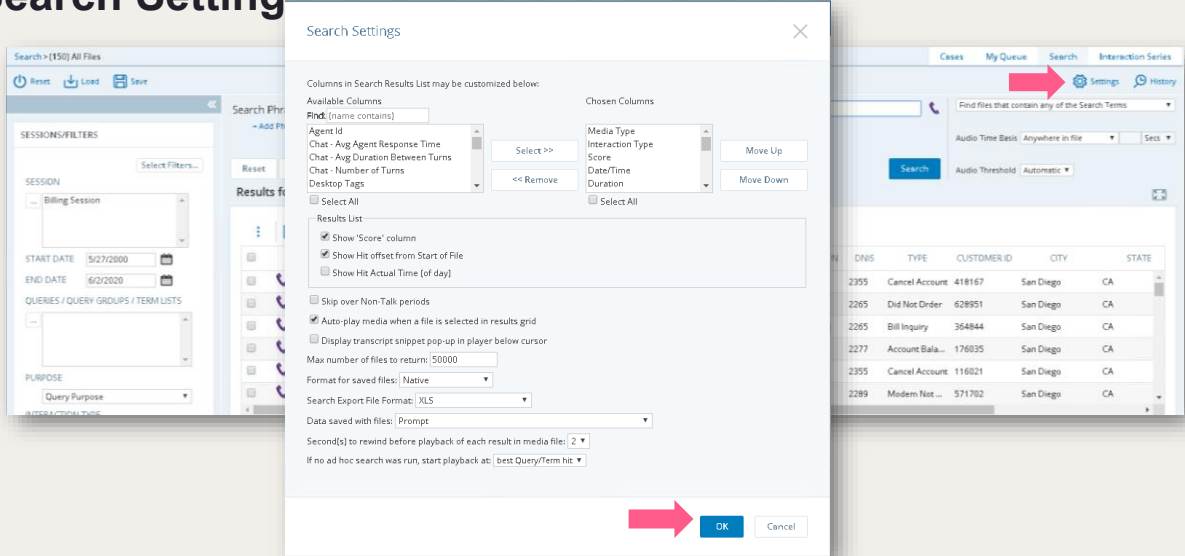
In this example, we added Cancel Service as a query and only 2 out of the 150 interactions in this session had these words matches. You can specify the language to use in the Search application and turn on or off Accessibility.

To specify account settings:

1. Click the Default Administrator (or logged in user) drop-down on the top right of the window.
2. Select the Account Settings link.
3. Select a Language.
4. Click OK.



Search Setting

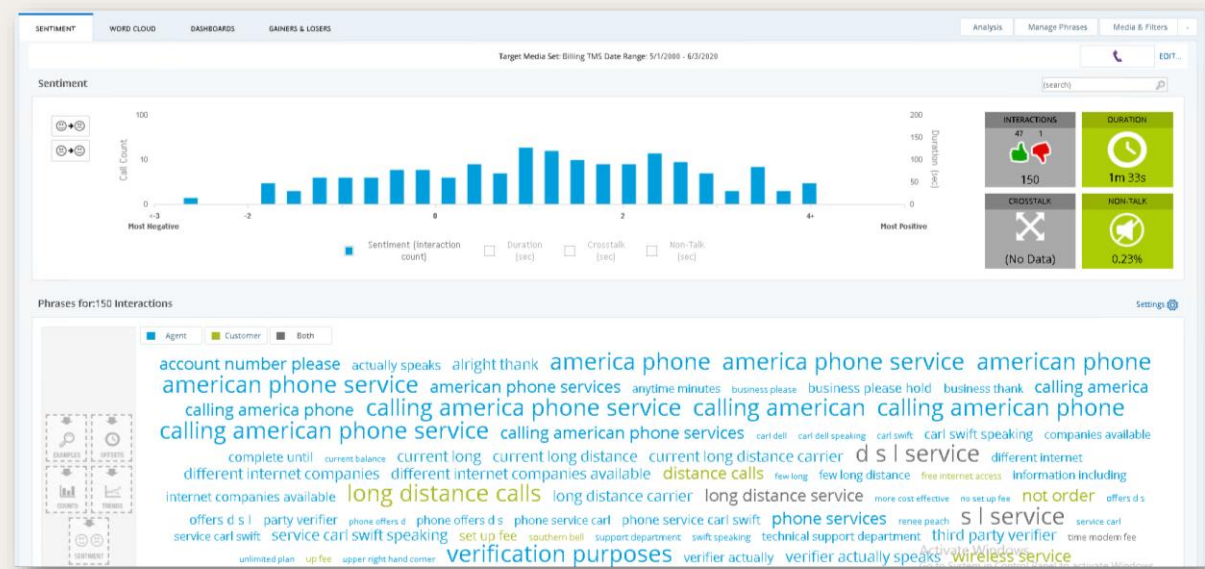


- You can customize the columns to display in your search results.
- Click Settings in the top right corner.
- Customize your Search Settings.

Discovery – Define TMS



Discovery – What You Don’t Know



Search uses “speech to phonetics” and Discovery uses “speech to text”. Discovery results are not based on query results, the insights provided are based on the transcriptions of calls. Therefore, we don’t need to build a query up front. Instead, the word cloud will simply show the phrases occurring on a specific set of calls.





Discovery is a process intensive flow - Phrase recognition consumes more CPU and memory resources than other Search Grid functions, therefore not all interactions will be sent for Discovery.

We use the TMS to reduce limit the number of calls sent to Discovery.



Define Target Media Set (TMS)

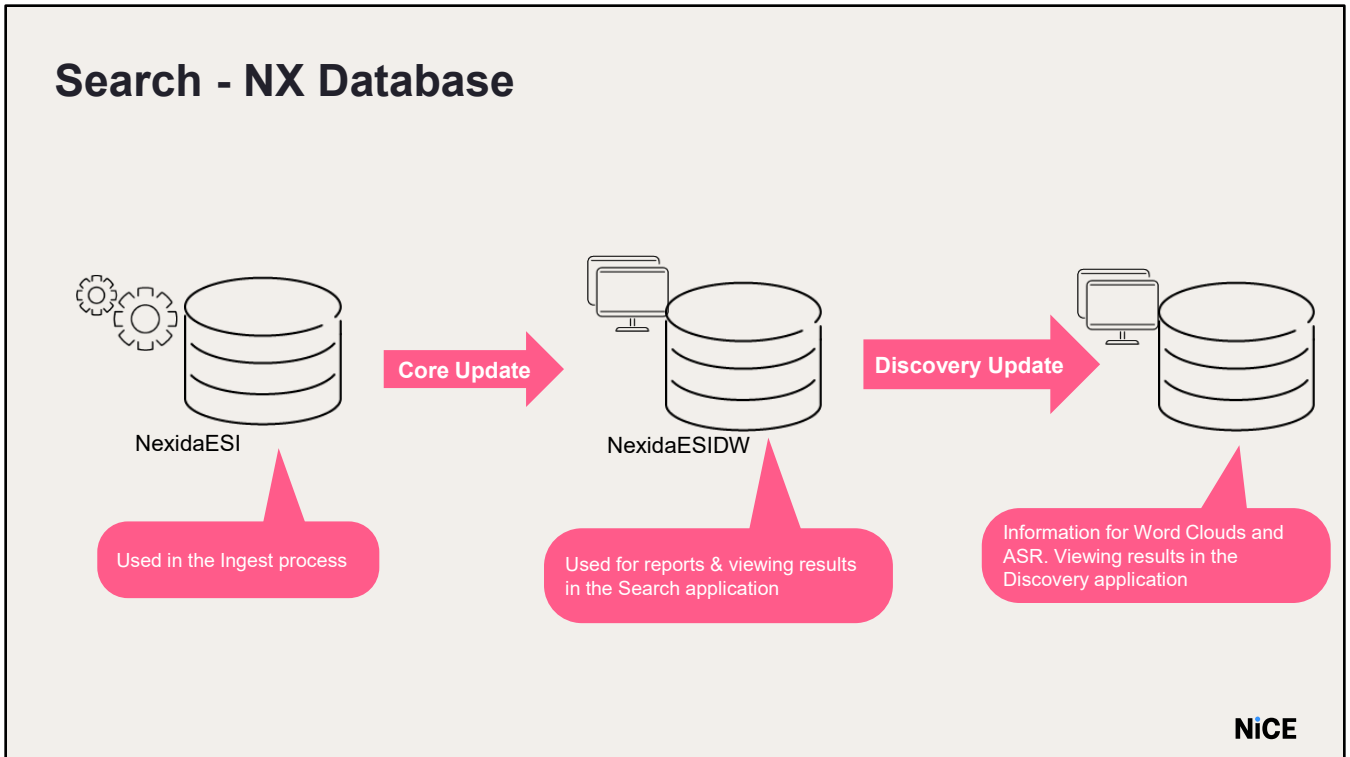
The screenshot displays the 'Define Target Media Set (TMS)' interface in NEXIDIA ANALYTICS. The sidebar on the left includes navigation options such as STATUS, SESSION MANAGEMENT, and TARGET MEDIA SETS. The main content area, titled 'Target Media Set Details', contains a form with the following fields:

- Name:** Billing TMS (highlighted with a red box)
- Session:** Billing Session (dropdown menu)
- Purpose:** Audio Query Purpose (dropdown menu)
- Description/Notes:** (text area)

Below the form, there are sections for 'SEARCH FILTERS' and 'METADATA FILTERS'. The 'REPORT METRICS' section at the bottom right lists various metrics for selection, including Average Duration (Sec), Average Interactions per Day, Average Non-Talk (H), Average Non-Talk (Sec), Average Non-Talk (Occurrences per Call), Average Offset from End of Call, and Average Offset from Start of Call. A red arrow points to this section.

- Go to Administration → Target Media Sets, then click New
- Important: You must have at least one Session with Read-Only or Full Control access rights defined before you can create a TMS. You can create a TMS without using any filters if desired.
- Ensure at least one session has Read-Only or Full Control access before creating a TMS
- Enter a name for the Target Media Set
- Select a session from the drop-down list
- Choose a query purpose (e.g., targeted listening for higher accuracy, lower volume)
- Select a metric (e.g., average interactions/day, non-talk time)
 - If Discovery shows no results, change the metric and use Total Interaction Count
- Interactions are scored by the phonetics engine based on detection accuracy

Search - NX Database



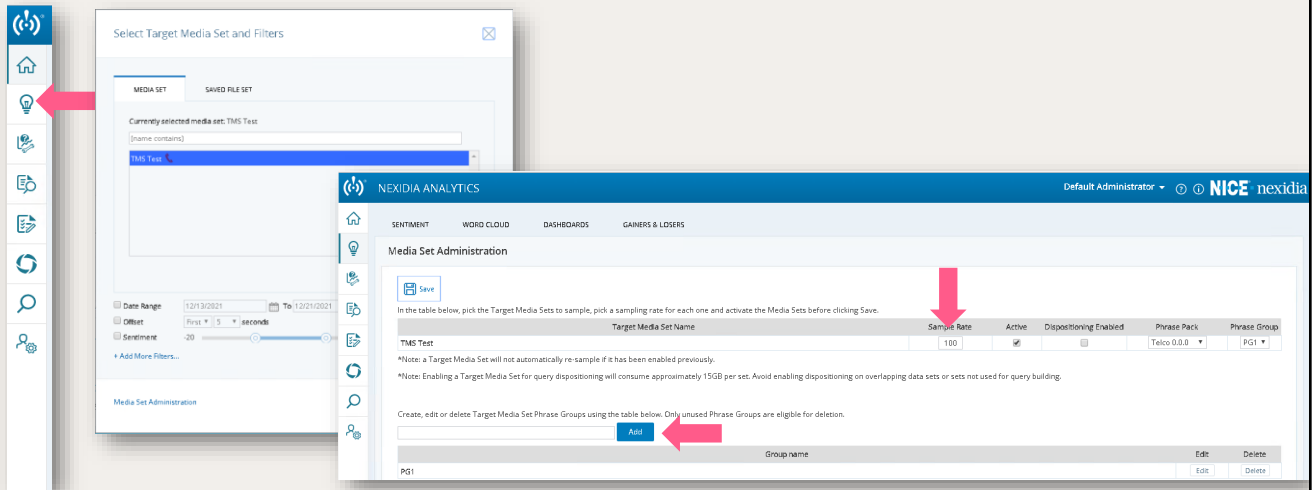
NexidiaESI - The results database records individual search term and query results against the processed media.

NexidiaESIDW – Provides rapid response at scale, an ETL (Extract

– Transform – Load) process is used to build a search database.

The search server accesses this database to identify any pre-existing results against user-requested searches.

Discovery Application



- Click the Discovery tab
- If you see a "No TMS is associated" error:
 - Restart the Nexidia Discovery Service in NX
 - Run NX-Core Update and NX-Discovery Update SQL jobs in NXDB1
 - Discovery Update is a sub job of the Core Update
- In the Select Target Media Set and Filters window:
 - Choose a Date Range (defaults to past 7 days)
 - If unchecked, all calls in the Media Set will be used
- Click the Media Set Administration button
- Enter the phrase group name and Add.
- Phrase Group is an alias for Phrase Pack; in training, it's named Discovery Group
- Click Add to include the group name in the list

- Phrase Groups control ignored phrases and custom respellings for all TMSs in the group
- These changes do not affect media sets in other Phrase Groups
- In the Discovery word cloud, you can:
 - Remove phrases
 - Change phrase spellings
- Changes are stored in the Phrase Group linked to the selected TMS
- Useful for organizations with multiple Lines of Business (LOBs) to tailor word clouds
- Example:
 - Media Sets A & B → Phrase Group One
 - Media Set C → Phrase Group Two
 - Ignoring a phrase in Group One affects A & B, but not C



Define the Population of Calls to Investigate

NEXIDIA ANALYTICS

Default Administrator

NiCE nexidia

SENTIMENTWORD CLOUDDASHBOARDSGAINERS & LOSERS

Media Set Administration

Save

In the table below, pick the Target Media Sets to sample, pick a sampling rate for each one and activate the Media Sets before clicking Save.

Target Media Set Name	Sample Rate	Active	Dispositioning Enabled	Phrase Pack	Phrase Group
TMS Test	100	☒	☐	Telco 0.0.0	PG1

*Note: a Target Media Set will not automatically re-sample if it has been enabled previously.

*Note: Enabling a Target Media Set for query dispositioning will consume approximately 15GB per set. Avoid enabling dispositioning on overlapping data sets or sets not used for query building.

Create, edit or delete Target Media Set Phrase Groups using the table below. Only unused Phrase Groups are eligible for deletion.

Add

Group name	Edit	Delete
PG1	Edit	Delete

If you don't see the Phrase Pack in the drop-down menu, restart the Discovery Service

Since the transcription is process intensive, not all calls may need to be indexed for Discovery.

Here we define the sampling percentage for the selected TMS. The possible values are 1 through 100.

Choose the Phrase Pack from the drop-down menu.

Choose the Phrase Group.

Mark the active column.

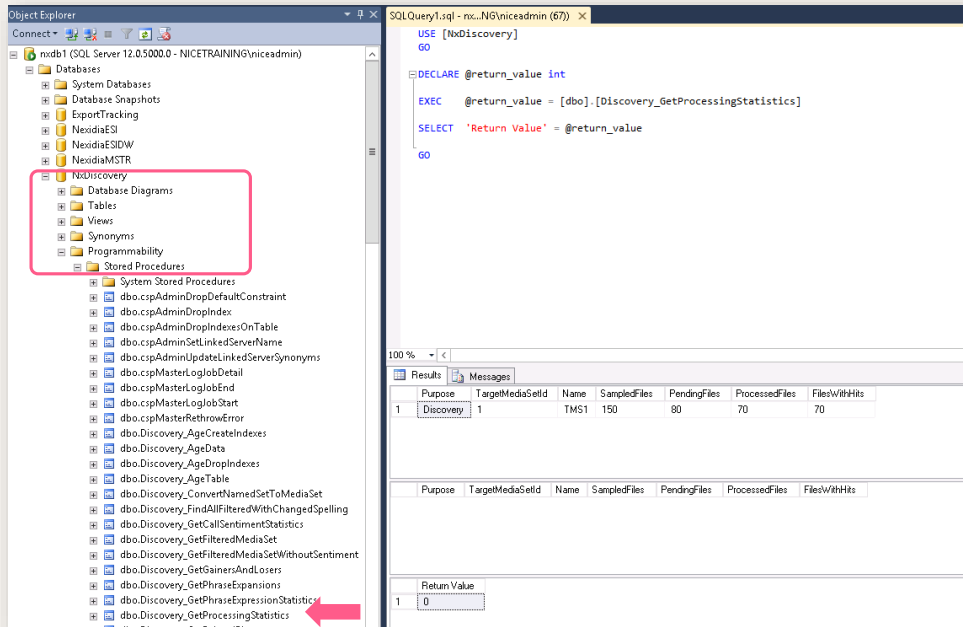
Click Save.

Note: This is the percent of interactions to be sampled from the selected TMS.

Click the Discovery/Sentiment tab again.



Discovery Progress



The screenshot shows the SQL Server Enterprise Manager interface on the left and a SQL Query window on the right. In the Enterprise Manager, the 'NdxDiscovery' folder under 'Databases' is highlighted with a red box. The 'Stored Procedures' folder under 'NdxDiscovery' is also highlighted, and a red arrow points to the 'dbo.Discovery_GetProcessingStatistics' procedure. The SQL Query window shows the following code:

```
USE [NdxDiscovery]
GO

DECLARE @return_value int

EXEC @return_value = [dbo].[Discovery_GetProcessingStatistics]

SELECT 'Return Value' = @return_value

GO
```

The Results tab shows the following data:

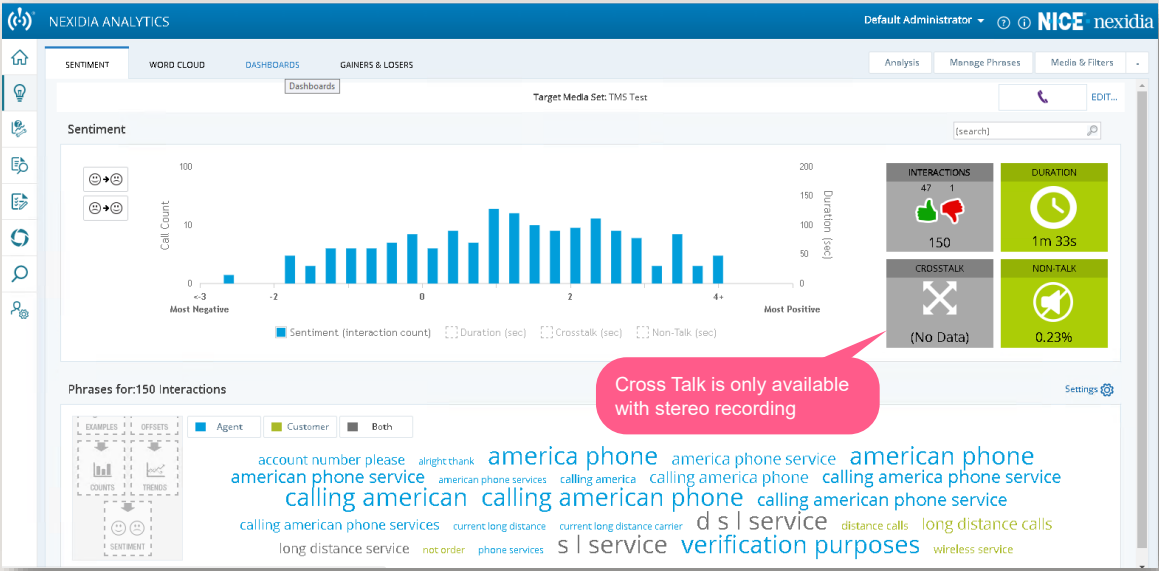
Purpose	TargetMediaSetId	Name	SampledFiles	PendingFiles	ProcessedFiles	FilesWithHits
1	Discovery	1	TMS1	150	80	70

The Return Value is 0.

To check the progress of the discovery task from the Discovery DB, run the `Discovery_GetProcessingStatistics` Stored Procedure.

- Tip: if you don't see any result - restart the discovery service, run the discovery update job, and try changing the Date Range on the Discovery Media & filters window.

Testing Discovery



We can see the result in the Discovery Application.

Right-click on a phrase to view the context menu – from here you can choose to remove the phrase from the phrase group, change the phrase spelling etc.

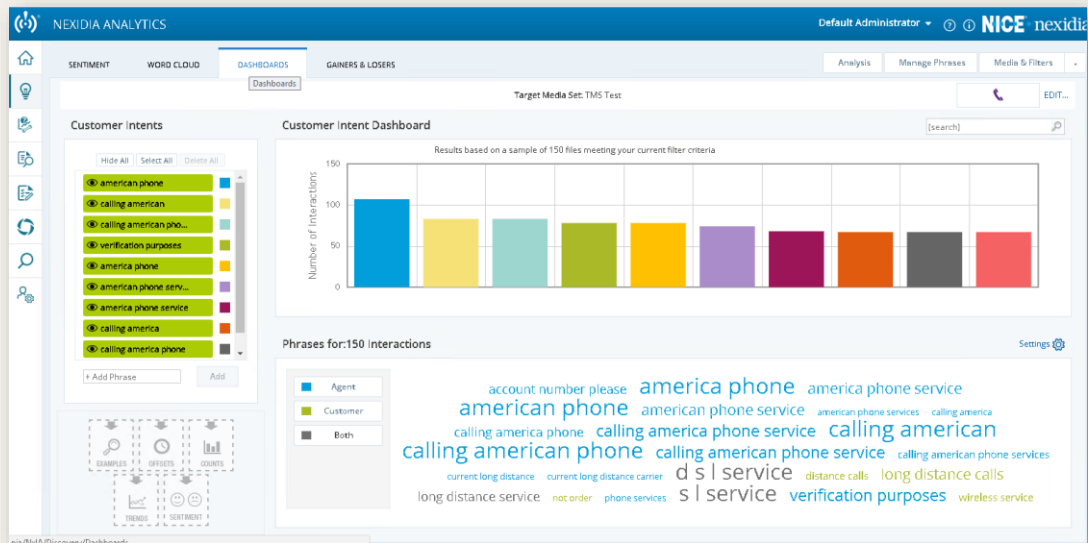
Double click on a phrase to view the transcription.

In the Phrase Pack there is a Stop Words file, which contains words that will appear in the transcription file but will not be part of the Word Cloud.

Cross Talk is only available with stereo recording



Discovery Dashboard



This is an example of a Dashboard.



Review Questions



Review Question #1

What is a Phrase Pack?

Phrase pack contains words that are not part of the standard language and required to create a phonetic index for these “unknown” words.

Phrase pack contains words that are not part of the standard language and required to transcribe these “unknown” words

Phrase pack contains words that are not part of the standard language and required to create sentiment modeling for these “unknown” words.

None of the answers are correct



Review Question #2

What is a Session?

Session defines which interactions we would like to analyze and what words & phrases we would like to search for in the interaction's content

Session defines the population of calls to be analyzed by the Discovery module

Session is a set of logical conditions and phrases used for performing searches on the interaction's content

None of the answers are correct



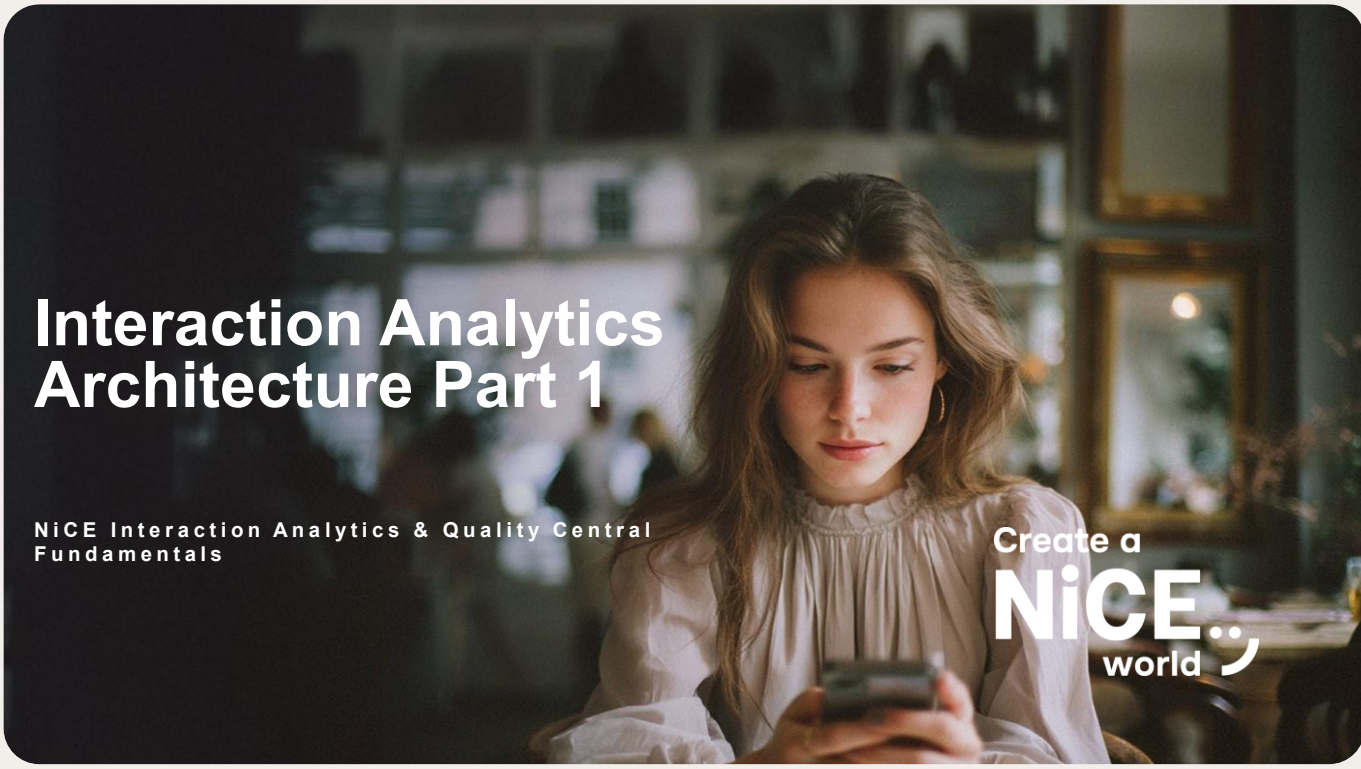
Do it Yourself

- Refer to exercise 1:
- Create a new session and point it to the “**Billing**” media folder, select the queries to apply to the interactions
- Run the session via the Session status tab
- Create a TMS and point it Session created in the previous step
- View results in the Discovery application





Thank You



Interaction Analytics Architecture Part 1

NiCE Interaction Analytics & Quality Central
Fundamentals

Create a
NiCE
world 

objectives

- Describe the Logs, Services and Databases in Interaction Analytics
- Describe the Ingest Flow
- Describe the Search Grid Functionality



Services And
Databases



Log Files

FATAL

ERROR

WARN

INFO

DEBUG

C:\ProgramData\Nexidia\Analytics\Logs

Name	Date modified	Type	Size
ManagementConsole.SupportLog.bt	6/5/2018 5:30 PM	Text Document	3 KB
ManagementConsole.SupportLog.bt.201...	6/3/2018 5:40 PM	2018-06-03 File	10 KB
ManagementConsole.SupportLog.bt.201...	6/4/2018 4:06 PM	2018-06-04 File	10 KB
Nexidia.Config.SupportLog.bt	6/4/2018 11:36 AM	Text Document	11 KB
Nexidia.Discovery.QueryService.Support...	6/4/2018 3:25 PM	Text Document	2 KB
Nexidia.Discovery.WebClient.SupportLog...	6/5/2018 2:29 PM	Text Document	62 KB
Nexidia.Discovery.WindowsService.Supp...	6/4/2018 3:25 PM	Text Document	121 KB
Nexidia.Enterprise.WebClient.SupportLog...	6/6/2018 3:00 AM	Text Document	9 KB
Nexidia.ForensicSearch.SupportLog.bt	6/6/2018 3:00 AM	Text Document	37 KB
Nexidia.Ingest.SupportLog.bt	6/4/2018 10:40 AM	Text Document	0 KB
Nexidia.Ingest.SupportLog.bt.1	6/4/2018 10:40 AM	1 File	25,601 KB
Nexidia.MicroStrategy.Integration.Supp...	6/6/2018 4:21 AM	Text Document	4 KB
Nexidia.QueryBuilder.SupportLog.bt	6/4/2018 12:55 PM	Text Document	11 KB
Nexidia.ReplayService.SupportLog.bt	6/3/2018 5:11 PM	Text Document	41 KB
Nexidia.ReportingConfiguration.Support...	6/3/2018 4:27 PM	Text Document	87 KB
Nexidia.RestApi.SupportLog.bt	6/6/2018 3:00 AM	Text Document	1 KB
Nexidia.Search.SupportLog.bt	6/3/2018 5:40 PM	Text Document	216 KB
NxCfgWiz.SupportLog.bt	6/4/2018 4:05 PM	Text Document	1 KB

Each Interaction Analytics application and service has its own support log, logs are located at C:\ProgramData\Nexidia\Analytics\Logs.

There are 5 logging levels. They are as follows (with the highest being at the top of the list):

1. FATAL
2. ERROR
3. WARN
4. INFO
5. DEBUG



Debug

Local Disk (C:) > Program Files > Nexidia > Nexidia Analytics > RestApi >

Name	Date modified	Type
bin	3/2/2018 10:40 PM	File folder
Global.asax	2/13/2018 6:15 PM	ASAX File
LoggingConfig	2/13/2018 6:15 PM	XML Document
Web	2/13/2018 6:15 PM	CONFIG File

Change Priority Value to "DEBUG"

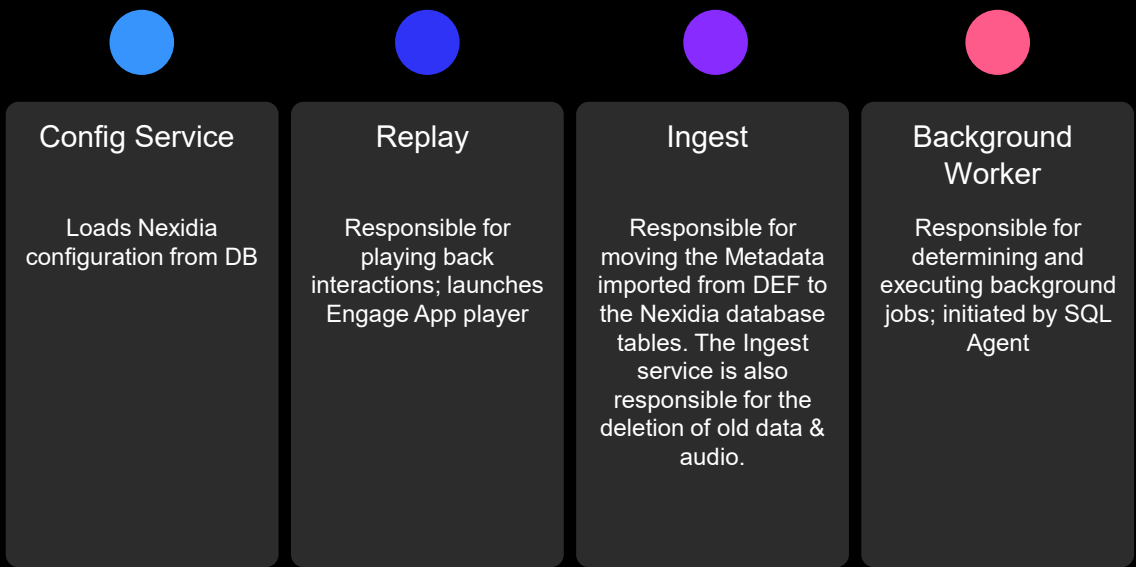
```
<log4net>
  <appender name="SupportFileLog"
    type="log4net.Appender.RollingFileAppender">
    <file value="{ProgramData}\Nexidia\Analytics\Logs
\\Nexidia.QcAdapter.SupportLog.txt"/>
    <appendToFile value="true"/>
    <rollingStyle value="Size"/>
    <maxSizeRollBackups value="8"/>
    <maximumFileSize value="5MB"/>
    <staticLogFileName value="true"/>
    <encoding value="utf-8"/>
    <layout type="log4net.Layout.PatternLayout">
      <conversionPattern value="%date %9timestamp %-6level %-
15.15thread %message [%ndc] %logger%n"/>
    </layout>
    <filter type="log4net.Filter.LevelRangeFilter">
      <param name="LevelMin" value="DEBUG"/>
      <param name="LevelMax" value="FATAL"/>
    </filter>
  </appender>
  <root>
    <priority value="WARN"/>
  </root>
  <logger name="QcAdapter.WebHost.Infrastructure">
    <priority value="WARN"/>
    <appender-ref ref="SupportFileLog"/>
  </logger>
  <logger name="QcAdapter.WebHost.Activity">
    <priority value="INFO"/>
    <appender-ref ref="SupportFileLog"/>
  </logger>
  <logger name="Persistence.Infrastructure">
    <priority value="WARN"/>
    <appender-ref ref="SupportFileLog"/>
  </logger>
  <logger name="Nexidia">
    <!-- Class Level Logging -->
    <priority value="WARN"/>
    <appender-ref ref="SupportFileLog"/>
  </logger>
  <logger name="ASP">
    <!-- Class Level Logging for .aspx, .ascx, and .asax
files-->
    <priority value="WARN"/>
    <appender-ref ref="SupportFileLog"/>
  </logger>
</log4net>
```

Each component has a separate folder in the install folder > Nexidia Analytics. Open the LoggingConfig.xml and change each Priority Value to "DEBUG". Do it for each component (Ingest, Search ,Replay and Discovery) to see the details





Core Analytics Services



Note: All services begin with the prefix "Nexidia" (e.g., *Nexidia Configuration*), except for MicroStrategy. While MicroStrategy includes multiple services, Nexidia only utilizes the Intelligence and Listening Services.

Ingest Service Overview:

The Ingest Service monitors for newly available media and initiates ingestion by submitting requests to the Search Grid. Once results are returned, it updates the database with call metadata and query outcomes.





Core Analytics Services

5

MicroStrategy

Used for Analysis and Reporting

6

Discovery Service

Submits the calls to the ASR process and collect the results. Presents the information on the Word Cloud

7

Search Service

Sends the request for the Search Grid to perform ad-hoc searches, and query the DB for metadata searches

Explore Other Services

Click on the appropriate service in the Services application to view detailed information and functionality for each service

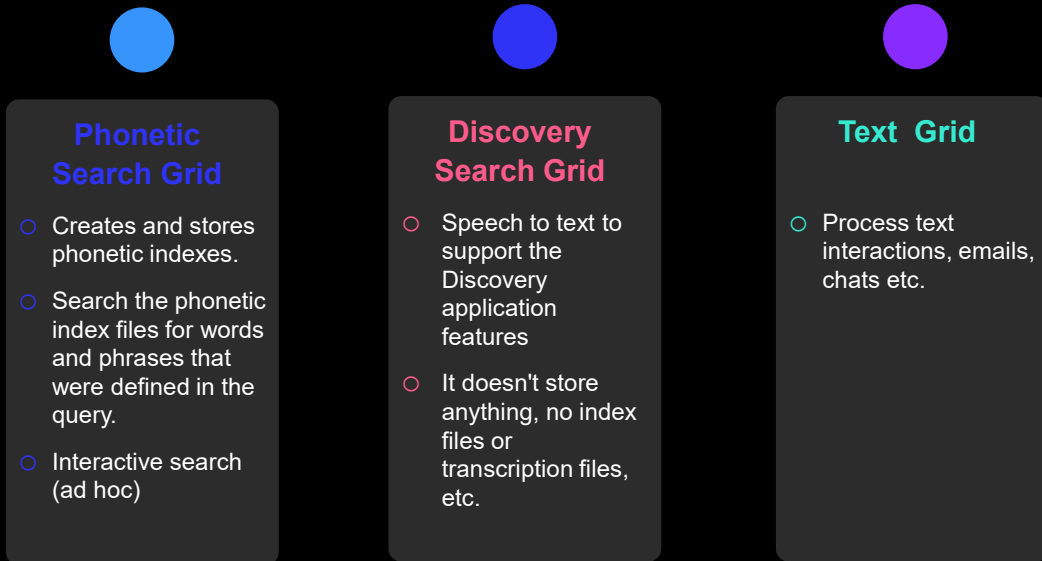


Search Grid





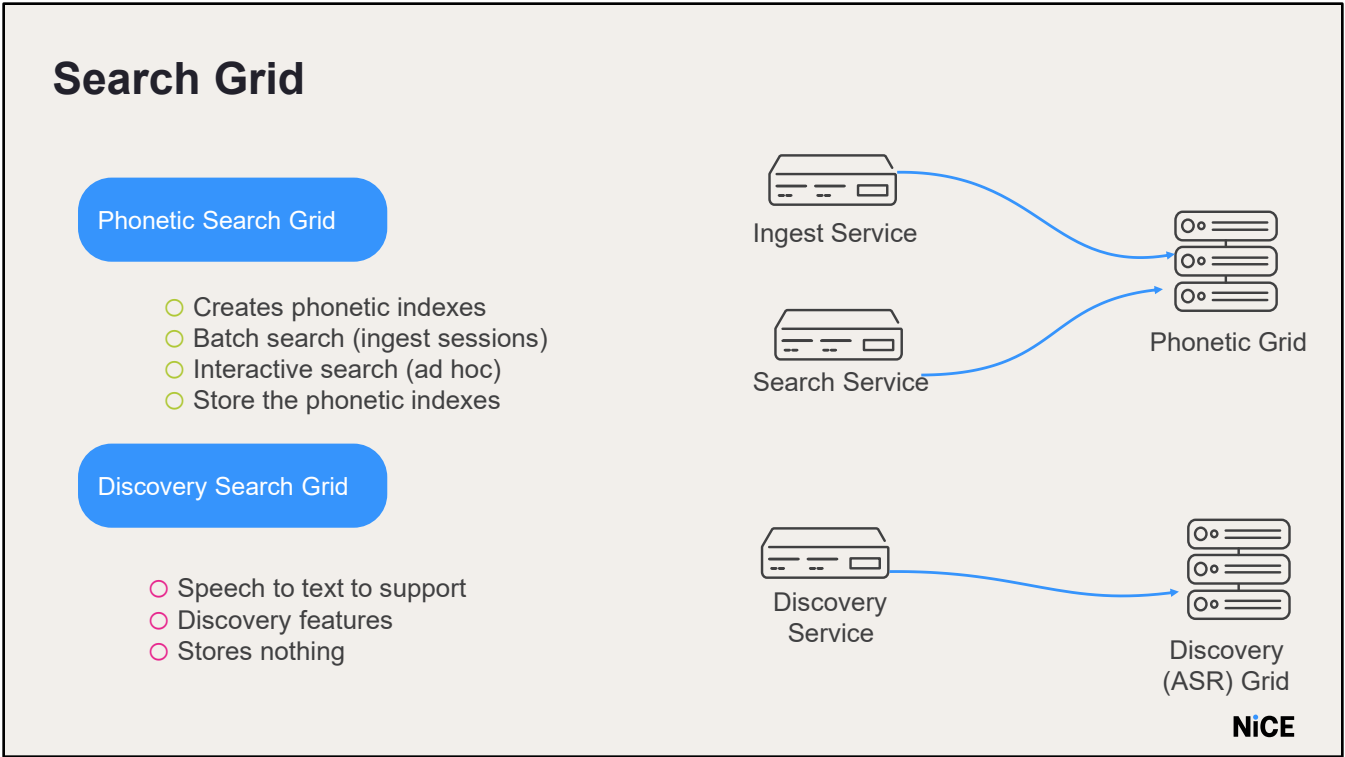
Search Grid



Search Grid is the analytics engine. It is a distributed system for analyzing and searching recorded audio.

There are three types of search grids





The Ingest service and the Search service (ad hoc) communicate with the Phonetics Grid.

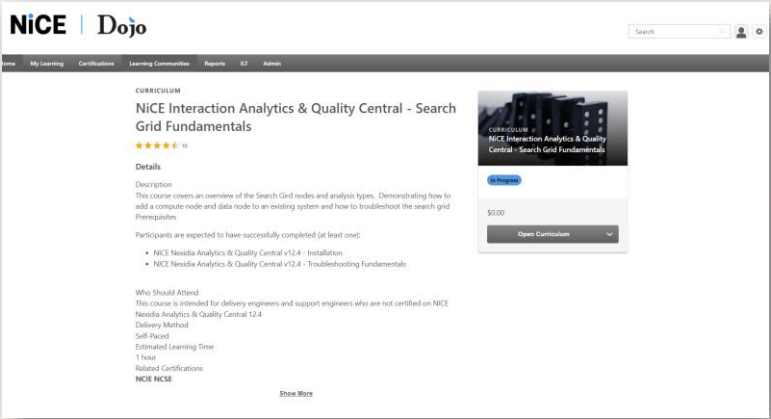
The Discovery service communicate with the Discovery Grid



Training

Search Grid Administration Guide

Available in DocHub



Search Grid Fundamentals Curriculum is available in NiCE Dojo.

Documentation on Search Grid is available in DocHub.
Search Grid Fundamentals Curriculum is available on NiCE DOJO

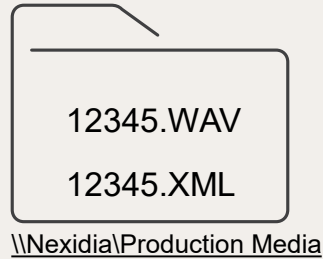


Ingest Flow



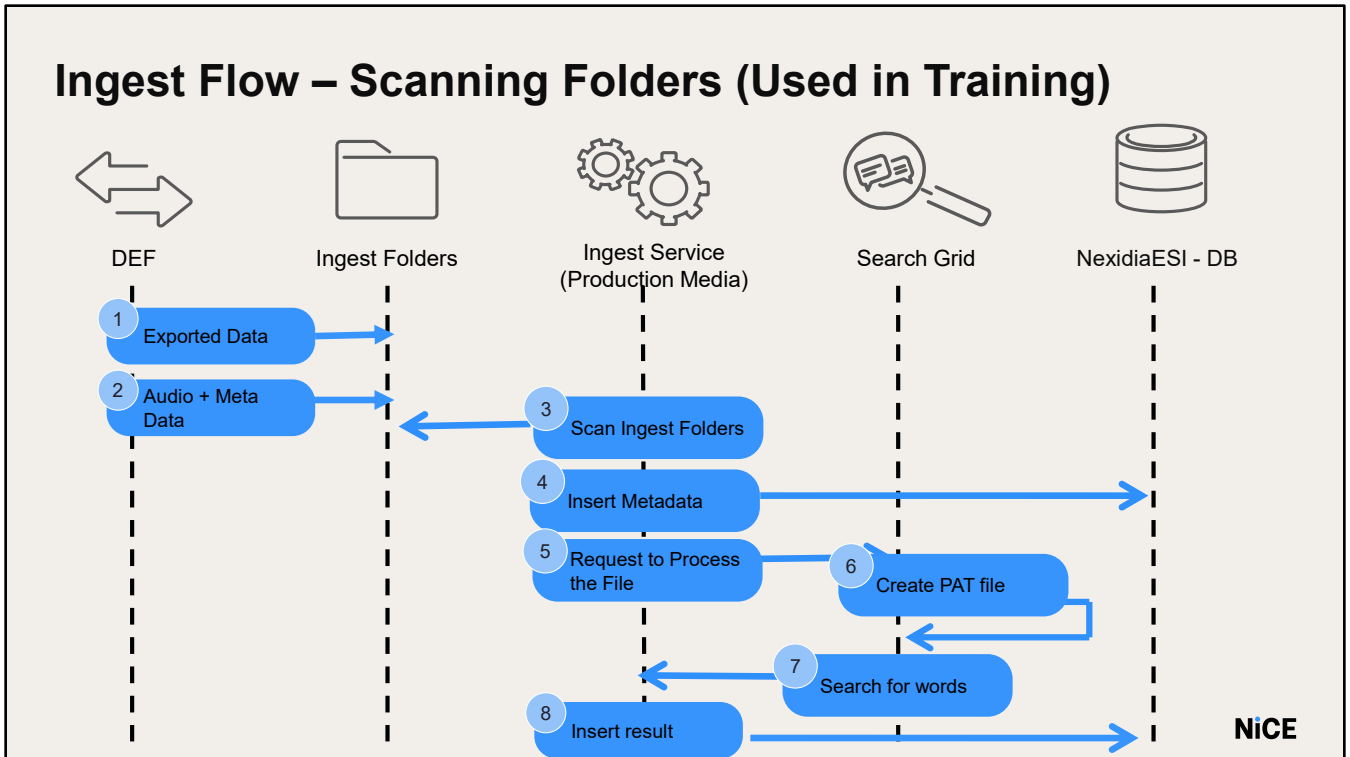
Ingest Flow

- Option 1 – Scanning Folder (Legacy)



Production Media folder contains both the audio file and the metadata file

Ingest Flow – Scanning Folders (Used in Training)



1. DEF automates the retrieval of media files and metadata from external sources.
2. DEF places the data in staging folders for media and metadata (under the Production Media root directory). The Ingest service scans the folders and looks for new files to process.
3. The ingest service inserts the interaction metadata into the DB.
4. The Ingest service scans the folders and looks for new files to process
5. The ingest service sends a request to the search grid to process the interaction (create PAT file & search for words and phrases). The PAT file is created only once when the interaction is ingested to system.
6. The search grid will create the phonetic index / PAT file and save it locally.
7. The search grid applies any search terms or queries to the PAT file. Results from those searches are sent to the ingest service,
8. The ingest service inserts them into the database.

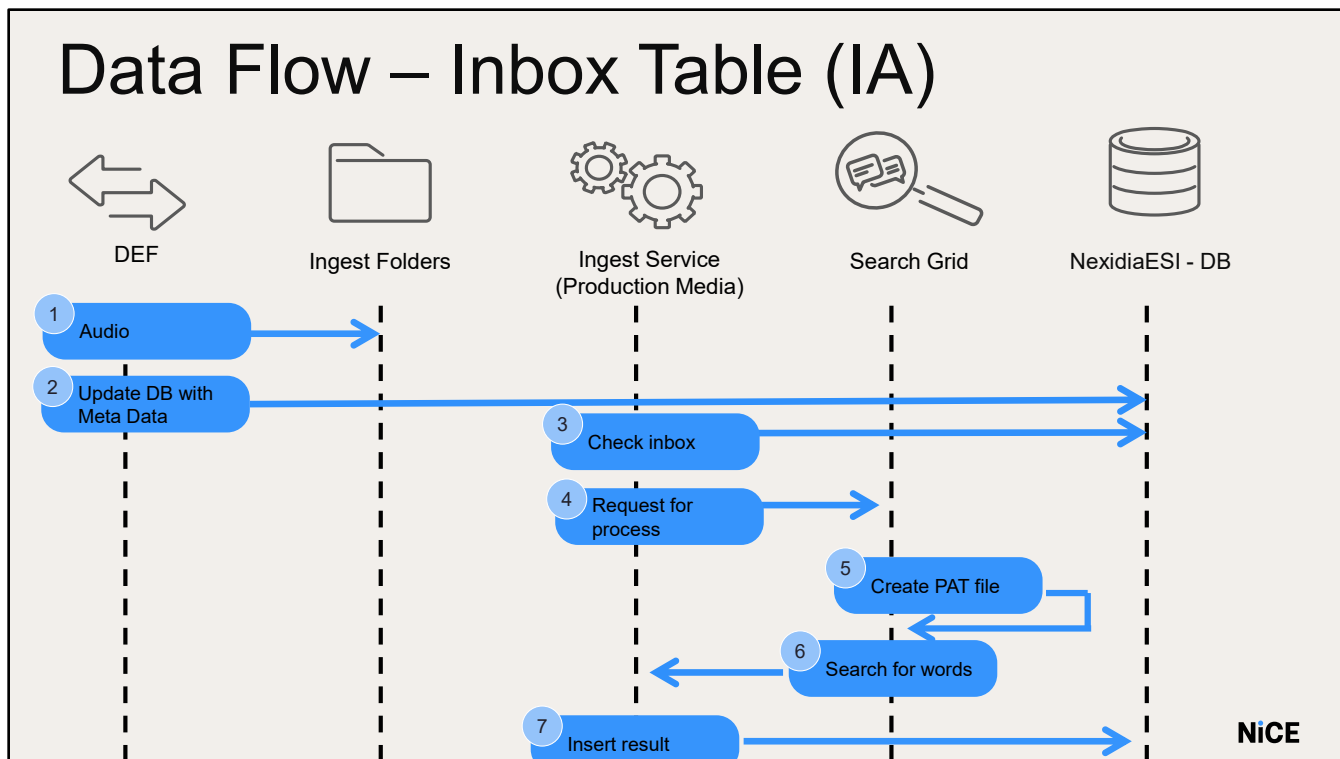
Ingest Flow

- Option 2 – Inbox Table



Production Media folder contains both the audio file and the metadata file

Data Flow – Inbox Table (IA)



DEF automates the retrieval of media files and metadata from external sources.

1. DEF places the audio in staging folders
2. DEF inserts the interaction's metadata to the database (NexidiaESI > NexidiaESI.SourceMediaInbox table)
3. Ingest service checks this table for new entries new interactions to process.
4. The ingest service sends a request to the search grid to process the interaction (create PAT file & search for words and phrases).
5. The search grid will create the phonetic index / PAT file and the transcription file and save them locally.
6. The search grid applies any search terms or queries to the PAT file. Results from those searches are sent to the ingest service.
7. The ingest service inserts them into the database.



Do it Yourself

- Refer to exercise 2:
- Create a new search term, “account 1.”
- Define a new session “**CS Tier 1**” that will use this term.
- The session will not ingest the files.

Why? Investigate the ingest support log file

Note: You can refer to the exercise book for guidance on how to create a search term



REVIEW
QUESTIONS



Review Question 1

What is the Source Media Inbox Table?

A table which contains all the interactions that were imported by the DEF server

A table which contains all the interactions that were imported by the DEF server, and were not yet processed by the ingest service



Review Question 2

How many times is the index (PAT) file created when an interaction is ingested into the system?

Only once
when the interaction
is ingested to system

Every time a
phonetic search is
performed



summary

- Describe the Logs, Services and Databases in Interaction Analytics
- Describe the Ingest Flow
- Describe the Search Grid Functionality





Thank You



Interaction Analytics Architecture Part 2

NiCE Interaction Analytics & Quality Central
Fundamentals

Create a
NiCE
world 

objectives

- Describe the Search Flow
- Describe the Discovery Flow
- Describe the Replay Flow
- Describe the Stereo Calls
- Describe the Data Aging Flow



AD-HOC SEARCH



Search Application

The screenshot shows the NiCE Search Application interface. On the left, there's a sidebar with 'SESSIONS/FILTERS' and 'JOB QUEUE'. The main area has a 'Search Phrase' input field and a 'Search' button. Below this is a table of search results for 'All Files'. A pink callout bubble points to the 'Search' button with the text 'Ad-Hoc Search- Send Request to SG'. Another pink callout bubble points to the 'Details' section with the text 'Metadata Search - Query the DB'. The 'Details' section shows a waveform and a table of search results.

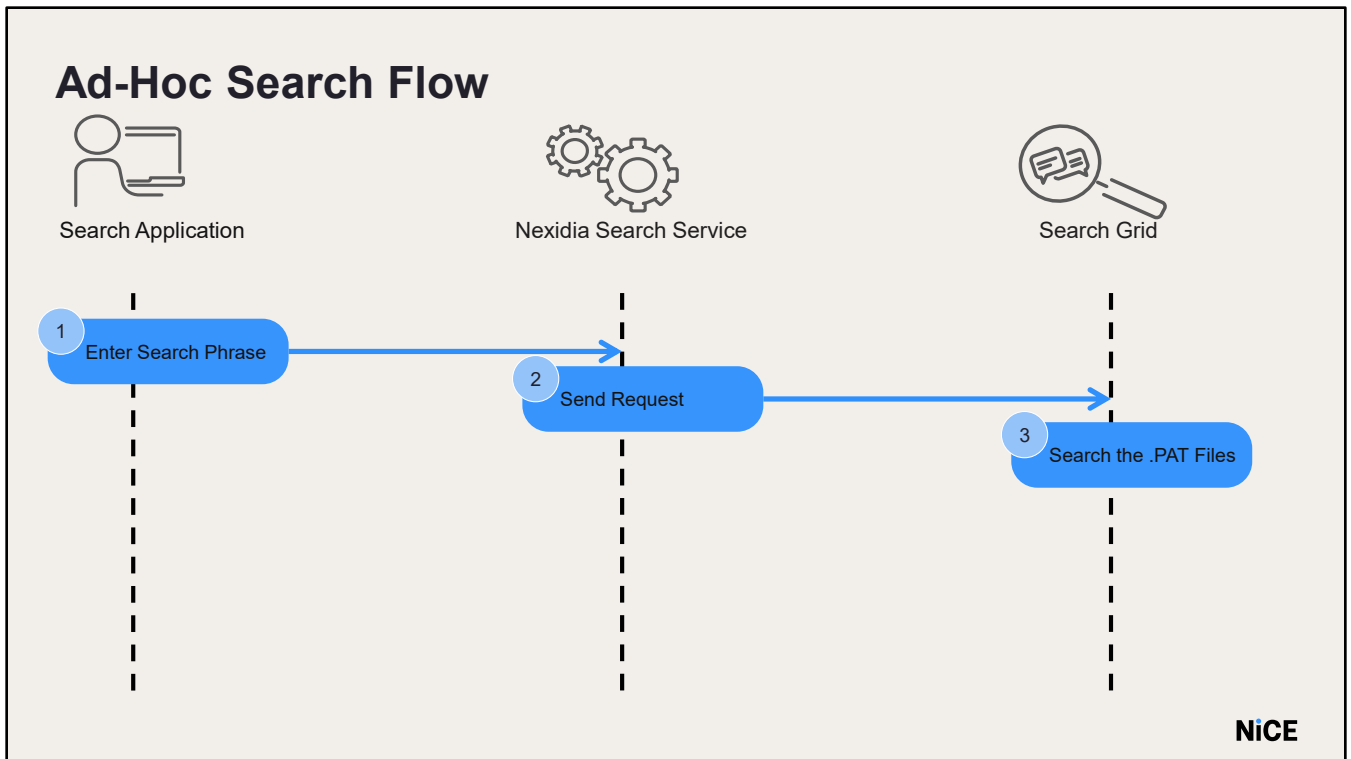
Ad-Hoc Search- Send Request to SG

Metadata Search - Query the DB

When we perform a metadata search, the application search service will query the DB for the result (ESIDW).

When we perform an ad-hoc search, Pat file needs to be queried. The application search service will send the request to the search grid.





1. In the Search application, the user selects the session of interactions to be queried and defines the phrase to be searched for in those interactions.
2. The search service sends the request (set of interaction+ search phrase) to the search grid.
3. The search grid looks for the search phrase in the interaction's PAT file
4. The results are returned to a temporary cache table and then displayed in the application.

If another user makes the same search within a relatively short time, then the results from the cache table are displayed. This table gets wiped periodically, and between logins, so search response varies depending on whether the results are cached.



Ad-hoc search - Flow

Nexidia.Search.SupportLog.txt

2017-06-14 06:33:05,354 504160278 **INFO** 220 methodExit:
method=**SearchService.BeginSearch**, elapsedTimeInMs=5346, searchId=c917ab97-456d-4b2a-9763-f5e81a9c3ac7 [(null)] Search.Infrastructure
2017-06-14 06:33:05,436 504160360 INFO 585 **Search Request Started:**
Algorithm=Conventional, **LanguagePackGuid**=7038b2ef-dcf8-495b-8537-d77b9bd35346, **Media To Search=391**, Operator=Any, TimeOperator=Anywhere, FilesExcludedDueToMaxSearchableFiles=False, Media Found In Cache=0, Media Found In Cache With Term Hits=0, CandidatesPerMediaHour=0.1,
Term=Id:1|**Term:hello**|Threshold:75 [SearchId=c917ab97-456d-4b2a-9763-f5e81a9c3ac7] Search.Activity

2017-06-14 06:33:08,231 504163154 **INFO** 220 methodEntry:
method=SearchService.**EndSearch** [SearchId=c917ab97-456d-4b2a-9763-f5e81a9c3ac7]
Search.Infrastructure

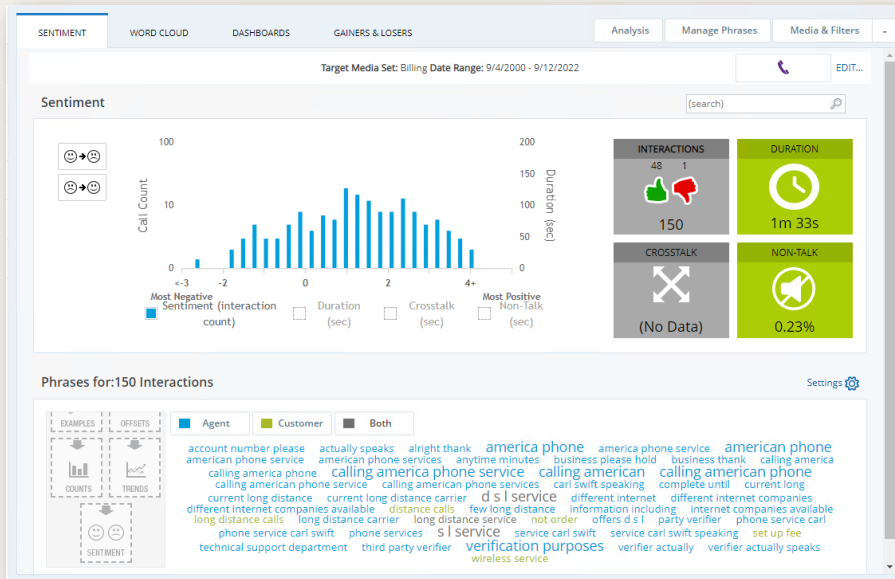
Located on **NIA** server
C:\ProgramData\Nexidia\Analytics\Logs



DISCOVERY FLOW

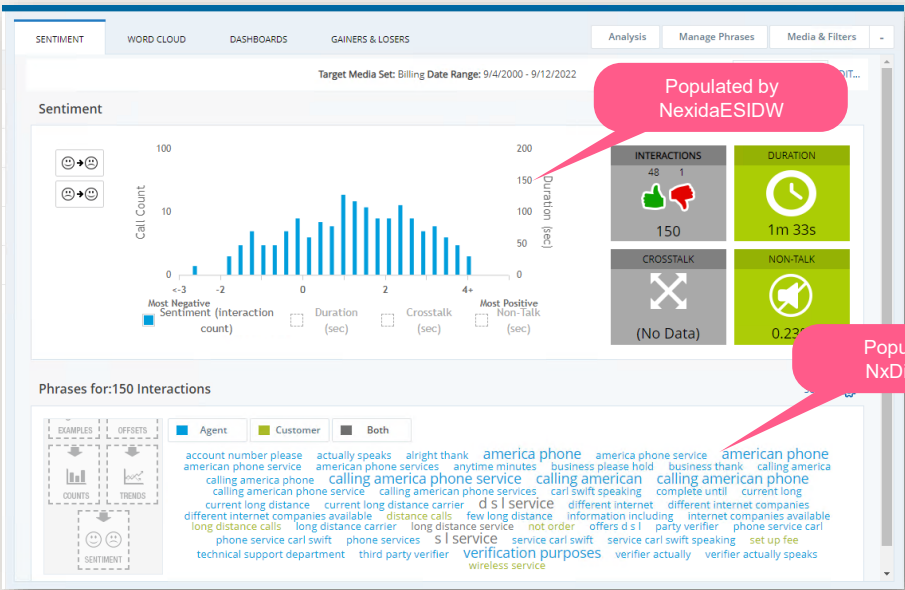


Discovery – What You Don't Know



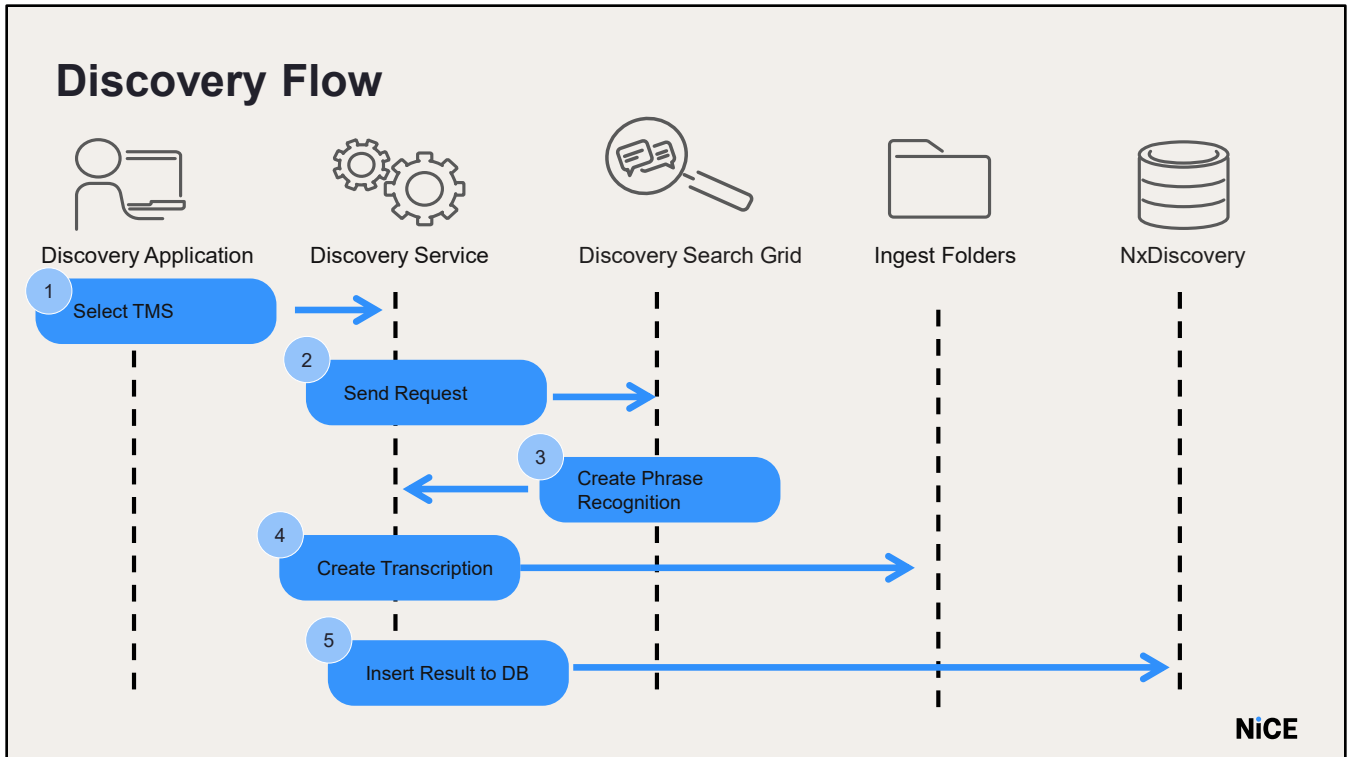
The search uses speech to phonetics and Discovery uses a speech to text. Discovery results are not based on query results; the insights that are provided are based on the transcriptions of calls. Therefore, we don't need to build a query upfront, and instead the words cloud will show the phrases occurring on a specific set of calls. Discovery provides insight into what we should be analyzing. Discovery starts with the industry phrase information, processes the audio transcriptions for the calls and results are generated, including sentiments, word count and phrase relationship. The more refined the phrase pack, the more accurate the results.

Discovery



1. Each part of the Discovery application is populated by different databases. The word cloud is populated with data from NxDiscovery database, and the sentiment graph is populated with data from
2. NexidaESIDW database. If there is data in sentiment graph but not in the word cloud this means that Discovery DB is not updated with the result, but the Nexidia DW is updated





1. In the Discovery application, the user selects a TMS to analyze.
2. The discovery service sends the request to the Search grid.
3. Discovery search grid, also known as ASR – creates a Phrase recognition index file and send it to the discovery service.
4. The discovery service – create transcription for the audio file using the Phrase recognition index file. The transcription files are stored in the ingest folder.
5. Discovery service inserts the result to the Discovery DB

Sentiment

WHAT IS SENTIMENT?

A machine learning (AI) model trained to measure whether a customer interaction is positive, negative, or neutral, on a relational scale.



Sentiment Detection

SPOKEN WORDS

- Language models identify positive and negative words and phrases
 - Positive phrases can off-set negative ones
- No problem
 this is ridiculous
 awesome
 I'm so annoyed

LAUGHTER DETECTION

- Indicates positive emotional state
 - Pervasive element
 - Can prevent false positives and unnecessary alerts
- 

CROSS TALK

- Customer and agent talk at the same time, interrupting one another



PITCH & TONE

- Intensity
- Pitch
- Jitter
- Shimmer
- Speaking rate



Sentiment detections comprise from 4 features :

Pitch and Tone or acoustic properties - These are things like how loud, how fast, or how agitated a person is speaking.

Laughter – a good indicator of positive sentiment regardless of what else may be detected during the call.

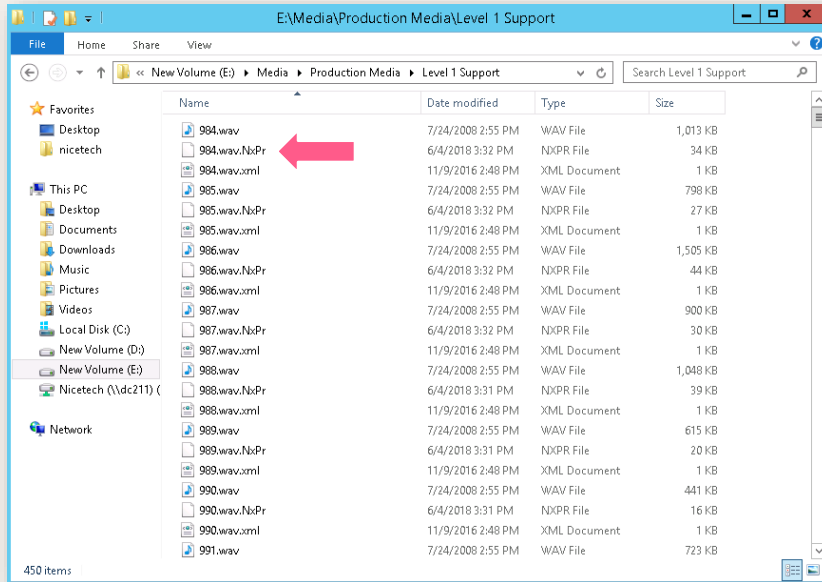
Cross-Talk – a reliable indicator of a negative emotional state if the two parties are talking over each other.

Spoken Words and Phrases – sets of training data are used to develop language models where the system can automatically identify positive and negative phrases specific to a customer's environment.

Note: Sentiment Detection is done by the Discovery service.



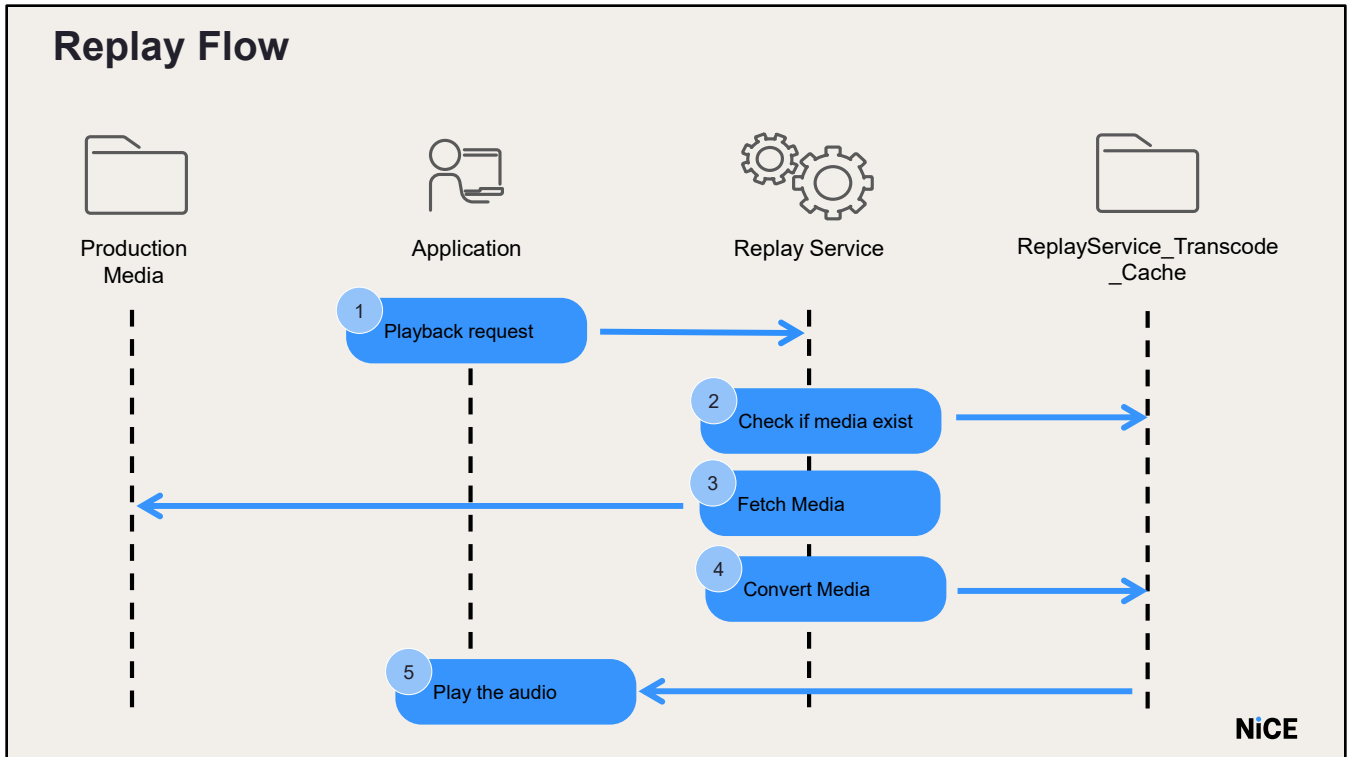
Transcription Files



The transcription files are stored in the ingest folder
 The file extension is.NxPr (Nexidia phrase recognition), this
 is a binary file.

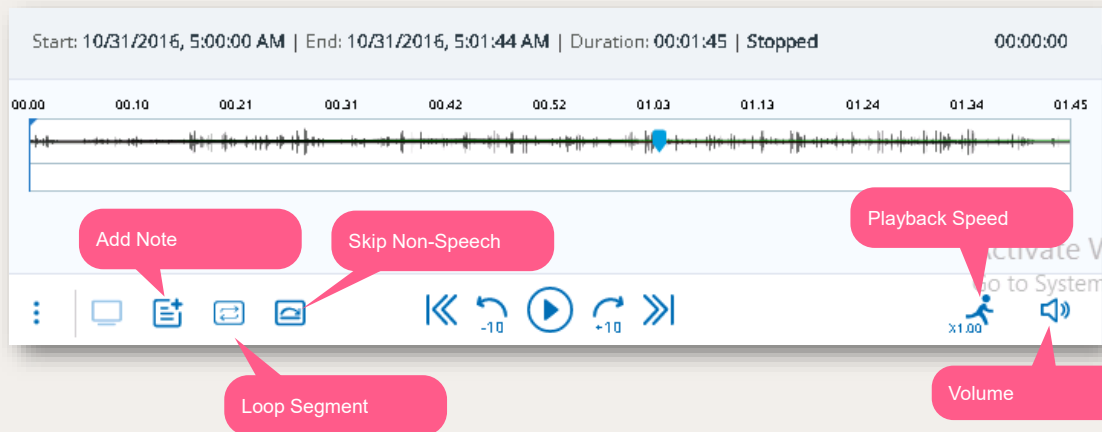
PLAYBACK FLOW





1. When the user clicks on the Play button, the request is sent to the replay service.
2. The replay service checks if the audio file already exists in the “replayService_Transcode_Cache.” If so, the replay service will play the audio file.
3. If the audio file doesn't exist in the “ReplayService_Transcode_Cache” folder, the replay service will fetch it from the production media location
4. The replay service will convert the audio file to MP4 format and will create a PNG file.

Player



- 1. Offset-Specific Notes – users can annotate specific points in the file, highlighting content of interest within the context where it occurred
- 2. Loop Segment - during playback, the user can specify a start and end point for a segment and have the playback loop within those points, allowing detailed investigation of the content therein
- 3. Enhanced Playback Speeds – In addition to the previous playback speeds available (0.5x, 1x, 2x), Nexidia Analytics v12.1 now also supports 4x and 8x speed
- 4. Integrated Volume Control – user can adjust (or mute) the volume directly in the player.



ReplayService_Transcode_Cache Folder

✓	ReplayService.Transcode.BitRate	Auto	Configuration for the bit rate (in bps) used when tran
✓	ReplayService.Transcode.Cache.CleanupInterval	60	The frequency at which the transcoded media cache :
✓	ReplayService.Transcode.Cache.ImportMP4	true	Determines if existing mp4 files should be copied din
✓	ReplayService.Transcode.Cache.MaxSize	10GB	Target maximum size for the service side transcoded
✓	ReplayService.Transcode.Cache.RetryFailureInterval	60	The number of seconds that should elapse before a f
✓	ReplayService.Transcode.Cache.		ed to stream media. If empty, a default port is

Folder Size

Name: ReplayService.Transcode.Cache.MaxSize
Value: 10GB
Default Value: 10GB

Notes: Target maximum size for the service side transcoded media cache. This cache is persis
The target size can be specified in Megabytes, Gigabytes, Terabytes and also as a per
Valid values are in the form of [size][units], where units is one of MB, GB, TB or %.
Examples:
100MB = 100 Megabytes.
10GB = 10 Gigabytes.
15% = Grow until there is only 15% free space left on the disk. This takes quotas int
The default is 10 Gigabytes.

RESTORE DEFAULT OK CANCEL

The cache by default is set to a limit of 10GB. It will fill the 10GB and then remove calls as needed to keep it at 10GB.

The size of the ReplayService_Transcode_Cache folder is configurable in the Management Console.

Essentially the folder LRU(Least Recently Used) clean triggered by the cache size exceeding its configured limits. The MaxSize can be configured in units of MB/GB/TB or as a % of available drive space



Do it Yourself

- Refer to Exercises 3
- Create a new session, in the "media file path" tab select the "**CS Tier2**" folder
- Login to the Search application, select the first interaction (interaction number 594), and play it.
- Go to the media folder located on E:\Media\Production Media\CS Tier 2 and delete this interaction(delete the 05212005-594.wav file).
- Go back to the Search application and try to play the interaction.
- Were you able to play the interaction? why?



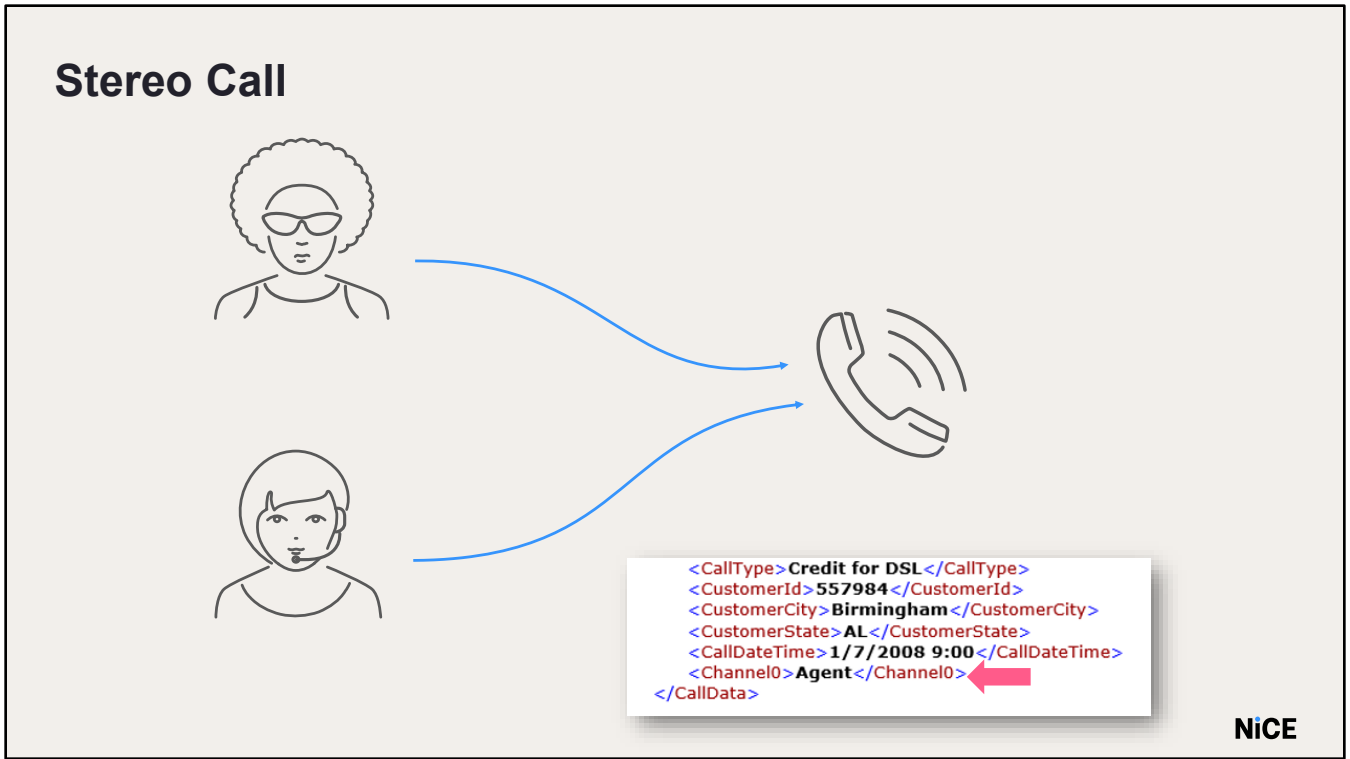
Do it Yourself

- Refer to Exercises 4
- Go to the media folder located on E:\Media\Production Media\CS Tier and delete the 05212005-593.wav file
- Login to the search application and try to playback interaction 05212005-593
- Were you able to play the interaction? Why?



STEREO CALLS





For Stereo calls, the customer and agent are recorded on separate channels.

Depending on the recording platform, there may be two recordings (one for agent, one for customer).

With NICE Engage, there will only be one NMF file.

For the stereo calls, there is new metadata filed named channel 0, which specifies who the speaker is on that channel.

Console Management

The screenshot displays the 'Nexidia Analytics - Management Console' window. The 'Configuration Parameters' tab is active, showing a table of parameters. A red arrow points to the 'Nexidia.Enterprise.SpeakerSeparatedProcessingEnabled' parameter, which is set to 'true'. Below this, a call analysis player is shown with a timeline and waveform. A pink callout bubble points to the player with the text 'The player displays SSA information'. The player interface includes a timeline from 00:00 to 01:10, a waveform, and a table of events.

Valid	Name	Value	Notes
✓	Nexidia.Enterprise.SpeakerSeparatedProcessingEnabled	true	Sets whether or not to process and display speaker separated call information. If this sy
✓	Scan.SpeakerSeparatedProcessingEnabled	true	Scan - (EVENT RE-PUBLISH NECESSARY AFTER CHANGE) - speaker separated calls enab

Start: 1/6/2008, 9:00:00 AM | End: 1/6/2008, 9:01:09 AM | Duration: 00:01:10 | Paused

Customer Agent

00:00 00:07 00:14 00:21 00:28 00:35 00:42 00:49 01:03 01:10

00:00:31 Angry Customers_8.6.0.14... 0.4923..

00:00:33 Angry Customers_8.6.0.14... 1.8578..

Activate Windows
Go to System in Control Panel to activate Windows.

Sets whether or not to process and display speaker separated call information.

The Nexidia.Enterprise.SpeakerSeparatedProcessingEnabled should be set to true to view Speaker Separation Analysis (SSA) If this parameter is set to false, the interaction will be treated as a mono interaction.

The player displays Speaker separation analysis information as well as the location and speaker of the query detection in the call duration.

AGING DATA



What Will Be Deleted



Interaction Aging

- Audio Files



Data Aging

- Database Records
- PAT Files
- Metadata Files

There are two processes that are responsible for data retention:

1. Interaction files will be deleted by the interaction aging process
2. Database records, PAT files, and metadata files will be deleted by the data aging process.

Both processes can have the same schedule (data & interactions file will be deleted at the same time), or they can be set to run at different times.



Interaction Aging

Run reports on the interaction

Perform Metadata search
& Phonetics search

Unable to
playback the interaction

Interaction Aging - Allows interactions to be aged out of the system (meaning the physical interaction file is deleted) on a shorter schedule than general data aging.

The advantage of this function is that it conserves storage space, and you can still report on this interaction. You cannot report on interactions after data aging occurs. After interaction aging is performed on a set of interactions, you cannot play them in the application, but the search application will allow you to perform a search on the interaction's metadata and perform phonetics search on the interaction's PAT file.



Interaction Aging

E:\Media\Production Media\111.wav

Media Folder before aging

E:\Media\Production Media\111.wav.nomedia

Media Folder after aging

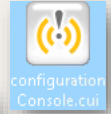
NiCE

When interaction aging is performed, the file is replaced with a .nomedia file extension.

For example: C:\test.wav becomes C:\test.wav.nomedia.



Configuring Aging



File Tools Help

Metadata Configuration Parameters

Filter ret

FIND/REPLACE >>

☐ Hide parameters with default values. Tag Filter

Valid	Name	Value	Notes
	Discovery.WordCloud.MaximumPhraseCount	2000	The maximum number of phrases to return for all pag
	DiscoveryService.MaxResultRetrievalThreads	20	The maximum number threads to retrieve search gric
	ForensicSearch.DrillToSearch.WebRoot	http://NIA/NxIA/Search/	URL for the DrillToSearch service to use for retrieving
	ForensicSearch.GridPagePreTranscodeFileCount	1	Search: The number of files to pre-transcode when lo
	ForensicSearch.MaxReportedHitsPerFile	50	Maximum number of stored reporting hits per file to
	ForensicSearch.ParameterSelectorExpandedDefault	true	Determines if Parameter Selector is expanded by defi
	Ingest.Retry.ErrorCodes	1, 10, 12, 30, 31, 40	Ingest Retry: Error codes for media that will be retriec
	Ingest.Retry.WindowSeconds	86400	Ingest Retry: How far back (in seconds) the job will lo
	Nexidia.DataAging.AudioDataRetentionDays	120	Data Retention Days for audio
	Nexidia.DataAging.AudioMediaRetentionDays	-1	Default media retention days for audio interactions. :

Media Retention = -1 means that the data & media have the same retention period

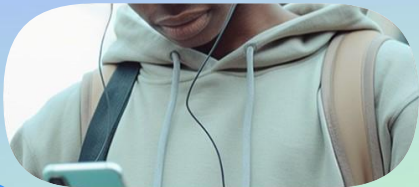
You can set your policy for aging by editing the applicable values in the Management Console:

Nexidia.DataAging.AudioDataRetentionDays

Nexidia.DataAging.AudioMediaRetentionDays



REVIEW
QUESTIONS



Review Question 1

Which one of the following is true?

The replay service
copies the interaction's
original audio file from the
Production Media folder
and places it in the
ReplayService_Transcode
_Cache folder

The replay service
converts the audio file into
MP4 and places it in the
ReplayService_Transcode
_Cache

Review Question 2

Which statement best describes the ad-hoc search flow?

The user submits an ad-hoc search request in the application, and the search service sends the request to the search grid, which performs a search on the PAT file

The user submits an ad-hoc search request in the application, and the search service queries the DB for the results

Review Question 3

What is the flow for Metadata Search?

A user submits a metadata search request in the application, and the search service sends the request to the search grid, which performs a search on top of the PAT file.

User submits an ad hoc search request in the application and the search service queries the DB for the results.

Review Question 4

You login to the Discovery application and see the sentiment results but not the word cloud.
What might be the reason?

The interaction's audio is too short

Discovery DB is not updated but the Nexidia DW is updated

Review Question 5

What is the purpose of Phrase pack?

Phrase pack contains Dictionary + words that are not part of the standard language and required to transcribe the interaction

Phrase pack contains words that are not part of the standard language and required to create a phonetic index for the interaction.

Review Question 6

What is the purpose of the Ingest Service?

Scans for new interactions to process send processing requests to the search grid and deletes old data & audio

Scans for new interactions to process, creates the PAT file for each new interaction and deletes old data & audio



Review Question 7

What is the difference between data aging & interaction aging?

After interaction aging - the user won't be able to perform those actions.
After data aging - the user will be able to perform searches & run reports on the interaction

After interaction aging - the user will be able to perform searches & run reports on the interaction.
After data aging - the user won't be able to perform those actions



summary

- Describe the Search Flow
- Describe the Discovery Flow
- Describe the Replay Flow
- Describe the Stereo Calls
- Describe the Data Aging Flow

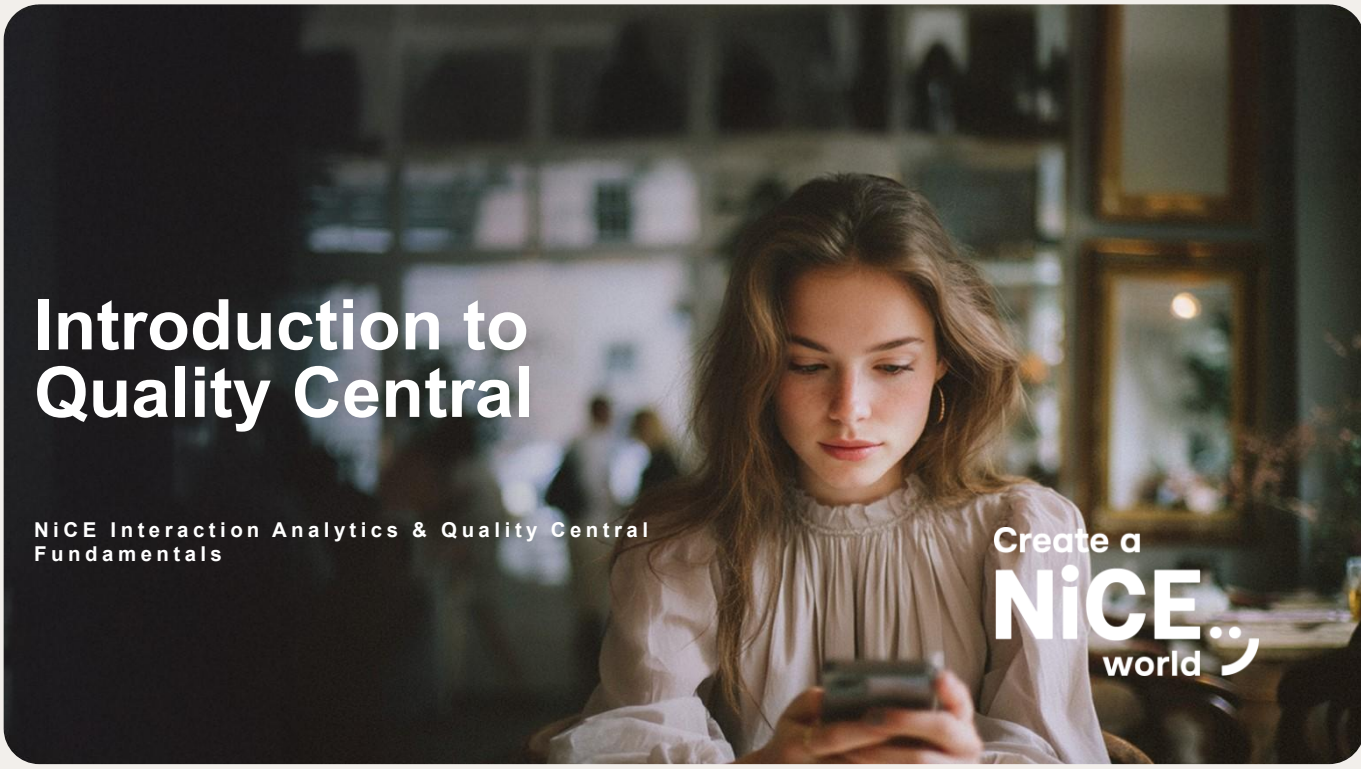


This should be similar to the Objectives / Agenda slide. Can also include screenshots for a visual element.





Thank You



Introduction to Quality Central

NiCE Interaction Analytics & Quality Central Fundamentals

Create a
NiCE
world 

objectives

- Evaluate the impact of Quality Management on NiCE customers
- Describe the steps involved in the Quality Management process
- Demonstrate how to use Quality Central effectively



NiCE QUALITY
CENTRAL (QC)



Why QC?

- Optimize\reduce Average Handle time
- Reduce hold time agents are better trained
- Increase in first call resolution (less repeat callers)
- Increase in up sales
- Increase in customer satisfaction



NICE Quality Central

NiCE

The main challenge of the call center managers is to reduce the cost of operation.

The main cost of operating a call center is employee wages, and it is being measured by cost per call (cost per minute of call).

By reducing the call average handle time, we are reducing the operational cost.

Using quality management and coaching our agents, we can save time and money to our organization by monitoring and improving key metrics. Eliminate repeat calls, Optimize Average Handle time, Reduce agent knowledge gaps

- Optimize\reduce Average Handle time
- Reduce hold time agents are better trained
- Increase in first call resolution (less repeat callers)
- Increase in up sales
- Increase in customer satisfaction



What is QC?



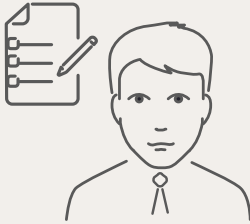
NiCE Quality Central



Quality Central is the process of evaluating Agent interactions with the customer to make sure they adhere to the company policies.

Usually a specific # of evaluations need to be performed each week or month. The Evaluations can be reported on and feedback shared with agent

Quality Management Participants



QA Evaluator



Agent



Team Lead

There are 3 main users in a Quality Management solution – QA Evaluator, Agent, and Team Lead.

QA Evaluator is responsible for monitoring the Agent interactions and providing feedback based on a pre-defined quality form.

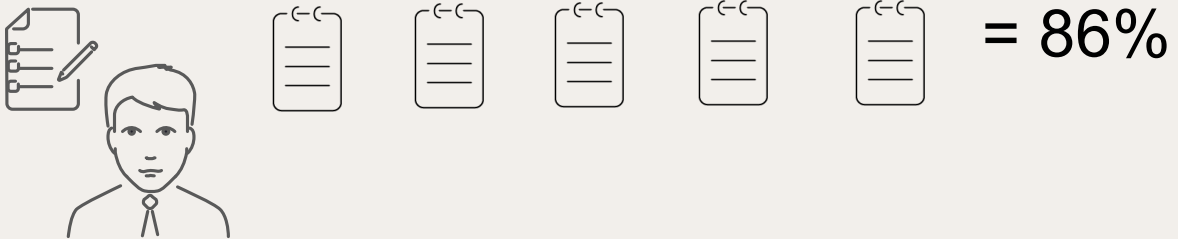
Agent interacts with the customer via phone calls, email, or chat.

Team Lead manages the Agent. They receive feedback from the QA evaluator about their performance.

NOTE: there may be other participants, such as Executive Management or System Admins, but they are not essential to the QM process.

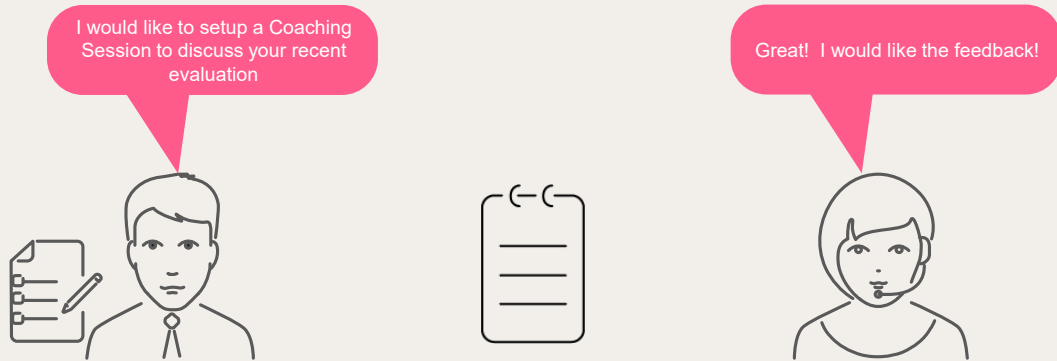


Performing Evaluations



Usually a specific # of evaluations need to be performed each week or month. The Evaluations can be reported on, and feedback shared with agent.

Reviewing and Coaching



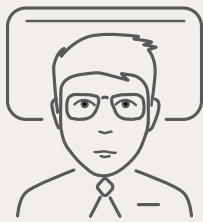
NICE

Evaluations can be shared with the Agents. Coaching Sessions can be set up between the Evaluator and Agent to discuss results and action plans to improve performance. These sessions may also involve the Team Lead.

Quality Central allows supervisors to send personalized coaching feedback – links to knowledge resources, instructions, and due dates – with the click of a button. Then they and the agent can track subsequent performance directly from the team's dashboard.



Reporting



Monthly Report

of Evaluations: 5

Scoring Trends

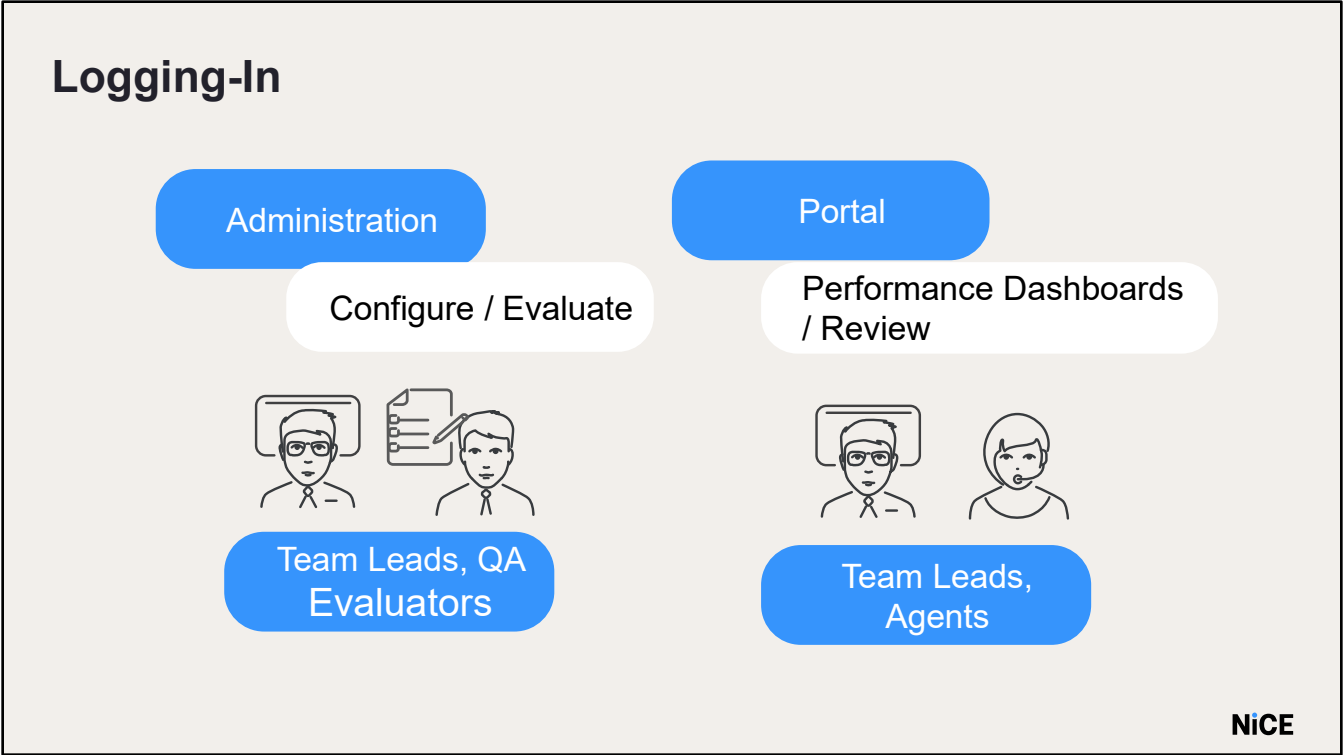


Reports can be run to monitor quality performance and other metrics using Scorecards.



USING QUALITY
CENTRAL





Quality Central has two interfaces the Administration (Main) and the Portal. NOTE: some customers may also have a System Admin role who would access the Administration portal



Logging-In

Main (Administration)

Log In

Login ID

Password

Log In

Version 12.4 PR2 (201805)

POWERED BY NICE
SOFTWARE

Used for system configuration

Portal

Log In

Login ID

phn01

Password

☒ I am an agent / customer service representative

☐ I am an evaluator / supervisor / manager

Log In

Version 12.4 PR2 (201805)

POWERED BY NICE
SOFTWARE

Used by Agents, Supervisors and Team Leads to review evaluations

NiCE

To configure the system, you first need to login to the Administrator interface. The default login is Superuser / password. NOTE: logging in to the Portal interface requires you to select the relevant role.



Using Quality Central

Find Calls

Evaluate Calls

Review Evaluations

View Dashboards

Report on QM Data

NiCE

Once the Metadata, Users / Agents, and Audio (if applicable) has been imported from Engage or another source system, the quality tasks can be performed using QC.



Using Quality Central – Find Calls

Portal

Dashboard for:

Current Capture Time:
Wednesday, September 6, 2017 9:47:55 PM

Prior Capture Time:
Not Available

ASSIGNED TO ME
Files across my queues assigned to me

0
PRIOR

IN MY QUEUES (TOTAL)
Count of files across all my queues

4

WORKFLOW	Queue	Total	Assigned to me	Max Age	Ages (Days)
Engage Quality	Team Lead	4	0	0	1 2 3 4 5 6+

Search (USE ALL FILES)

NOTIFICATIONS - There are no notifications at this time

Workflow Filters

Results for: All Files

INTERACTION TYPE	PRIORITY	DATE/TIME	DURATION	WORKFLOW	QUEUE	STATUS
Audio		9/29/2017 6:46:37 ...	00:00:00	Engage Quality	Team Lead	Open
Audio		9/29/2017 6:46:36 ...	00:00:00	Engage Quality	Team Lead	Open
Audio		9/29/2017 6:46:36 ...	00:00:00	Engage Quality	Team Lead	Open
Audio		9/29/2017 6:46:36 ...	00:00:00	Engage Quality	Team Lead	Open

Items: 1 - 4 of 4

STARTS AT

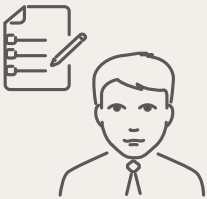
Calls can be found by using the Dashboard and selecting calls that have been assigned to you either directly or via the Queue. Calls can also be searched for using a manual search.

Regardless of the method used to find the calls, they will be displayed with the same format in the interface.



Using Quality Central – Evaluate Calls

Portal



Evaluation Form

Template: Engage Quality CHANGE Grant access to: [Name Contains] Auto-suggest when possible

☒ Allow Agent access [when Publish... Agent: **Britt, Peter** (11945) [4] Send for Self-Evaluation

CATEGORIES	ORDER PROCESS	POINTS	NOTES
Introduction 10 / 10			
Order Process (in progress)			
Closing [not started]			
Overall Score: 30.00 / 30 - (100.00%)			

MEDIA FILE

05212005-385.wav

This form references 1 media file

METADATA

Agent: Peter Britt

Date/Time: 8/29/2017 6:46:36 PM

Evaluator: Steven Mills

QUESTION ANSWER POINTS NOTES

Placed Order Correctly * Yes No n/a 10 / 10 Verified Order #

View as single page Notes for Order Process Notes for entire Evaluation

File: Upload a file... (Maximum size = 2 MB)

< Previous: Introduction Next: Closing >

HISTORY SAVE SAVE AND CLOSE PUBLISH PRINT CANCEL

NiCE

Once a call is selected, the Evaluator opens and completes the evaluation form. They will have options to allow agents to view the Evaluation when published. The evaluations can be performed through both the interfaces, Main and Portal.



Using Quality Central- Review Evaluations

Evaluation Portal

ID Contains

From 11/04/2022 to 12/04/2022

Row Contains

☐ Self-Evaluations only

View

Print

Evaluate

ID	EVALUATOR	TEMPLATE	AGENT	SCORE	AGENT SCORE	PUBLISHED	ACKNOWLEDGED ON	NOTES	COACHING REQUEST
1009	David Voss	EngageQuality	Pete	50.00 / 50					
1008	David Voss	EngageQuality	Pete						
1007	David Voss	EngageQuality	Pete						
1006	David Voss	EngageQuality	Pete						
1005	David Voss	EngageQuality	Pete						
1004	David Voss	EngageQuality	Pete						

CATEGORIES

Introduction

10 / 10

Order Process

10 / 10

Closing

10 / 10

Overall Score

30.00 / 30 (100.00%)

MEDIA FILE

05212005_367.png

This form references 1 media file

METADATA

Agent:

Peter Brim

Date/Time:

8/29/2017 6:46:37 PM

INTRODUCTION

10 / 10

N/A

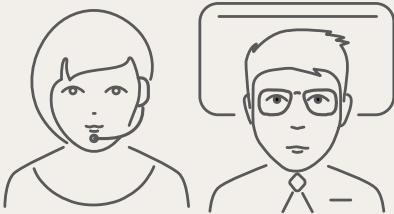
QUESTION	ANSWER	POINTS	NOTES
Used Proper Greeting	Yes No File	10 / 10	Said Company Name

View as single page

Notes for Introduction

Notes for entire Evaluation

Request Coaching...

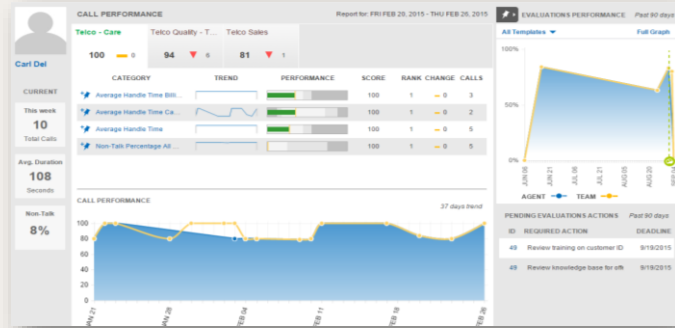


When the Evaluation is saved and published, it becomes available for review by the Agent using the Evaluation Portal.



Using Quality Central- View Dashboards

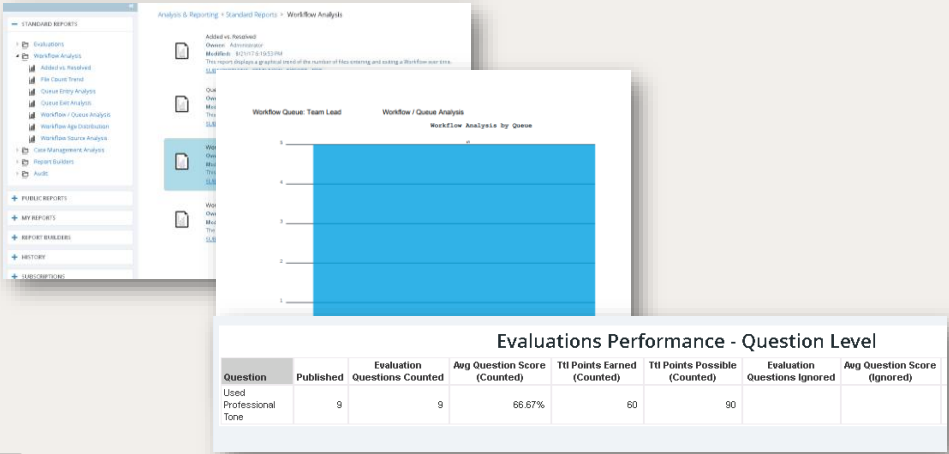
Evaluation Portal



There are scorecards for Evaluators / Supervisors and scorecards for Agents. The result for each of them can be found in the Evaluators Portal Dashboard

Using Quality Central - Report on QM Data

Portal



Data from Quality Central can be analyzed and reports generated about Evaluations, Workflow processes, and QC audit reports. Reports may be shown as graphs or grids.



REVIEW
QUESTIONS



Review Question 1

Who are the 3 main participants in the Quality Management Process?

Evaluator, Agent &
Administrator

Team Lead, Agent &
Evaluator



Review Question 2

True / False –

Agents do not have access to the Dashboards.

True

False

Review Question 3

True / False –

The evaluations can be performed through both the interfaces, Main and Portal

True

False

summary

- Evaluate the impact of Quality Management on NICE customers
- Describe the steps involved in the Quality Management process
- Demonstrate how to use Quality Central effectively





Thank You



QC System Configuration

NiCE Interaction Analytics & Quality Central Fundamentals

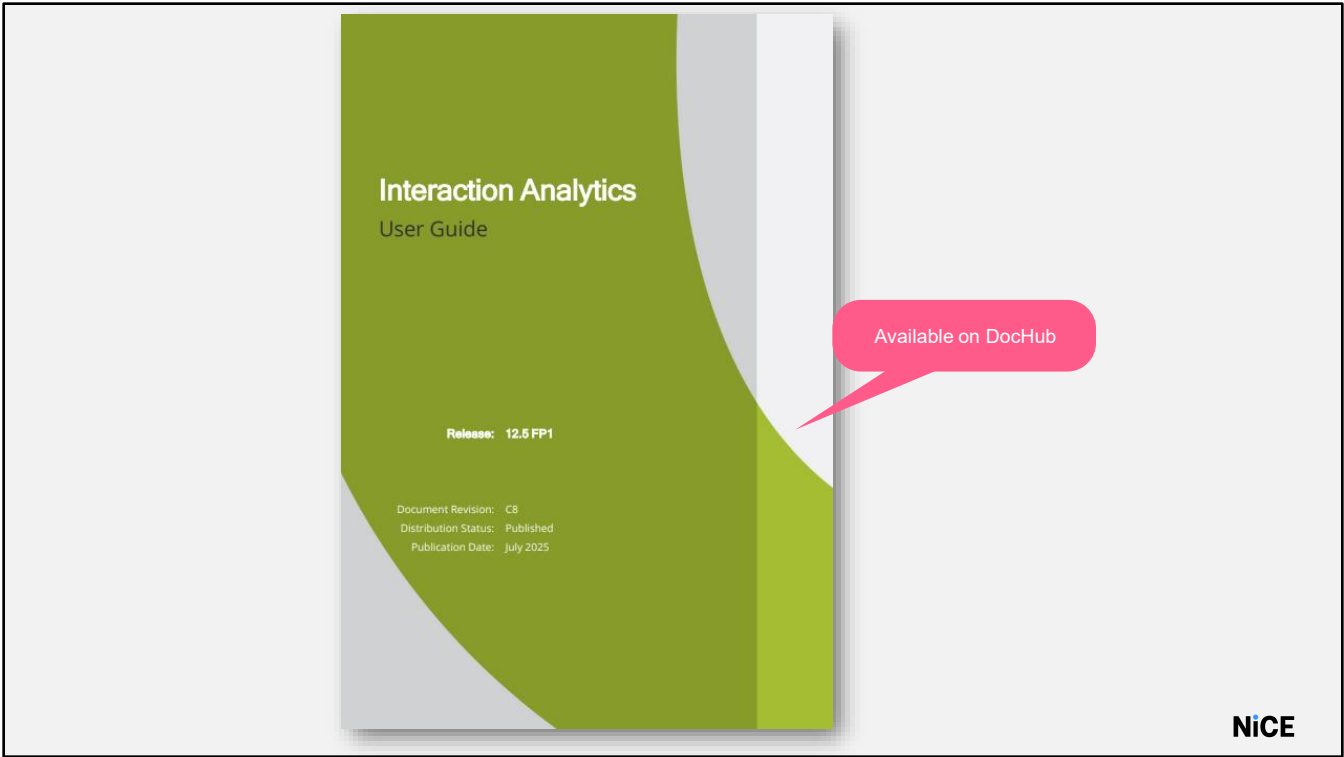
Create a
NiCE
world 



objectives

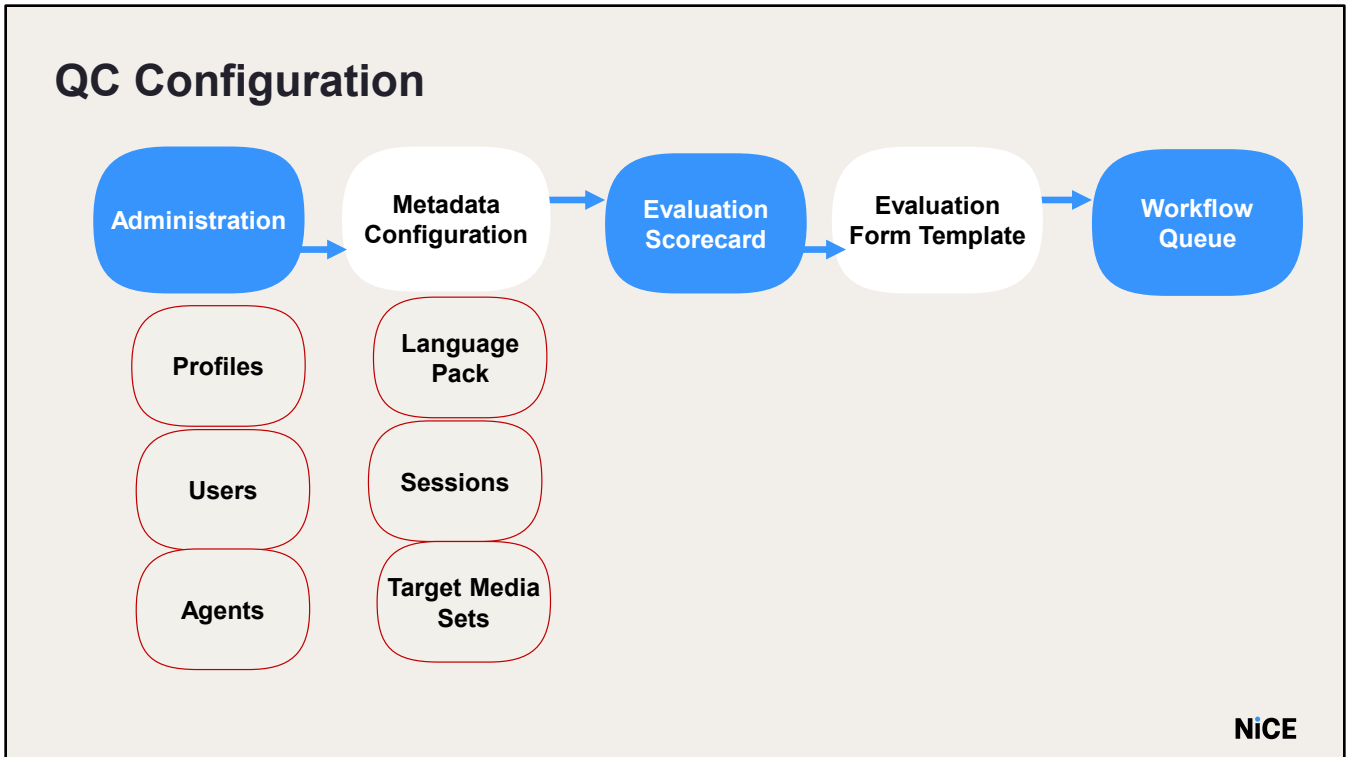
- Log into the Main and Portal interfaces
- Configure the Quality Central Application
 - Profiles
 - Users
 - Agents
 - Sessions
 - Target Media Sets
 - Scorecards
 - Evaluation Form Template
 - Workflows





QC Configuration

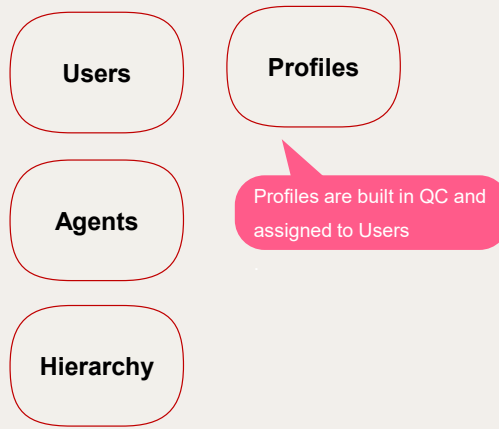




The QC Configuration focuses on who is being evaluated / doing the evaluations; what is being evaluated; and how it is being evaluated. The Workflow is what assembles these pieces together and automates the QM process.

Some configuration steps need to be performed before the workflow can be used

Administration



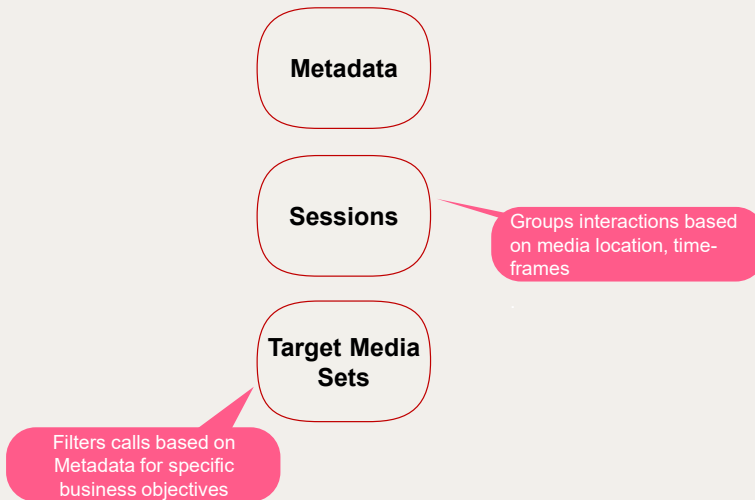
If the source system is Engage, then DEF imports agent, user, and the organizational hierarchy into QC.

Please note that agents, users, and the organizational hierarchy will not be imported from a 3rd party system.

If the source system is a 3rd party system, users and agents need to be manually imported or manually created

NOTE:. In training, we aren't using DEF so Users and Agents will be manually imported

Metadata Configuration

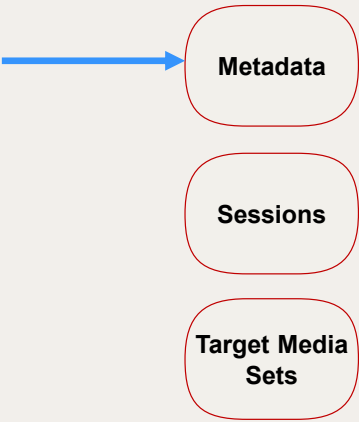


Metadata is imported into Quality Central using DEF. NOTE: data can be imported from Engage or another 3rd-party system.

A session includes parameters that define how the application searches the interactions.

Target Media Sets are used to manage the interactions and puts them into Workflow Queues.

Evaluation Scorecards



Scorecards define the acceptable range for metrics

Average Call Duration		Score
0 - 180	Great	100
181-240	Very Good	80
241-360	Good	60
360+	Poor	0

For example, a scorecard can define a metric for Average Call Duration of less than 360 seconds would be defined as good. Anything above would be considered “out of bounds”. “Good” scores can be further classified using categories.



Evaluation Scorecards

Scorecards

SCORECARDS

Default Scorecard

Properties: Default Scorecard

PROPERTIES

Creation Date: 9/21/2017 5:49:14 AM
Created by: Administrator, Default - (superuser)
Description: Default Scorecard provided by the administrator

Modified Date: 9/25/2017 1:43:04 AM
Last Modified by: Administrator, Default - (superuser)

USERS/PROFILES

Show Inactive Users

USERS/PROFILES	ACCESS
SuperUser (Profile)	Full Control
Team Lead (Profile)	Full Control

Contents

DISPLAY NAME	METRIC	TARGET MEDIA SET	GOALS	INCLUDED	WEIGHT	DISPLAY	LIMIT
Average Duration	Average Duration (Sec)	Engage Media Target	Good Out of Bounds 0 - 360 360 -	Yes	5 (33.3%)	Numeric	1080 or higher
Average Interactions	Average Interactions per Day	Engage Media Target	Good Out of Bounds 0 - 19				

 Yes | 5 (33.3%) | Numeric | 57 or higher || Average Non-Talk | Average Non-Talk | Engage Media Target | Good Out of Bounds 0 - 5 |

5 -

 Yes | 5 (33.3%) | Numeric | 15 or higher |

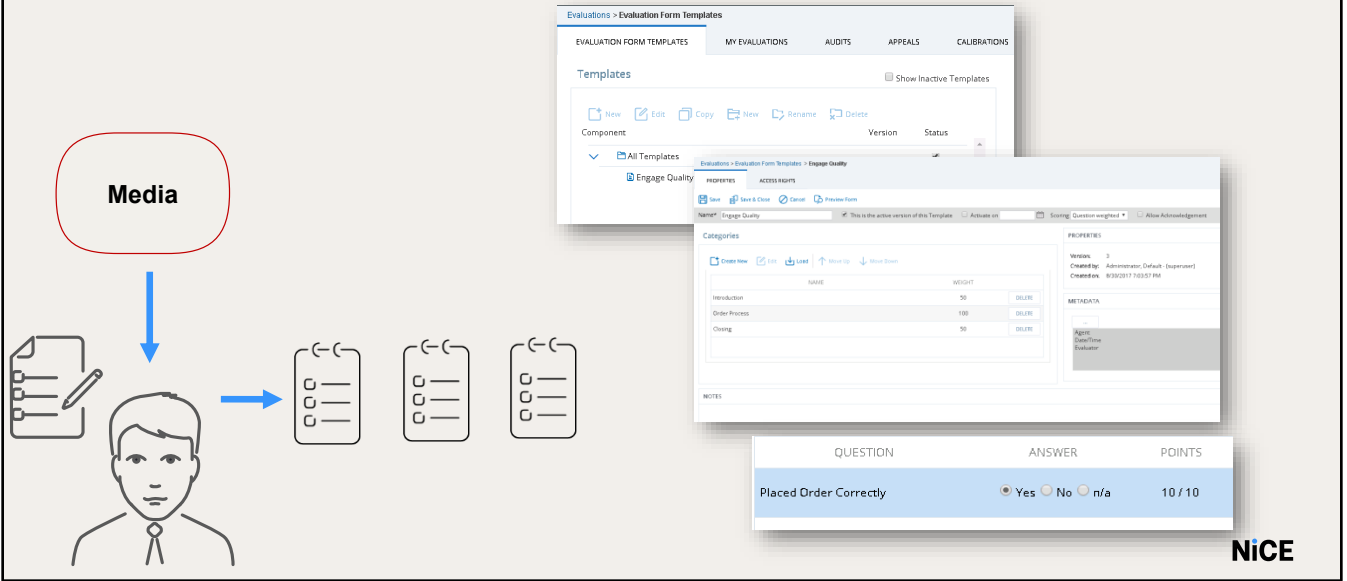
CATEGORY	TREND	PERFORMANCE	SCORE	RANK
Average Handle Time Bill...			100	1
Average Handle Time Ca...			100	1
Average Handle Time			100	1
Non-Talk Percentage All ...			100	1

- Average Agent Chat Turn Response Time (sec)
- Average Chat Turns per Day
- Average Duration (Sec)
- Average Duration Between Chat Turns (sec)
- Average Interactions per Day
- Average Non-Talk (%)
- Average Non-Talk (Occurrences per Call)
- Average Non-Talk (Sec)
- Average Offset from End of Call
- Average Offset from Start of Call
- Average Sentiment
- Exists
- Max Interactions per Day
- Min Interactions per Day
- Percentage of Interactions

Scorecards are used to monitor an agent’s performance against site and team objectives. You can select categories which are contained in the metadata available in QC. This data can then be displayed in the user and agent dashboards.

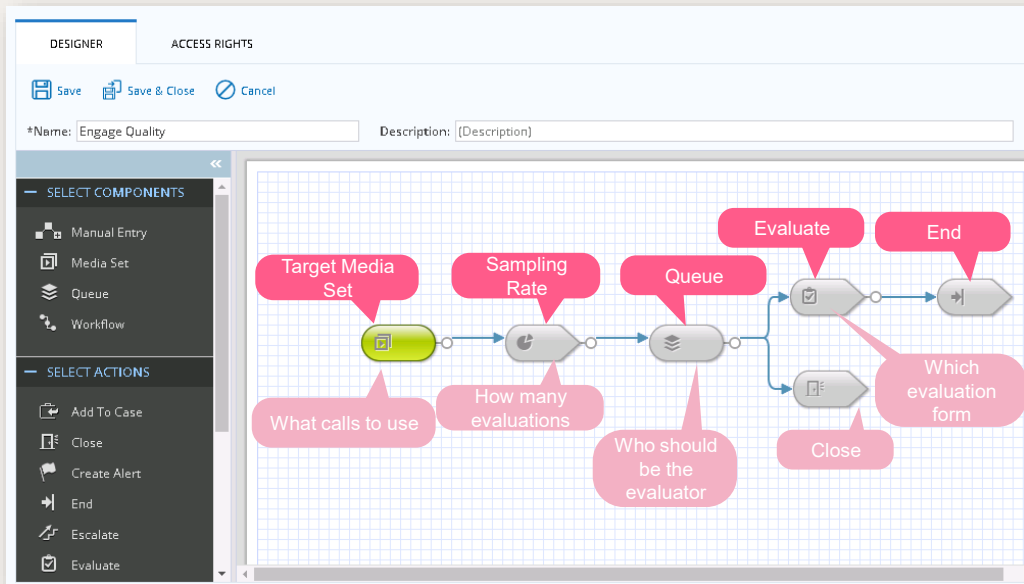


Evaluation Form Template



The Evaluation Form Template is used to create the Evaluations done by the Supervisor. The Supervisor reviews the interaction media and creates Evaluations based on the Template Form to provide feedback to the Agent. The Evaluation Form may have multiple categories each with their own questions / values.

Workflow Queue



NiCE

One of the major selling points for Quality Central is being able to automate the QM process. QC does this by using Workflows. Calls are automatically assigned to Evaluators using specific criteria.



Logging-In

Administration

Configure / Evaluate



Team Leads, QA
Evaluators

Portal

Performance Dashboards
/ Review



Team Leads,
Agents

Quality Central has two interfaces the Administration (Main) and the Portal. NOTE: some customers may also have a System Admin role who would access the Administration portal

To configure Quality Central, use the Administration interface. Login: Superuser / Password: password



Profiles



Profiles

Privileges

	PRIVILEGE	HAS PRIVILEGE
ANALYSIS & REPORTING	Analysis Consumer	✓
	Analysis Analyst	✓
	Report Administrator	✓
	Evaluate - Reports	✓
	Workflow - Reports	✓
	Search Unbounded	✓
SEARCH & REPLAY	Save Media Files	✓
	Assign Media Files	✓
	Annotate Files	✓
	View Annotations	✓
	Modify / Delete Annotations	✓
	View File History	✓
	Workflow Viewer	✓
	Workflow - Manually Add Files	✓
	Evaluate - Access All Calibrations	✓
	Evaluate - Access All Evaluations	✓

Profile

Team Lead

Evaluator



Team Lead



Evaluator

Access to Quality Central is granted via privileges. Privileges are grouped into profiles. Profiles are then assigned to individual Users.

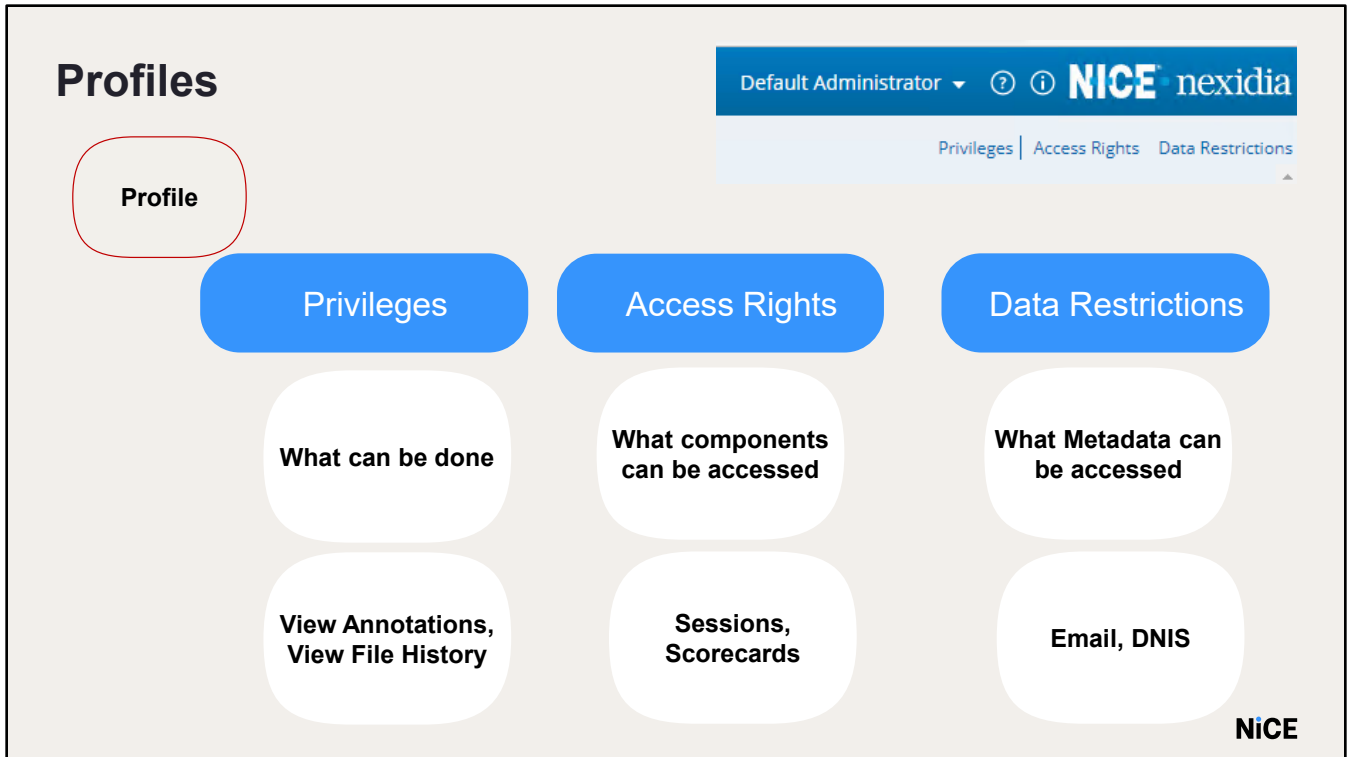
NO

There are a group of pre-defined profiles in the system. These can be used, or new ones created. NOTE: Superuser and Supervisor can't be edited.

To create new profiles: Click New

TE: Agents do not need to be assigned Profiles.





The Profile has different layers of access using Privileges, Access Rights, and Data Restrictions.

Access Rights -determine which sessions can a user see (sessions point to media and associated queries).

Data restrictions are based on metadata, and the user can only see interactions with metadata matching the restriction

I.e., site =Denver

Used to define what media users have access to based on metadata. Only media where the metadata is populated with the correct value will be viewable

Profiles

Save Save & Close Cancel

DETAILS

Name: Evaluator

Description:

☒ Create reporting user group ☐ This is a restricted access Profile

☒ Create workflow queue

Creator:

Creation Date:

Last Modified by:

Last Modified Date:

Select to make profile available to be used in the workflows

USERS ASSIGNED TO THIS PROFILE ☐ Show Inactive Users

+ Add Remove

PRIVILEGES

Tags - Assign Tags to Files	☒
Tags - General Administration	☒
Tags - View Tags in Applications	☒
Target Media Sets Administration	☒
User Administration (Basic)	☒
User Administration (Data Access & Restr)	☒
User Administration (Extended Basic)	☒
User Administration (Full)	☒
Workflow Activation	☒
Workflow Administration	☒
Case - Access Cases	☒
Case - Create Cases	☒
Case - Delete Cases	☒
Case - General Administration	☒

If you want to be able to assign interactions via the Workflow Queues, the “Create Workflow Queue” must be selected

NOTE: Users who evaluate calls will need access to Sessions, Scorecards, and Evaluation Form Templates. These components will be covered later in the lesson.

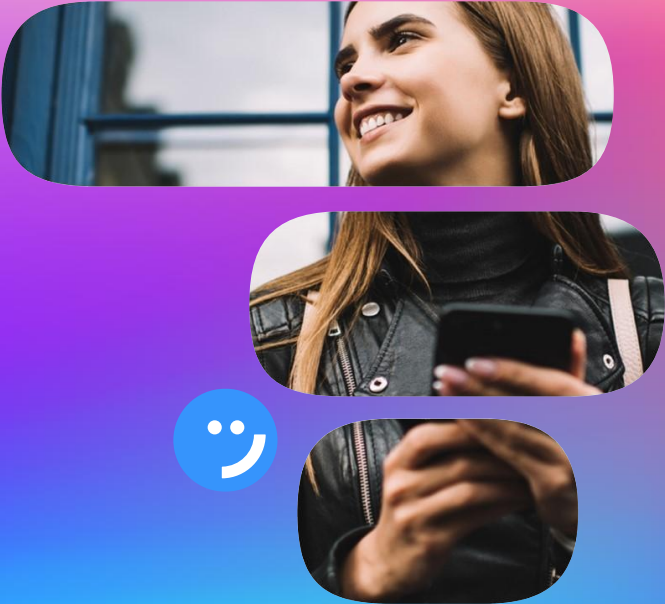
You will need to build 2 profiles (Evaluator, Team Lead). Grant all privileges to each account.

Do it Yourself

- Refer to exercise 5 - Create Profiles
 - This section creates profiles for the QC Users. You will need to create 2 profiles:
 - **Team Lead**
 - **Evaluator**
- 1) Login to the Main Portal (<http://nia/NxIA/main>). NOTE: use the Superuser account to login (password: password).
 - 2) For the purposes of testing, it is recommended to grant all the privileges to each account.
 - 3) The Evaluator Profile should be added to the Workflow Queue. The Team Lead should NOT be added to the Workflow Queue.
 - 4) NOTE: access to the Scorecards, Sessions, and Templates will be created in later sections.

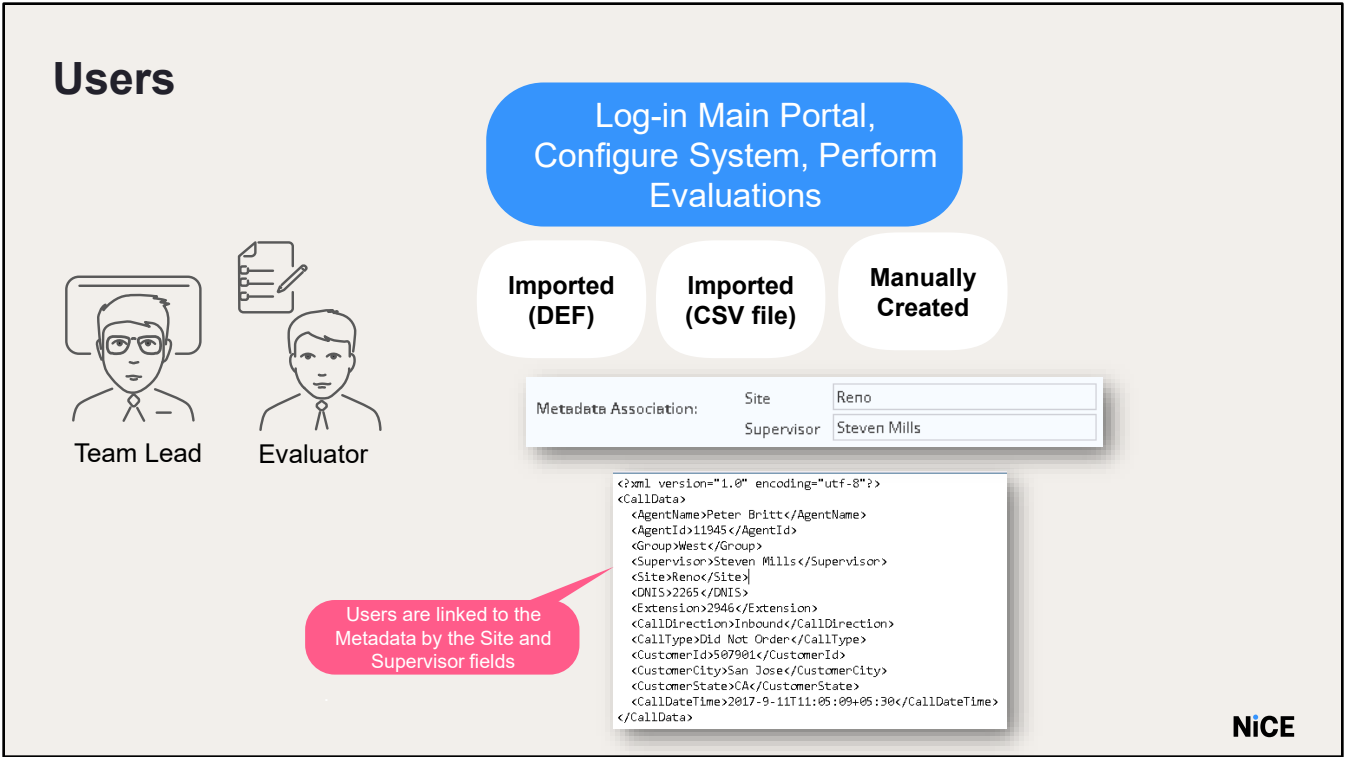


Users





NiCE



Each person who accesses Quality Central needs to have an account created. These can be created manually, imported, or imported via DEF (from Engage). The result of each method will be the same.

NOTE: some employees may be Users and Agents. They will access the Main and Evaluation Portals.

In our training environment, we will import users using a CSV file. The Site and Supervisor information must match the information in the Interaction Metadata. If it does not match, then the interaction may not be properly assigned to the proper team in the Workflows.

Users imported via DEF will automatically appear in the list.

Enter User Details

In the lab, we will use Internal Authentication (Forms Authentication).

Site / Supervisor is the link to the Interaction Metadata



Importing Users

User List

New...

Import...

Load Restrictions

Export...

Activate

Deactivate

Unlock

Find User: (Name Contains)

	LOGIN ID	FIRST NAME	LAST NAME	PROFILES	LAST LOGIN	ACTIVE	AUTH
☐	superuser	Default	Administrator	SuperUser	10/26/2022 7:30:30 PM	Yes	

David Voss

Evaluator

Evaluate Interactions;
assign to Workflows

Password: nicecti

Steven Mills

Team Lead

Review Evaluations;
Run Reports

Password: nicecti

Import/ Update Users

Users [2].ms.csv

Browse...

File Format:

If the LoginID being imported already exists in the system:

Overwrite the existing data with the imported data for that user

Keep the existing data - do not import data for that user

Summary - This import will result in:

New users created: 1

Existing users updated: 1

Records not imported: 0

Details...

Note: Login IDs must be unique.

Upload

Cancel

For importing users, click Import and browse the appropriate .csv file.

In our Lab, we will use Users.csv to import users .

Click Upload.

NOTE: if you click to edit the accounts, you may receive an error about the Site / Supervisor. Click No on the prompt.



- Refer to exercise 6- Create Users
 - This section creates the users for QC. These Users will be associated to the Profiles created in the previous section. These users will be created using the Import process using a CSV file.

NOTE: In a production environment with DEF / Engage, the Users will be automatically populated.

- Steven Mills
- David Voss
- 1) Import the Users.csv file located in the D:\nicetech\NIA 12.5\Users and Agents
- 2) The password for these accounts is 'nicecti'



Agents



Agents



Log-in Main Portal,
Configure System, Perform
Evaluations

Imported
(DEF)

Imported
(CSV file)

Manually
Created

Agents are linked to the
Metadata by the AgentID
field

*Agent ID

11945

```
<?xml version="1.0" encoding="utf-8"?>
<CallData>
  <AgentName>Peter Britt</AgentName>
  <AgentId>11945</AgentId>
  <Group>West</Group>
  <Supervisor>Steven Mills</Supervisor>
  <Site>Reno</Site>
  <DNIS>2265</DNIS>
  <Extension>2946</Extension>
  <CallDirection>Inbound</CallDirection>
  <CallType>Did Not Order</CallType>
  <CustomerId>507901</CustomerId>
  <CustomerCity>San Jose</CustomerCity>
  <CustomerState>CA</CustomerState>
  <CallDateTime>2017-9-11T11:05:05:30</CallDateTime>
</CallData>
```

Agents are the employees whose interactions are recorded. Similar to Users, there are 3 methods to create Agents. Each method has the same end result.



Agents

New/Edit Agent in Inventory

Photo

Select Photo (Max size: 2 MB)

*Evaluations Portal Login ID

pbritt

*First Name

Peter

*Last Name

Britt

*Agent ID

11945

*Account type

Internal Authentication

*Password

*Re-enter Password

*Active

Yes

☒ User must change password on next login

E-mail

Verify E-mail

External User ID

Unique Agent ID

Site

Reno

Supervisor

Steven Mills

Agent Groups

Generate

Ok

Cancel

Agent Details

Agent ID must match the Metadata Agent ID

Assign Password

Complete the relevant info about the Site / Supervisor

Agent Groups.

Agent can also be added to a group

Administration > Agent Groups

STATUS

Session Status

Server Status

Job Status

SESSION MANAGEMENT

QUERIES

QUERY GROUPS

TERM LISTS

TARGET MEDIA SETS

EVALUATION SCORECARDS

Metrics

Scorecards

INTERACTION SERIES

WORKFLOWS

AGENTS

Agent Inventory

Agent Groups

PORTAL ADMINISTRATION

Player Metadata

Agents - Portal Privileges

KNOWLEDGE BASE

We have to manually create agents and create inventory

NOTE: there are other ways to assign passwords. For example, when an evaluation is sent for review, they agent can receive an email. This will prompt them to change the password at that time. The Generate option can also be used to create generic passwords (for agents who don't login or are prompted to change their passwords). Click to Load the CSV file. Navigate to the CSV file using the Browse option. Click Upload. After uploading the Agent, edit and add a password. NOTE: you can not uncheck "User must change password on next login". When you login with the agent you will need to change the password again.

Click Agent Groups. Click New.

In the Update Agent Group tab, give a Name, select Agents and click Add.

NOTE: groups don't support sub-groups



- Refer to exercise 7- Create Agent
 - This section creates the Agents for QC. These Agents are associated to the Audio that will be used in the lab. These Agents will be created using the Import process using a CSV file.

NOTE: In a production environment with DEF / Engage, the Agents will be automatically populated.

- Carl Swift
- 1) Import the AgentInventory.csv file located in the D:\nicetech\NIA 12.5\Users and Agents
 - 2) The password for this account is not automatically created via the import. It needs to be added manually afterwards.
 - 3) Uncheck the “user must change password on next login” option
 - 4) Create Agent Group



Session & Target
Media Sets



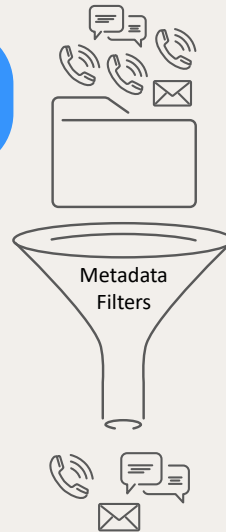
Do it Yourself

- Refer to exercise 8 - Create Session
- This section associates a session with the audio to be used for Quality Central.
 - 1) Use the Billing Session and point it to the Billing Media folder, which contains the Audio.
 - 2) The Session must be active to be used for Quality Central.



What is a Target Media Set?

A filtered set of media used to measure agent performance



The initial media is defined within a session. It's also possible to filter the media using metadata filters.

The remaining (filtered) media is the Target Media Set and is used to measure agent performance (quality).

By using the TMS, we select which set of calls will be analyzed.

NOTE: It's not possible to create a new TMS without selecting which session it will be based on.

Target Media Sets

Target Media Set Details

Save

Cancel

*Name:

Engage Media TMS

*Session:

EngageSession

Purpose:

Audio Query Purpose

Provide a logical name

Select Session

SEARCH FILTERS

Include files that match

any of these

NAME	TYPE
------	------

Select...

Exclude files that match

any of these

NAME	TYPE
------	------

Select...

METADATA FILTERS

+ Add/Remove Filter(s)

(example: Interaction Duration (seconds) is greater than 30)

Filter the media by specific metrics

REPORT METRICS

☒ Average Duration (Sec)

☒ Average Interactions per Day

☐ Average Non-Talk (%)

☐ Average Non-Talk (Occurrences per Call)

☐ Average Non-Talk (Sec)

☐ Average Offset from End of Call

☐ Average Offset from Start of Call

☐ Average Sentiment

☐ Error

Make metrics from the media available in Reports

Enter the name of the Target Media Set and select the Session from the drop-down.

NOTE: you must have access to at least one session with Read / Write permissions.



Do it Yourself

- Refer to exercise 9 - Create Target Media Sets
 - This section defines what media from the Session should be used for Quality Central.
 - 1) Create a New Target Media Set and point it to Session created in the previous example.
 - 2) In the 'Filters and Report Metrics' select the Average Duration & Average Interactions per Day metrics

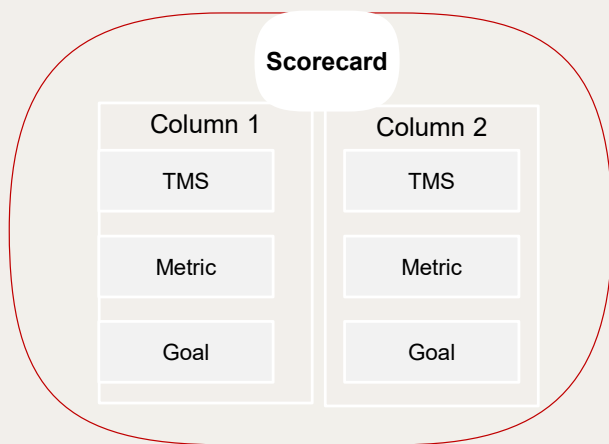


Scorecards



What is a Scorecard?

A tool to monitor agent performance relative to user-specified goals defined by column criteria



Column may also be referred to as Category or Quality Initiative.

- Scorecards enable to monitor agent performance relative to user-specified goals, based on one or more columns.
- The columns determine:
 - The media that is searched (the target media set)
 - The criteria that is used when searching for results (metric)
 - And the standard that is used to interpret the results (goals)
- Each column can run on a different set of interactions and provide different metric results.

NOTE:

- At least one target media set must be defined in the system in order to create a scorecard.
- Column may also be referred to as category or Quality Initiative. We will now go over each component of a column.



Types of Scorecards

Evaluation
Scorecards

Created in Main

Based on
Metadata Metrics

QC Scorecards

Created in Portal

Based on
Evaluations



QC Scorecards will be covered in the
Using QC lesson



What is a Metric?

A measurable value defining the criteria used when searching a target media set

Metric Name	Description
Average Agent Chat Turn Response Time (Sec)	Identifies the agent response between chat turns in the Target Media Set for this individual and returns the average.
Average Chat Turns per Day	Indicates the number of chat turns in the Target Media Set for this individual divided by the number of days in the Date Range for which this individual has at least one chat.
Average Duration (Sec)	Identifies the Duration of each call in the Target Media Set for this individual and returns the average.
Average Duration Between Chat Turns (Sec)	Identifies the duration between chat turns in the Target Media Set for this individual and returns the average.
Average Interactions per Day	Indicates the number of interactions in the Target Media Set for this individual divided by the number of days in the Date Range for which this individual has at least one call.
Average Non-Talk (%)	Identifies the percentage of non-talk in each call in the Target Media Set for this individual and returns the average. Calculation is the sum of the non-talk percentages divided by the number of calls.
Average Non-Talk (Occurrences per Call)	Identifies the number of non-talk occurrences in each call in the Target Media Set for this individual and returns the average.
Average Non-Talk (Sec)	Identifies the number of seconds of non-talk in each call in the Target Media Set for this individual and returns the average.
Average Offset from End of Call	Identifies the average offset (in seconds) of the last hit for the Target Media Set within the media file from the end of the file.
Average Offset from Start of Call	Identifies the average offset (in seconds) of the first hit for the Target Media Set within the media file from the beginning of the file.

Metric Name	Description
Average Evaluation Score	Identify average evaluation score for individual user and team associated to the user in a defin...

Metrics decide what will be measured – for the calls in the session that match the TMS criteria.

For example: Average Interactions per day or Average Duration (sec).

Most of the available metrics and their description are displayed under Metrics menu (under Evaluation Scorecards).

Additional metrics regarding evaluation score are displayed in the Evaluations Portal, under QC Scorecards.

NOTE: the default Scorecard is created during the installation, but it does not contain any columns for the scorecard. The default scorecard can be updated or a new one created.

Select the type of metric to include in the scorecard using the drop-down. Select the Target Media Set to query using the metric. You have additional options for creating the goal.

Evaluation Scorecards

PROPERTIESACCESS RIGHTSTHRESHOLDSAGENTS' DASHBOARD

SaveSave & CloseCancel

*NameEvaluation Scorecard

☒ Use in Portal Dashboards

* Indicates a required field

Click to show the Scorecard in the Dashboard

COLUMNS

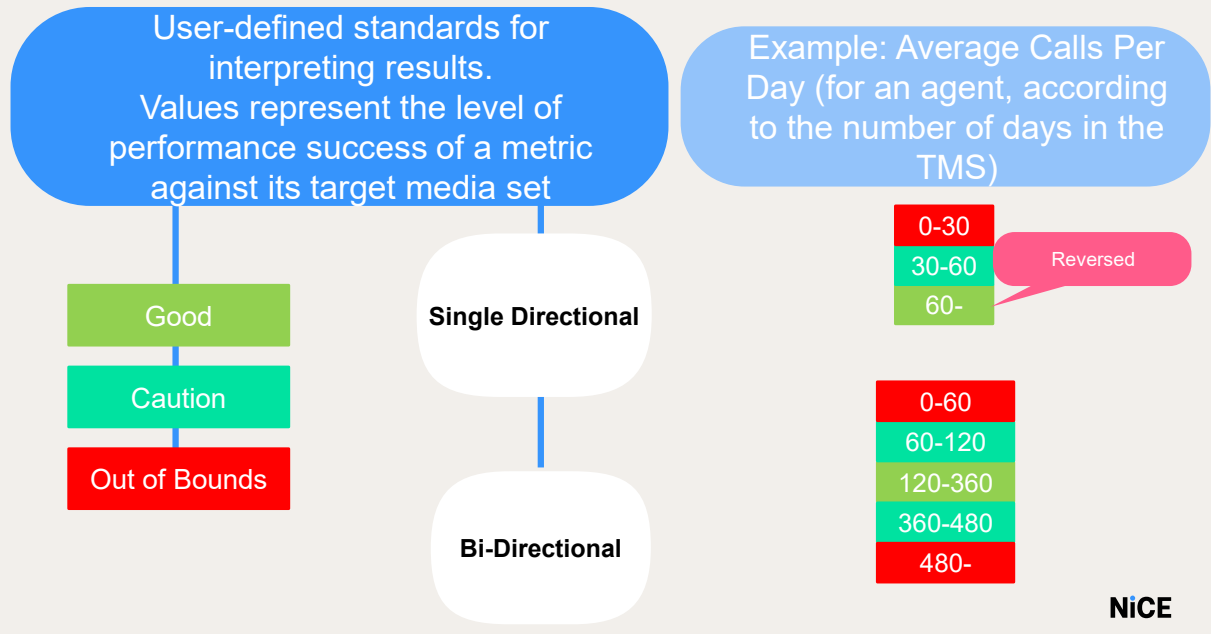
NewLoadRemoveMove UpMove Down

	DISPLAY NAME	METRIC	TARGET MEDIA SET	GOALS	INCLUDED	WEIGHT	DISPLAY	LIMIT (0 POINTS)
--	--------------	--------	------------------	-------	----------	--------	---------	------------------

Provide a logical name for the scorecard. Select Use in Portal Dashboards if you want the scorecard to be seen by agents. Click New to add a new column to the Scorecard.



What are Goals?



Goals are user-defined standards for interpreting results.

By using goals, we decide what are the desired scores for those calls. These standards are labeled in 3 levels (good, caution, out of bounds).

The values represent the level of performance success of a certain metric against its target media set (acceptable or unacceptable performance).

Each goal should be defined by type: single directional or bi-directional.

- **Single Directional:** refers to the ability of the goal to be tracked on a single scale.

Any value below the provided limit is good while the lower the better in a normal legend, or the higher the better in a reversed legend.

- **Bi-Directional:** refers to the ability of the goal to be monitored as a range. This type includes 5 levels of goal values, where the middle is the good level.

For example: the 'Average Calls per Day' metric measures the number of interactions in the TMS for an agent divided by the number of days in the Date Range for which the agent has at least one interaction. See the above example for possible values in each level and for each goal type.

Defining a Goal

*Goals: (interactions)

Type: Single Directional ▼

0 2 4

☐ Reverse

☒ Include "Caution" state

Goal Thresholds/Legend

☐ Good	0 - 2 interactions
☐ Caution	2 - 4 interactions
☐ Out of Bounds	4 - interactions

(lower is better)

Note: Scores equal to Goal boundaries are always assigned the category that benefits the individual.

*Goals: (interactions)

Type: Bi-Directional ▼

0 2 4 6 8

☒ Include "Caution" state

Goal Thresholds/Legend

☐ Out of Bounds	0 - 2 interactions
☐ Caution	2 - 4 interactions
☐ Good	4 - 6 interactions
☐ Caution	6 - 8 interactions
☐ Out of Bounds	8 - interactions

Note: Scores equal to Goal boundaries are always assigned the category that benefits the individual.

A goal is defined while creating or updating a column in a scorecard.

We will go over all steps of scorecard creation on the next slides.

NOTE:

- Make sure the goal type is correct.
- The Caution level may be removed by unchecking the Include box.
- To switch between the order of the legend, select the Reversed box.

Evaluation Scorecards

COLUMNS

NewLoadRemoveMove UpMove Down

☐	DISPLAY NAME	METRIC	TARGET MEDIA SET	GOALS	INCLUDED	WEIGHT	DISPLAY	LIMIT (0 POINTS)
☐	Average Duration	Average Duration (Sec)	Engage Media TMS	Out of Bounds 0 - 2 Caution 2 - 4 Good 4 - 6 Caution 6 - 8 Out of Bounds 8 -	☐	(0.0%)	n/a	☐ or lower☐ or higher

The weight for the column relative to others in the scorecard

The Max/Min score(s) for each column for earning points

If used for the scorecard, score calculation is displayed in the Portal Dashboards

How the individual column value is presented

Make sure to select Included if you want the data included on the Dashboard. NOTE: individual Metrics can be included / excluded.

NOTE: There is also a setting for Limit – which sets the Min / Max score for the column.



Scoring Thresholds in the Portal

1. Categories and Colors

Number of categories: 5

0

2

4

6

8

100

Display Type

☐ Numeric (normalized) score

☐ Stars ★★ ★★

☒ Harvey Balls ○●○●○●

2. Labels

From - To

Category Label

Min - 2

Lowest

○

2 - 4

Low

○

4 - 6

Moderate

○

6 - 8

High

○

8 - Max

Highest

○

* Scores matching boundary values are categorized to the benefit of the individual

The scorecard display was set to Numeric score

The scorecard display

Agent performance compared to the goal for each category. Each threshold is displayed by different category label color

liance

Sales

100

18

CATEGORY

TREND

PERFORMANCE

SCORE

Avg Calls Per Day

Min Interactions per Day

Max Interactions per Day

75

100

40

NiCE

Grant the appropriate Access Rights to individuals or profiles. NOTE: access can also be granted at the profile or individual level.

NOTE: Thresholds and Agents Dashboard tabs will only appear if “Use in Portal Dashboards” is selected. Use the Thresholds tab to indicate how to configure the scoring thresholds/categories for displaying the overall scorecard score.

Perform the following:

1. Set Categories Configuration:
- Select the number of categories. The default is set to 4 categories.
 - Input a numeric threshold value for each category. The total of the threshold must be in the range of 0 to 100.
 - Select a color for each range by clicking the color picker button.
 - Select how to display the scores – as values or icons.

2. Set Labels:

Input Category Label for each category. The label is displayed as tooltip when the user hovers the mouse over a Scorecard/Tab in the Dashboard.

When done, click Save & Close.

How is data displayed on the QC Portal? The performance and trends are shown per scorecard as compared to the last 7 days. The above example shows the difference in the display.



Evaluation Scorecards

PROPERTIESACCESS RIGHTS

THRESHOLDSEVALUATION DASHBOARD

SaveSave & CloseCancel

* Indicates a required field

Select the Agents (or Groups) to have access to this Scorecard

Import configuration from (select a Scorecard)

Individual Agents

All SitesAll Supervisors(Name Contains)

Berni, David - (Berni)

Berni, Mary - (Berni)

Mohan, Kathy - (Mohan)

Ram, Ram - (Ram)

Swati, Carl - (Swati)

Add >

Add All >

< Remove

< Remove All

Agent Groups

☐ Support

Selected Agents

(Name Contains)

Selected Agents (via Group)

Scorecards

Properties: Evaluation Scorecard

SCORECARDS

Default Scorecard

Evaluation Scorecard

PROPERTIES

Creation Date: 9/29/2017 8:55:17 PMModified Date: 9/29/2017 10:32:53 PM

Created by: Administrator, Default - (Supervisor)Last Modified by: Administrator, Default - (Supervisor)

Description:

USERS/PROFILES

Show Inactive Users

USERS/PROFILES

ACCESS

Evaluator (Profile)Full Control

SuperUser (Profile)Full Control

Contents

DISPLAY NAME	METRIC	TARGET MEDIA SET	GOALS	INCLUDED	WEIGHT	DISPLAY	LIMIT
Average Duration	Average Duration (Sec)	Engage Media Target	Goal: 5,288 (lower is better) Out of Bounds: 5,300	Yes	5 (33.3%)	Numeric	1000 or higher
Avg Interactions Per Day	Average Interactions per Day	Engage Media Target	Goal: 10 (higher is better) Out of Bounds: 5, 10	Yes	5 (33.3%)	Numeric	0 or lower
Avg Non-Talk	Average Non-Talk (%)	Engage Media Target	Goal: 0 (lower is better) Out of Bounds: 20, 0	Yes	5 (33.3%)	Numeric	100 or higher

Select the Agents who will see the Scorecard displayed in their Dashboard and click Add. This shows the completed Evaluation Scorecard.

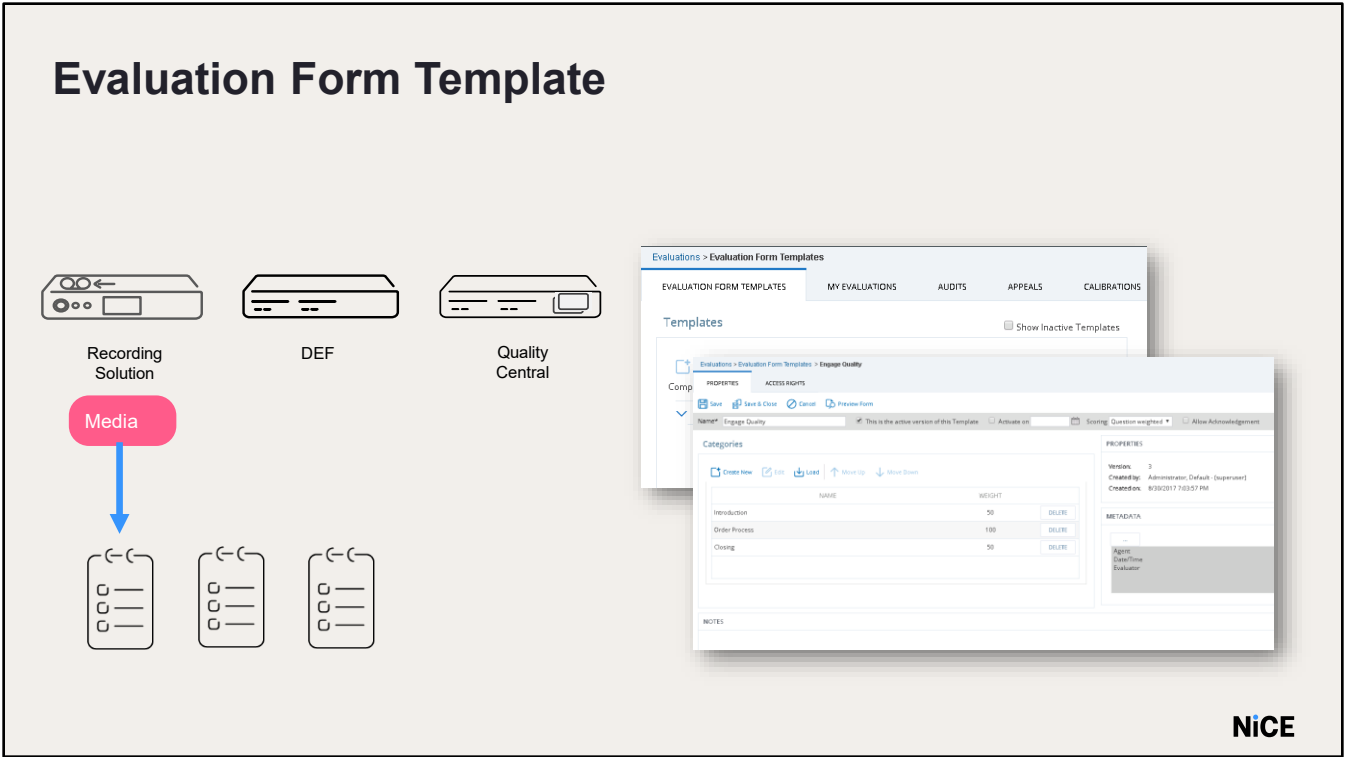


- Refer to exercise 10 - Create Evaluation Scorecard
 - This section creates an Evaluation Scorecard which can be used to measure specific performance metrics.
 - 1) Create a Scorecard and provide a logical name.
 - 2) Select to "Use in Portal Dashboard".
 - 3) Include 2 column:
 - Average Duration and set the goals to 80.
 - Average Interactions per Day set the goals to 3 and check the 'reverse' option (0-3 out of bounds)
 - 4) Grant full control access to the Team Lead profile.
 - 5) Select an appropriate threshold for the Scorecard.
 - 6) Grant access to the Scorecard to the agent.

Evaluation Form Templates



Evaluation Form Template

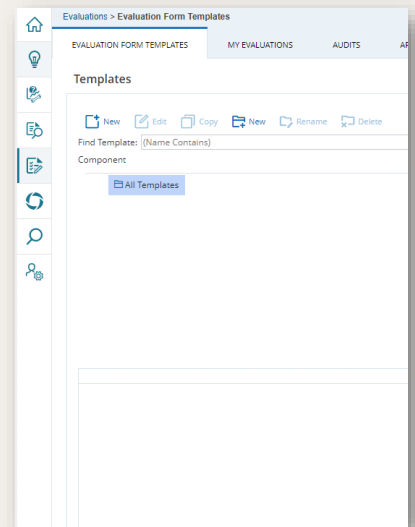


The Evaluation Form Template is used to create the Evaluations done by the Evaluator. The Evaluator reviews the interaction media and creates Evaluations based on the Template Form to provide feedback to the Agent. The Evaluation Form may have multiple categories each with their own questions / values.



Evaluation Form Template

- Navigate to the Quality Central module and go to Evaluation Form Templates
- Highlight "All Templates" and click New to create a new form
- Click the Settings option to configure specific form settings
- Enable the "Allow Agent Acknowledgement" setting
- Create a new category and assign a weight to it
- Click "Introduction" and then select "Create New"
- When finished, click Save
- Confirm by clicking OK on the "Activate Template" alert box
- Click Preview Form to view how the form will appear in production



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NOTE: there are different options of activating the form on a specific date and whether to use Category or Question weighted.

NOTE: there are 2 ways to score a form – category or question-weighted. Once the Category is created, you will need to add questions. There are different options for the types of question you may use. Question types can be mixed and matched. Assign points to the relevant answers. Use the default option if you want an answer to be pre-populated. You can choose to hide this question from an Agent during a review; it will be visible to all other users.

NOTE: data will not be saved when the form is previewed.



Evaluation Form Template

NICE QUALITY CENTRAL

Default Administrator

NICE

Evaluations > Evaluation Form Templates

EVALUATION FORM TEMPLATESMY EVALUATIONSAUDITSAPPEALSCALIBRATIONSSELF-EVALUATIONSNOTIFICATIONS

Templates

NewEditCopyNewRenameDelete

Find Template: (Name Contains)

ComponentVersionStatus

All TemplatesEngageQuality1

Show Inactive Templates

Content

IntroductionPoints = 10

QUESTION TEXT	TYPE	POINTS
Verified Billing Information	Yes, No, N/A	10

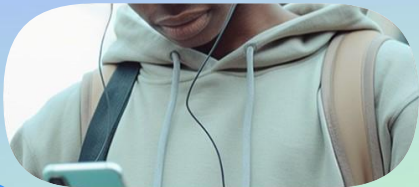
NiCE

This is the EngageQuality Template.

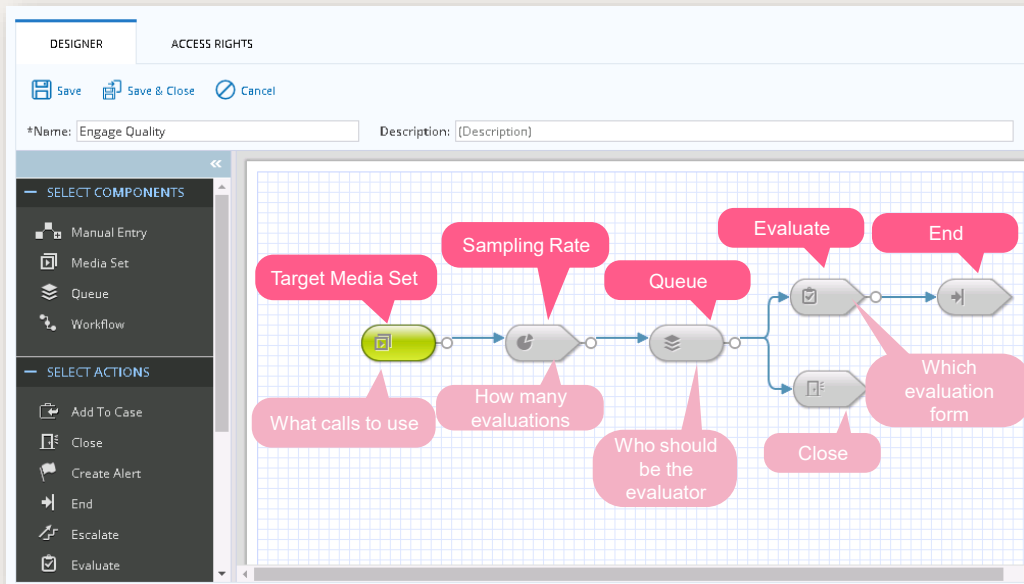
Do it Yourself

- Refer to exercise 11- Create Evaluation Form Template
 - This section creates an evaluation form template which will be used by the Evaluator to conduct review of agent performance.
 - 1) Create a new Evaluation Template and provide a logical name.
 - 2) Create a new category and add a new question.
 - 3) Save the template.

Workflows



Quality Central Configuration - Workflows



NiCE

One of the major selling points for Quality Central is being able to automate the QM process. QC does this by using Workflows. The workflow defines and controls all the relevant steps required to complete the evaluation process:

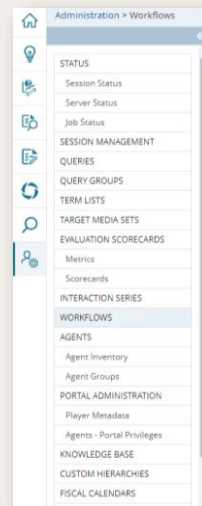
1. Identify the set of calls (population of calls) that can be used to evaluate the agents (using TMS).
2. Set the sampling rate how many calls from the population of calls are sent to the evaluators (number of evaluation per agents)
3. Assign the evaluators- which evaluators can perform the evaluation
4. Which evaluation form to use
5. Reevaluation or calibration if needed

The End step is not configurable but is required to complete the workflow. The Close step states if Notes are required to complete the workflow or how it should be tagged. NOTE: tags are not covered as part of this course. See the QC Getting Started Guide for more details.



Workflows

- Navigate to Workflows and click "Create a New QC Workflow"
- Enter a logical and descriptive name
- Choose a Start Date that is earlier than the date of the calls to be ingested
- Select the appropriate Target Media Set
- Assign a queue to the workflow
- Evaluate Step: Assign a specific Evaluation Form to the workflow — this will be the default form used by evaluators
- End Step: No action is required
- Close Step: Enable the "Allow interaction replacement" option



Navigate to **Workflows** and click **Create A New QC Workflow**. NOTE: Workflows can also be built from scratch, but this is not covered as part of this course.

Enter a logical name.

NOTE: if you attempt to save and the workflow has not been completed, you will receive an error.

Select the appropriate **Target Media Set**. Click the refresh button to verify how many calls will be selected from this set.

NOTE: You will have to run the SQL Job (NIA-Core-Update) around 5x to see the call records in the Workflow because the ETL job step is scheduled to run once every 5 times the NIA-Core-Update Job Pool is run. Depending on where it is in the cycle it may need to be run 5 times, but it could be 1.

Determine how many of the available interactions will be processed as part of the workflow. Unless the workflow is defined to sample every X days in a week/month, the interactions are sampled on each workflow

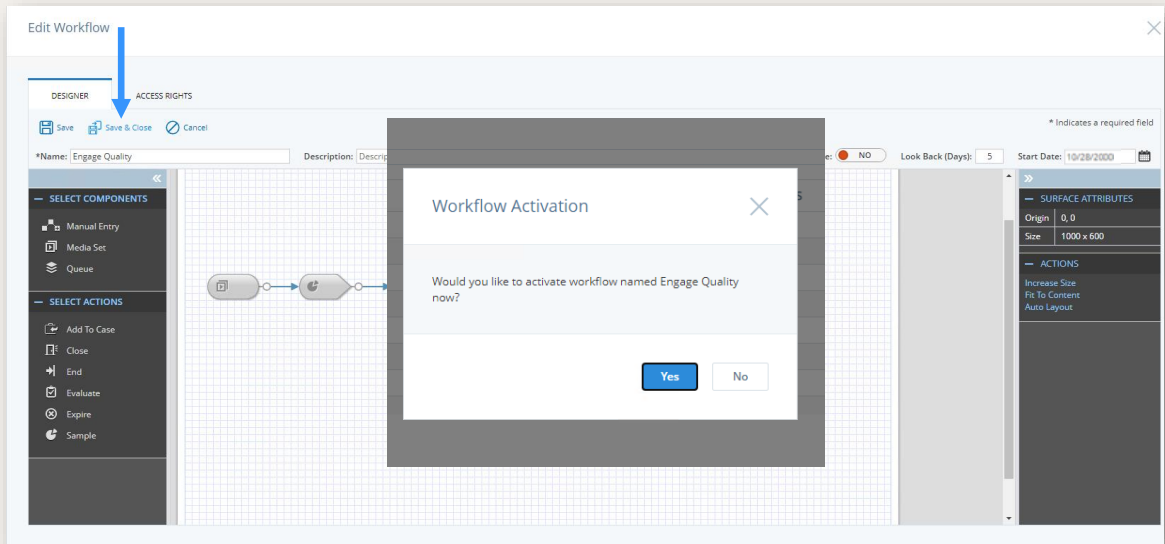
Assign a specific Evaluation form to the Workflow. This is the default form the Evaluators will use.

NOTE: Selecting the **"Allow interaction replacement"** option at the close step – this allows the evaluators to request a new interaction to evaluate.

For other Options, please see the QC QCI Getting Started guide available on DocHub



Workflows



This is a diagram of a completed workflow.

Once you click **Save & Close**, you will get a prompt for activating the Workflow. Click **Yes**



- Refer to exercise 12 - Create Workflow
 - This section creates a workflow to distribute interactions to review to the users with the Evaluator profile.
 - 1) Create a new QC Workflow and provide a logical name.
 - 2) Assign the Target Media Set created previously to this workflow.
 - 3) Select the number of calls to be sampled.
 - 4) Assign the queue to the Evaluator Profile (it should be assigned to the Queue, but not to a user).
 - 5) Select the Template created previously to the workflow.
 - 6) For the Close step, leave the defaults.

summary

- Logging into the Main and Portal interfaces
- Configuring the Quality Central Application
 - Profiles
 - Users
 - Agents
 - Language Pack
 - Sessions
 - Target Media Sets
 - Scorecards
 - Evaluation Form Template
 - Workflows





Thank You



Using Quality Central

NiCE Interaction Analytics & Quality Central
Fundamentals

Create a
NiCE
world 

objectives

- Use the Quality Central Application
 - Find Calls
 - Evaluate Calls
 - Review Evaluation Form
 - View Dashboard
 - Report on QM Data
 - Additional Functionality



USING QUALITY
CENTRAL



Using Quality Central

Find Calls

Evaluate Calls

Review Evaluations

View Dashboards

Report on QM Data

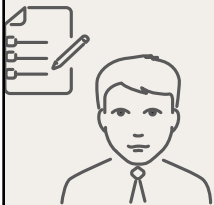
NICE

Once the Metadata, Users / Agents, and Audio (if applicable) has been imported from Engage or another source system, the quality tasks can be performed using QC. These are the basic functions to be defined in the Quality Central application.

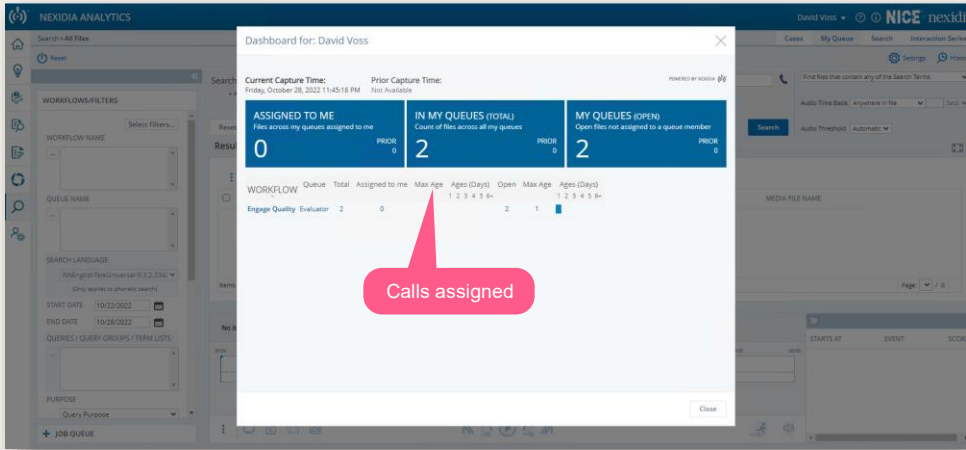


Find Calls

Portal



QA
Evaluator

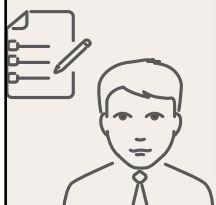


The Evaluator(dvoss) logs into the Main portal and finds calls assigned to them based on the workflow queue. NOTE: the dashboard shows up on initial login. Calls can be searched directly by selecting the Session.Calls can be searched by selecting the workflow. Highlight a call and the audio will begin playing (if available).



Evaluate Calls

Portal



QA
Evaluator

The screenshot shows the NEXIDIA ANALYTICS interface. On the left, there are filters for WORKFLOWS/FILTERS, WORKFLOW NAME (Engage Quality), QUEUE NAME (Evaluator), SEARCH LANGUAGE (N-EnglishTactUniversal-9.3.2.234C), START DATE (10/22/2020), END DATE (10/28/2022), and PURPOSE (Query Purpose). The main area displays 'Results for: All Files' with a table of call records. The table has columns: INTERACTION TYPE, DATE/TIME, DURATION, WORKFLOW, QUEUE, STATUS, ASSIGNED TO, and MEDIA FILE NAME. Two records are shown, both for 'Engage Quality' workflow and 'Evaluator' queue, with status 'Open'. Below the table, there is a 'No item selected' message and a timeline view.

INTERACTION TYPE	DATE/TIME	DURATION	WORKFLOW	QUEUE	STATUS	ASSIGNED TO	MEDIA FILE NAME
Audio	8/19/2022 12:14:3...	00:00:19	Engage Quality	Evaluator	Open		1_7129987917355745287_712998792160...
Audio	8/19/2022 12:14:1...	00:00:14	Engage Quality	Evaluator	Open		1_7129987827181432068_712998784000...

NICE

Once the call has been identified as a good candidate to be reviewed, the User clicks the Evaluate button to open the form.

NOTE: the call must be highlighted and not check-marked for it be evaluated.

NOTE: the Evaluation will open in a new window so pop-up blockers should be disabled.

Once the form opens, answer the relevant questions.

Grant access to Team Lead (smills).

Select Auto-suggest option.

By default, agent will have access to the Evaluation via the Portal.

To add an Agent, click none. Choose the appropriate agent and click OK

If required, click the next section and continue through the form.

Enter relevant information in the notes section.

Click Publish when the form has been completed.



Do it Yourself

- Refer to exercise 13 - Use Quality Central
- These exercises lets you see how the system works for evaluators, team leads, and agents. For this exercise, access the system using the Main (<http://nia/NxIA/Main>) and the Portal (<http://nia/NxIA/Portal>).
- **Evaluator**
 - 1) Login to the Main interface using the Evaluator User (dvoss account).
 - 2) Add a filter for "Agent" in search and search for interactions for "Carl Swift". Highlight an interaction and verify there is audio.
 - 3) Evaluate 1 or 2 interactions using the defined Evaluation Form Template. Make sure the evaluation can be acknowledged by the agent.
 - 4) Publish the evaluation (make sure it is accessible to the Agent).



Review Evaluations

Evaluation Portal

ID Contains From 10/01/2022 to 10/31/2022 Row Contains ☐ Self-Evaluations only

View Print Evaluate

ID	EVALUATOR	VERSION	TEMPLATE	AGENT	SCORE	AGENT SCORE	PUBLISHED	ACKNOWLEDGED ON	NOTES	COACHING REQUEST	COACHED	STATUS	FO
	David Voss	2	EngageQuality	Peter Britt	50.00 / 50 (100.00%)		10/31/2022		Verify Billing Information			Published	

- Once the evaluation is **saved and published**, it becomes available for the agent to review
- Click the **Evaluation ID** to open the form
- Agents can click **"Request Coaching"** on the form, enter relevant details in the pop-up window, and click **OK**
- Evaluators can add **coaching notes** during the evaluation process
- Click the **Coaching** button to view coaching details
- After reviewing the form, agents can click the **Acknowledge** button
- Use the **History** button to view the evaluation's activity log



Agent



Team Lead

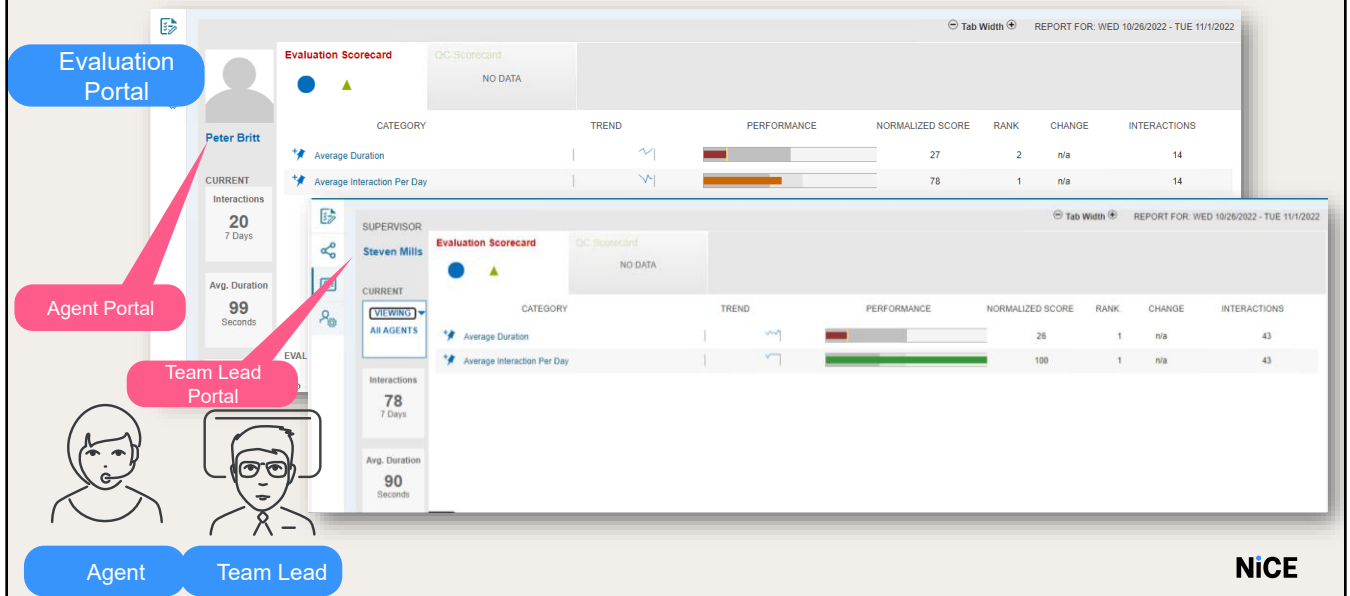
NiCE

When the Evaluation is saved and published, it becomes available for review by the Agent using the Evaluation Portal. NOTE: Evaluations can be reviewed by Users using the Main portal or the Evaluations Portal. Click the ID

Evaluators can add coaching notes during the evaluation



View Dashboards



There are scorecards for Evaluators / Supervisors and scorecards for Agents. The result for each of them can be found in the Evaluators Portal Dashboard. NOTE: Team Leads will also have an option for looking at all agents / specific agents.



Do it Yourself

- Refer to exercise 14- Use Quality Central
- **Team Lead**
 - 1) Login to the Portal interface using the Team Lead User (smills account).
 - 2) Verify evaluations created by the Evaluator are available.
 - 3) Verify the Evaluation Scorecard shows data in the dashboard.

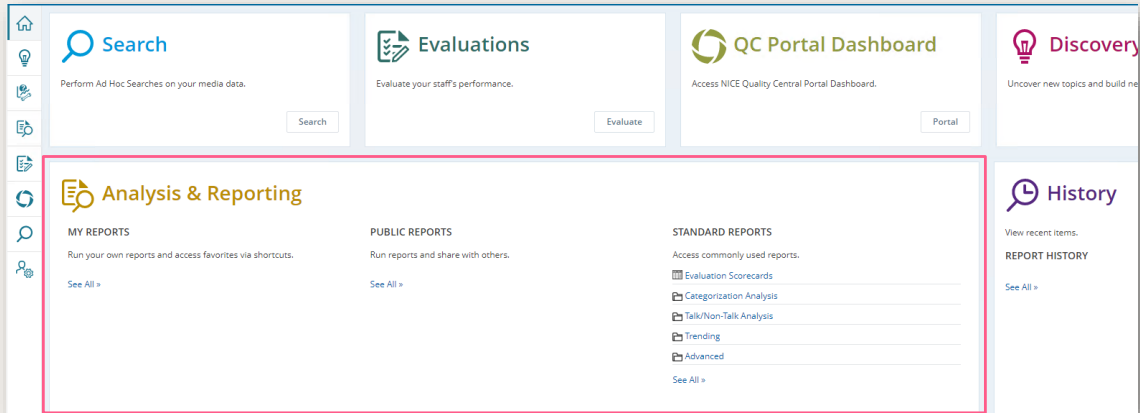


Do it Yourself

- Refer to exercise 15 - Use Quality Central
 - **Agent**
 - 1) Login to the Portal interface (<http://nia/NxlA/Portal>) using the Agent account (cswiftaccount).
 - 2) Verify evaluations created by the Evaluator are available. Acknowledge at least one evaluation.
 - 3) Verify the Evaluation Scorecard shows data in the dashboard.



Report on QM Data



There are pre-defined Standard Reports. Each category has several report types available.

Click the title to drill down to report criteria.

Click **Evaluations -> Evaluations Performance**

The report can be run on-demand by clicking the title. Other options for running the report are available – subscriptions, send now, export (to XLS, etc), or PDF.

For this report, the only required fields are the Evaluation Form and date range.

Do it Yourself

- Refer to exercise 17 - Use Quality Central Exercise
 - **Team Lead**
 - 1) Navigate to the QC Reports
 - 2) Run an Evaluation Performance Report



summary

- How to use the Quality Central Application
 - Find Calls
 - Evaluate Calls
 - Review Evaluation Form
 - View Dashboard
 - Report on QM Data
 - Additional Functionality





Thank You

QC Architecture

NiCE Interaction Analytics & Quality
Central Fundamentals

Create a
NiCE
world 

objectives

- Describe the Ingest Flow
- Describe the Playback Flow
- Additional Features



Ingest Flow



Databases



NexidiaESI

Production DB - App Configuration (Users, Agents, Workflows, Call Data, ingest



NexidiaESIDW

Data Warehouse - Stores information used in reports, from the data populated via SQL jobs from QC, Evals DBs and Discovery DB



NexidiaEvaluations

Information about Evaluations; moved to DW

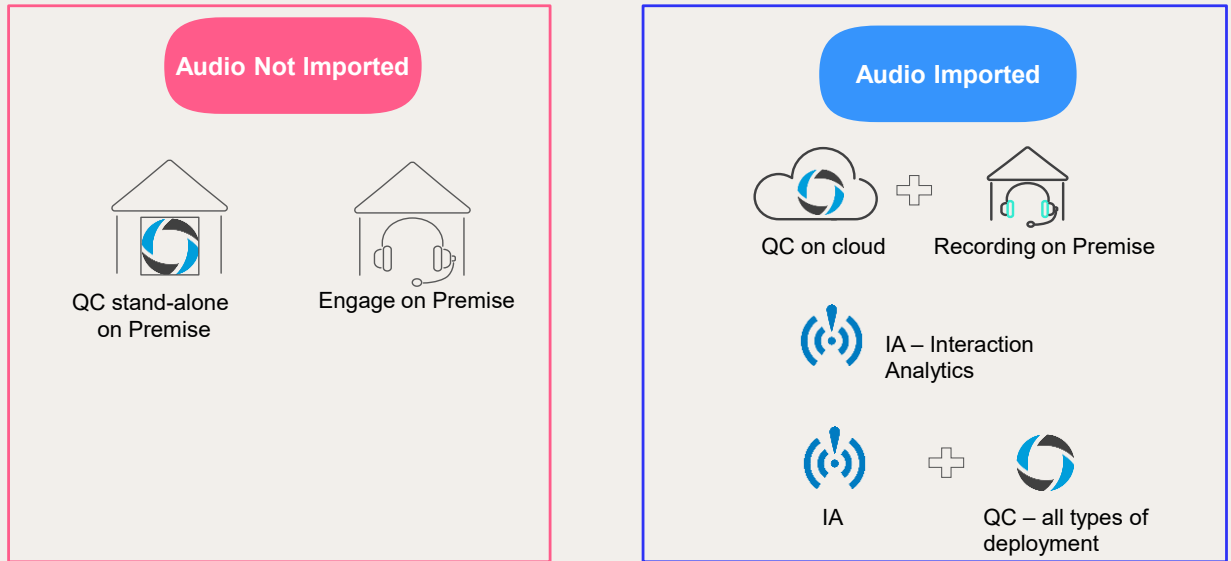


NxDiscovery

Information about Words cloud



DEF and Audio Import



NiCE

• NICE Recording On Prem + Quality Central On Prem

No replication of media

Integrated playback from Engage

Any Recording On Prem + Quality Central in Cloud

Replicate media from on prem to cloud

Playback directly from QC storage

Any Recording in Cloud + Quality Central in Cloud

Replicate media from cloud to cloud

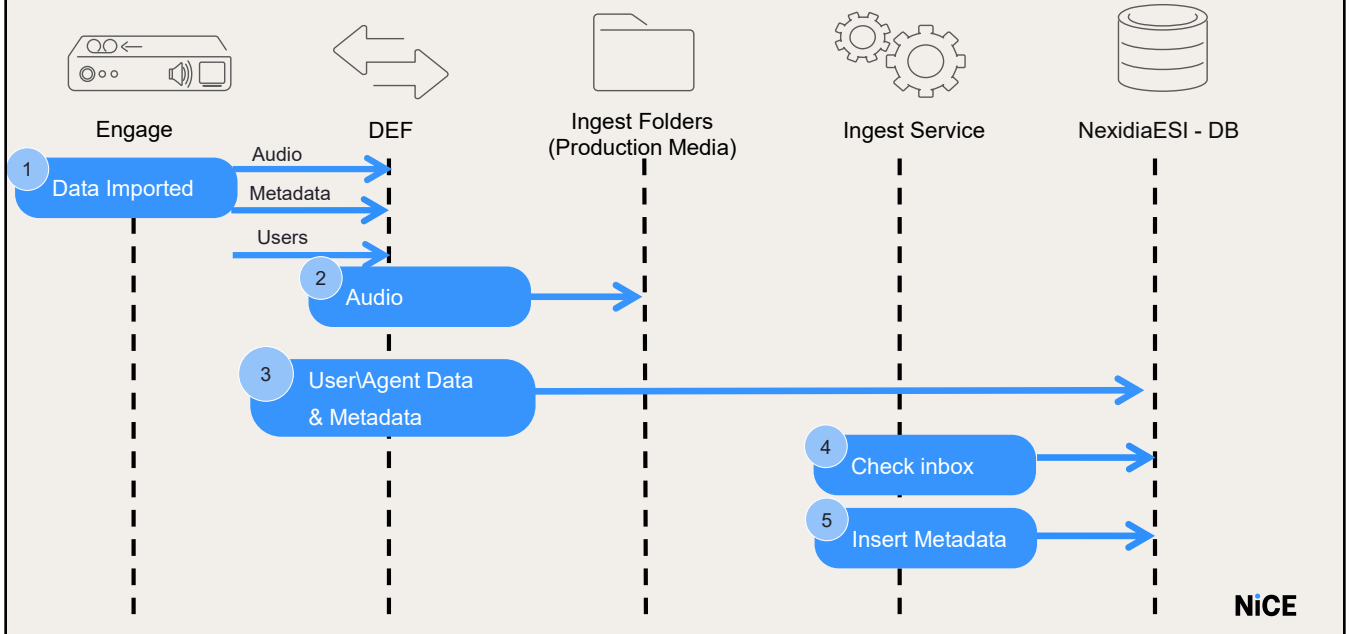
Playback directly from the cloud

The only case where DEF will not import the audio files from the recording system is when we have a QC stand-alone system where both the QC system and the Engage system are installed on-premise at the customer site.

If IA is installed, then the audio must be imported into Interaction Analytics since its have to be processed by the analytics engines.



Ingest Flow (QC) – Inbox Table



1. DEF is used to import information from the Engage system – the following information is imported from Engage: users and agents data, interaction’s metadata, and audio files. NOTE: In QC stand-alone system where both the QC system and the Engage system installed on-premise, DEF will not import the audio files. NOTE: Users / Agents are not imported from 3rd-party Recorders.
2. DEF places the audio in staging folders – only if it is not QC standalone on primes
3. DEF inserts the interaction’s metadata to the database (NexidiaESI > NexidiaESI.SourceMediaInbox table) and the agent’s \user’s data to NexidiaESI>table.....
4. Ingest service checks the inbox table for new entries new interactions to process.
5. The ingest service inserts\process the interaction metadata into the NexidiaESI DB.

Ingest Flow

If NIA is installed audio must be imported into Interaction Analytics

Users / Agents are NOT imported from 3rd-party solutions

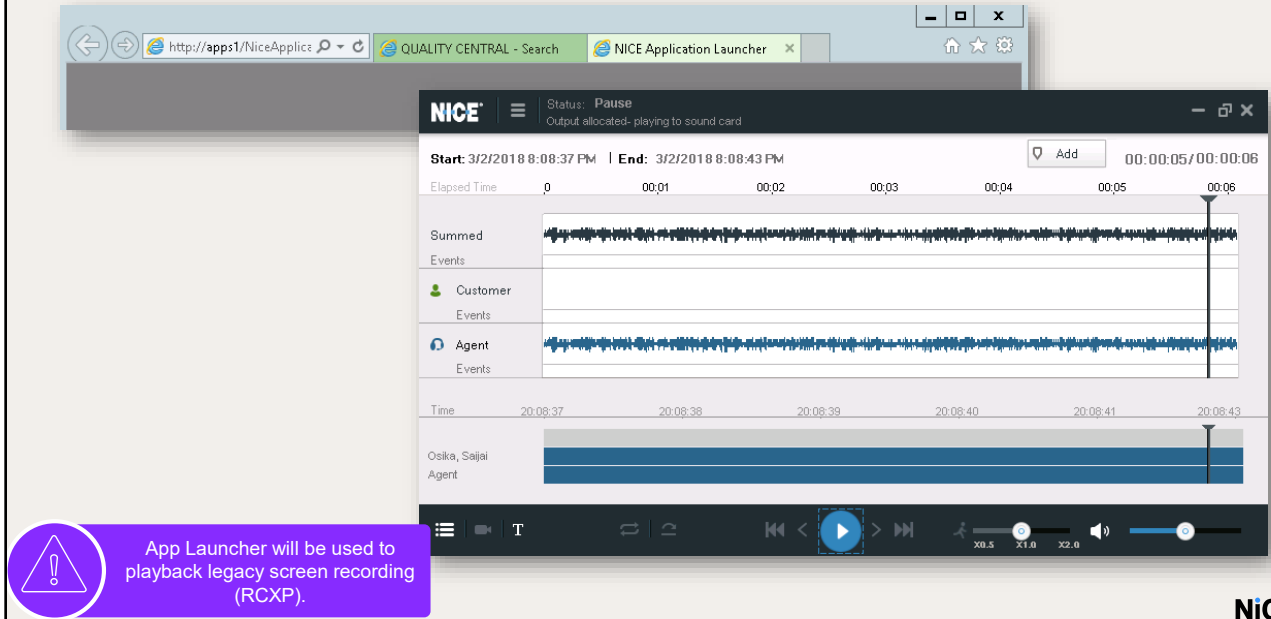
In QC stand-alone system installed on-premise, audio files will not be imported into QC server



Playback Flow



Engage App Launcher



The screenshot displays the Engage App Launcher interface. At the top, there is a browser window with tabs for 'http://apps1/NiceApplica...', 'QUALITY CENTRAL - Search', and 'NICE Application Launcher'. Below this, the main application window is titled 'NICE' and shows a 'Status: Pause' message. The interface includes a timeline with 'Start: 3/2/2018 8:08:37 PM' and 'End: 3/2/2018 8:08:43 PM'. A 'Summed' audio track is visible, along with a 'Customer' and 'Agent' section. The 'Agent' section shows a timeline with 'Time' markers from 20:08:37 to 20:08:43. A 'Osika, Sajjai Agent' section is also present. A purple callout box with a warning icon states: 'App Launcher will be used to playback legacy screen recording (RCXP)'. The bottom right corner features the 'NiCE' logo.

Quality Central offers 3 methods to playback from Engage-Nexidia Player, NICE Player Pro, Engage App Launcher. This appears in a separate tab with an independent player. NOTE: only Internet Explorer is supported for the Engage App Launcher.

Engage App Launcher

The screenshot displays the Engage App Launcher interface. At the top, there's a search bar and a table of media files. Below the table, there's a 'NOTES & TAGS' section and a call playback timeline.

MEDIAFILENAME	MEDIA TYPE	DURATION	DATE/TIME	AGENT	AGENTID	SUPERVISOR	SITE
732332210102777145.	Audio	00:10:08	4/1/2019 8:20:28 PM	Joe Bevel	11980	Stephanie Ervin	Atlanta
732330983626978762.	Audio	00:15:20	4/1/2019 9:09:28 PM	Joe Bevel	11980	Stephanie Ervin	Atlanta
playlistHold.mp4	Audio	00:00:41	3/1/2006 9:00:00 AM	Patricia Genovese	11990	Stephanie Ervin	Atlanta
Wrap_Up.nmf	Audio	00:01:01	3/1/2006 9:00:00 AM	Patricia Genovese	11990		
7324373973890749925.	Audio	00:10:10	3/1/2006 9:00:00 AM	Patricia Genovese	11990		
7324373973890769627.	Audio	00:02:33	3/1/2006 9:00:00 AM	Patricia Genovese	11990		

Below the table, there's a 'NOTES & TAGS' section with a 'Notes' field and a 'Find Tags' field. At the bottom, there's a call playback timeline showing the duration of the call and the segments played.

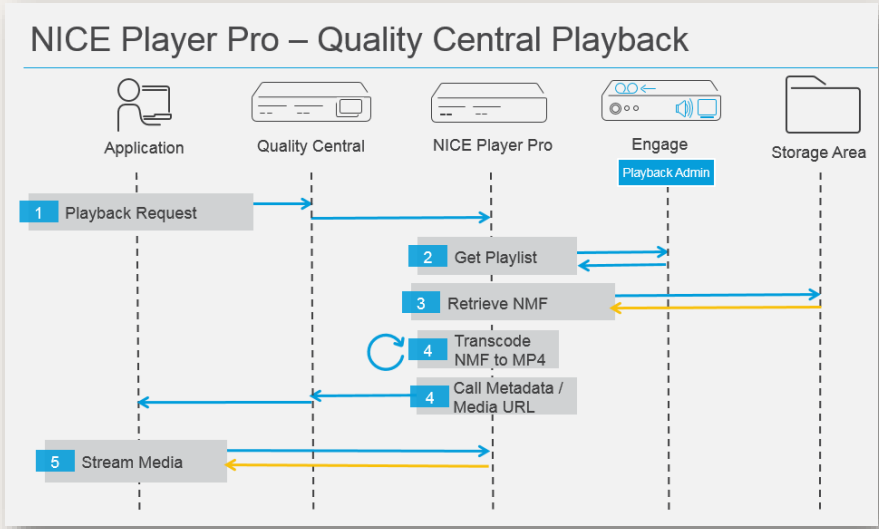
If the call is not eligible to be played with Player Pro and App Launcher is enabled, button is shown

App Launcher reduce the Media storage size needed on Interaction Analytics servers – the audio is located on the Engage server and not on the Interaction Analytics server. In cases we need to export the audio files from Engage to Interaction Analytics the audio files will be saved on the Interaction Analytics sever for a short time and as a result we don't need Large storage area for the audio files.



There is a Web-Based Training for the Engage App Launcher available on NiCE Dojo

NICE Player Pro



NPP allows for faster playback than using the Engage App Launcher. It can be integrated with NICE Quality Central and Engage.



Nice Player Pro

SUPERVISOR

Stephanie Ervin

TEAM PERFORMANCE

NPP Scorecard

CURRENT

VIEWING All AGENTS

CATEGORY

Local Cache Session

This Week

15

Total Interac...

Avg. Duration

431

Seconds

Non-Talk

34%

NPP

100

80

60

40

20

00

Click to show screen

NICE Quality Central - Media Player - Google Chrome

Not secure | dv-air-nxdb/NxlA/Portal/Evaluations/MediaPlayer.aspx?SharedMediaSetId=68&SupervisorKey=38&AgentKey=-1&...

NPP Scorecard : Local Cache Session

6 results for File Set Name PortalDashboardNameSet_156678b0-95c4-4753-96a7-e11144f0d800

Agent Name	Agent Id	Ext. Id	From	To	MediaFilename	View	
MEDIAFILENAME	MEDIA TYPE	DURATION	DATE/TIME	AGENT	AGENTID	SUPERVISOR	SITE
7323322210102777145.	Audio	00:10:08	4/1/2019 3:20:28 PM	Joe Bevel	11980	Stephanie Ervin	Atlanta
7323320983626978762.	Audio	00:15:20	4/1/2019 3:02:28 PM	Joe Bevel	11980	Stephanie Ervin	Atlanta
playlchHold.mp4	Audio	00:00:41	3/1/2006 9:00:00 AM	Patricia Genovese	11990	Stephanie Ervin	Atlanta
Wrap_Up.nmf	Audio	00:01:01	3/1/2006 9:00:00 AM	Patricia Genovese	11990	Stephanie Ervin	Atlanta
7324373973890749925.	Audio	00:10:10	3/1/2006 9:00:00 AM	Patricia Genovese	11990	Stephanie Ervin	Atlanta
7324373375500769627.	Audio	00:02:33	3/1/2006 9:00:00 AM	Patricia Genovese	11990	Stephanie Ervin	Atlanta

Showing items 1 - 6 of 6 | Page 1 of 1

NOTES & TAGS

Notes

ADD

Find Tags

ADD

CLEAR

4/1/2019, 3:18:48 PM | Duration: 00:15:20 | Paused

Toggle Segments | 00:00:00

Toggle Screen

Add Note

Loop Segment

Skip Non-Speech

⏮

⏪

⏩

⏭

⏸

⏹

⏻

⏼

FOLLOW-UP ACTIONS

Outstanding Requests

1

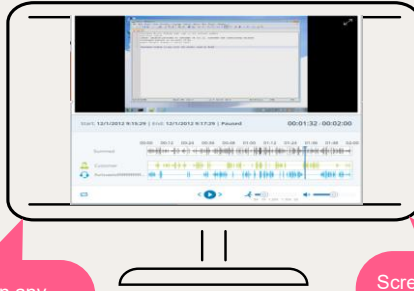
TARGET DATE



Player Pro

Browser-based (HTML5)

Zero footprint on desktop



Supports any browser on any device

Screen recording – support only H.264 codec

NiCE

Player Pro serves as a way to stream interactions recorded by Engage, from Engage Replaces the need to use App Launcher to be able to view the screen recording of a call

With the caveat that the call's screen was recorded using the browser-friendly H.264 codec

Integration is completely seamless to the end user

Calls from Player Pro get played directly in the Interaction Analytics media player in Search/QC/etc.





There are Web-Based Trainings for the NICE Player Pro solution available on NICE DOJO.



Additional Features



Additional Features

Dossier

- An interactive display for showcasing and exploring business data
- Designed to help users interpret and analyze data effectively

Explore

- A tool for exploring interaction data and uncovering insights
- User-friendly interface for defining query searches using content and metadata

**Real Time
Interaction
Guidance**

- Uses speech and predictive analytics to score agent behaviors
- Provides real-time alerts to agents for behavior adjustments
- Helps improve call outcomes and customer experience



Additional Features

**Total
Access**

- An interactive display for showcasing and exploring business data
- Designed to help users interpret and analyze data effectively

**Enlighten
XO**

- Enhances self-service applications using real agent-assisted interactions
- Analyzes audio and chat conversations to extract valuable insights

**Auto
Discovery**

- A topic clustering tool for organizing interaction data
- Automatically clusters interactions into related topics
- Maps tagged queries to clusters to improve coverage and accuracy



Additional Features

Data
Export
Utility

- Facilitates data consumption from Interaction Analytics
- Designed for integration with corporate data warehouses and analytics platforms



summary

- Describe the Ingest Flow
- Describe the Playback Flow
- Additional Features





Thank You

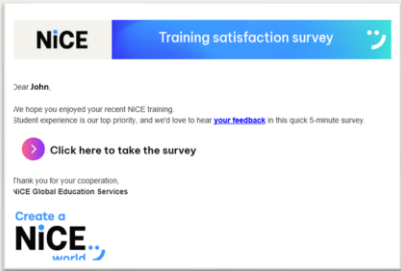
Course Closing

NiCE Interaction Analytics & Quality Central
Fundamentals

Create a
NiCE..
world ☺



Course Survey

A screenshot of the "Training satisfaction survey" form. The header shows the NiCE logo and the title "Training satisfaction survey" with a smiley face icon. The first question is "How satisfied are you with the overall training experience?". Below it is a 5-point scale: 1 (Very Dissatisfied), 2 (Dissatisfied), 3 (Neither Satisfied nor Dissatisfied), 4 (Satisfied), and 5 (Very Satisfied). The second question is "On a scale of 1 to 5, how satisfied are you with the following aspects of the training course:". Below this is a table with 5 columns corresponding to the scale points and 5 rows of aspects. A purple callout bubble points to the scale, stating "Scores range from 'Very Dissatisfied' (left) to 'Very Satisfied' (right)".

	Very Dissatisfied	Dissatisfied	Neither Satisfied nor Dissatisfied	Satisfied	Very Satisfied
The instructor's knowledge of the subject matter	1	2	3	4	5
The instructor's delivery, attitude and behavior	1	2	3	4	5
The course content and structure (agenda, presentations, documents, etc.)	1	2	3	4	5
The balance's between lectures, demos, and hands-on activities	1	2	3	4	5
The labs used for demos and exercises	1	2	3	4	5

Please leave us comments on how we can improve the learning experience for this course

You should have received a course survey in your email from CSATTeam@nice.com. Please take a few moments to take the course survey before we proceed.

If you did not receive the course survey in your Inbox, check your Junk email folder. If you did not receive the survey, please let your instructor know so that he/she can have it resent to you.



Written Exam Instructions

Examination Instructions

Welcome to your online test.

As you have only ONE ATTEMPT to pass this test, please ensure the following points are followed before you start:

1. You have covered all learning materials and feel confident to pass the test

2. You are in a quiet environment

3. Once started, this test will run for exactly 40 minutes so try to avoid any disruption

Test name: **NiCE Nexidia Analytics & QC v12.4 - Fundamentals - Theoretical Exam**

Number of questions: **20**

Allotted time: **40 minutes**

Passing score: **80**

This course and subsequent test have been designed to ensure that you have the relevant knowledge to fulfill your role with the utmost success. We therefore ask you to complete the test in the appropriate environment and answer the questions openly and honestly.

Use the assessment page to respond to or mark questions for follow-up. When finished, click the Summary button to review the answers and the questions marked for follow-up. Note that changes cannot be made once the assessment has been submitted. Upon completion, click the Submit button to save the answers.

Good Luck!

Warning:

Please do not use your Browser buttons to navigate in the Test.

Please use the navigation buttons at the bottom of each page.

Your test answers will not be recorded if you navigate using the Browser buttons.

Cancel

Continue

Read the Written Exam instructions carefully

Use all your resources from the course including the lab and other NiCE documents

Complete all assigned tasks

Email the instructor with your final score

After clicking the Launch Test link, an Examination Instructions page will appear. Make sure you read the Examination Instructions before beginning the written exam. Especially heed the warning regarding navigation. Do not use the browser forward and back buttons during the test. Always use the navigation buttons at the bottom of each test page. If you use the browser buttons, your test answers will not be recorded.





Written Exam Instructions

Complete the exam in the allotted time. Make sure to submit the exam before the 2:00 countdown

Test - NiCE Nexidia Analytics & QC v12.3 - Troubleshooting Fundamentals - Theoretical Exam

Time remaining
36:10

Use Mark for follow up to return to questions later

☐ Mark for follow up

Use the navigation at the bottom of the page and not the browser. Click Summary to review your answers before submitting them

Save / Return Later

Summary

You must complete the exam in the allotted time. The clock in the upper right hand corner will specify how much time you have remaining.

NOTE: Make sure to submit your test prior to the 2:00 minute warning. If not, the test session may expire.

Do not get hung up on a single question. If you don't know the answer, make your best guess and move on. Use the Mark for follow-up box to check questions that you want to go back later and reconsider. These may be questions you have answered or not answered. You can come back later and try to answer the question again.

Use the buttons at the bottom of the test pages to navigate through the exam. The summary button will take you to a page where you can see all questions marked for follow-up and all unanswered questions.





Practical Exam

NiCE Nexidia Analytics and Quality Central – Practical Exam

General Instructions

Students will complete the steps to configure the NiCE Nexidia Analytics and Quality Central solution. The software and License keys will be provided by the Trainer. The systems already have the Operating System, SQL, and NIA+QC installed.

Please feel free to refer to any documentation, manuals, or notes to assist in completing the Practical Exam

Duration: 2 hours

You have been assigned a project to configure the NIA + QC solution at a customer site. They have provided test calls, users, and agents to import in the system to verify functionality. The system should be configured according to the customer specifications outlined below. Make sure to successfully import the calls using the QC Template workflow.

Read the Practical Exam document carefully

Complete all assigned tasks

Notify the instructor when complete

You will receive the Practical Assessment document from the instructor. Before beginning please read the entire document carefully. Complete all assigned tasks and notify the instructor when everything has been completed.





Thank You



Contact us: training@nice.com | Visit us: [NiCE Dojo](#)

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